

Nostr Book of NIPs

TODO

Abstract

This is the abstract.
It consists of two paragraphs.

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Introduction

Welcome to the Nostr Books of NIPs, which is a streamlined guide to the Nostr Implementation Possibilities (NIPs). Instead of sticking to the original numerical order, I've grouped similar NIPs together to make them easier to understand and more practical to use. Whether you're new to Nostr or a seasoned participant, this reorganized format should help you get a better grip on how things work and what's being developed.

I want to be clear: I didn't write the NIPs. All the credit goes to the original authors and contributors of these notes. My contribution has been to sort these NIPs into a flow that makes sense and brings out the connections between them, making everything more accessible.

Each section of this book kicks off with a short introduction to give you a heads-up on what to expect from the NIPs that follow. The goal is to make the technical details a bit friendlier and the big ideas a bit clearer, so more people can join in, understand, and contribute to the Nostr community.

Thanks for picking up this book! I hope it helps you navigate the exciting waters of Nostr more easily and encourages you to dive deeper into this innovative project. Let's explore and build the future of decentralized communication together!

Git Commit

To provide readers with the most up-to-date information, this page showcases the latest git commit from the nostr-protocol/nips repository on GitHub. This commit log offers a snapshot of the most recent changes, updates, and enhancements made to the Nostr Improvement Proposals (NIPs). By incorporating this information, readers can gain insight into the ongoing development and evolution of the Nostr protocol, ensuring they are informed about the latest contributions and modifications from the community. This inclusion underscores the dynamic nature of the project and highlights the collaborative efforts driving its progress.

```
commit 19f7d6652601ca32a9960d30e1954f9e3b0476b1
```

```
Author: Henrique Albuquerque <18596542+henrialb@users.noreply.github.com>
```

```
Date: Sat Dec 13 13:49:19 2025 +0100
```

```
Fix typo (#2158)
```

Nostr Overview

...

NIP-01

Basic protocol flow description

draft mandatory

This NIP defines the basic protocol that should be implemented by everybody. New NIPs may add new optional (or mandatory) fields and messages and features to the structures and flows described here.

Events and signatures

Each user has a keypair. Signatures, public key, and encodings are done according to the Schnorr signatures standard for the curve secp256k1.

The only object type that exists is the event, which has the following format on the wire:

```
{
  "id": <32-bytes lowercase hex-encoded sha256 of the serialized event data>,
  "pubkey": <32-bytes lowercase hex-encoded public key of the event creator>,
  "created_at": <unix timestamp in seconds>,
  "kind": <integer between 0 and 65535>,
  "tags": [
    [<arbitrary string>...],
    // ...
  ],
  "content": <arbitrary string>,
  "sig": <64-bytes lowercase hex of the signature of the sha256 hash of the serialized event data, which is the same as the "id" field>
}
```

To obtain the event.id, we sha256 the serialized event. The serialization is done over the UTF-8 JSON-serialized string (which is described below) of the following structure:

```
[
  0,
  <pubkey, as a lowercase hex string>,
  <created_at, as a number>,
  <kind, as a number>,
  <tags, as an array of arrays of non-null strings>,
  <content, as a string>
]
```

To prevent implementation differences from creating a different event ID for the same event, the following rules MUST be followed while serializing: - UTF-8 should be used for encoding. - Whitespace, line breaks or other unnecessary formatting should not be included in the output JSON. - The following characters in the content field must be escaped as shown, and all other characters must be included verbatim: - A line break (0x0A), use \n - A double quote (0x22), use \" - A backslash (0x5C), use \\ - A carriage return (0x0D), use \r - A tab character (0x09), use \t - A backspace, (0x08), use \b - A form feed, (0x0C), use \f

Tags

Each tag is an array of one or more strings, with some conventions around them. Take a look at the example below:

```
{
  "tags": [
    ["e", "5c83da77af1dec6d7289834998ad7aafbd9e2191396d75ec3cc27f5a77226f36", "wss://nostr.example.com"],
    ["p", "f7234bd4c1394dda46d09f35bd384dd30cc552ad5541990f98844fb06676e9ca"],
    ["a", "30023:f7234bd4c1394dda46d09f35bd384dd30cc552ad5541990f98844fb06676e9ca:abcd", "wss://nostr.example.com"],
    ["alt", "reply"],
    // ...
  ]
}
```

```
],
// ...
}
```

The first element of the tag array is referred to as the tag *name* or *key* and the second as the tag *value*. So we can safely say that the event above has an *e* tag set to "5c83da77af1dec6d7289834998ad7aafbd9e2191396d75ec3cc27f5a77226f36", an *alt* tag set to "reply" and so on. All elements after the second do not have a conventional name.

This NIP defines 3 standard tags that can be used across all event kinds with the same meaning. They are as follows:

- The *e* tag, used to refer to an event: ["e", <32-bytes lowercase hex of the id of another event>, <recommended relay URL, optional>, <32-bytes lowercase hex of the author's pubkey, optional>]
- The *p* tag, used to refer to another user: ["p", <32-bytes lowercase hex of a pubkey>, <recommended relay URL, optional>]
- The *a* tag, used to refer to an addressable or replaceable event
 - for an addressable event: ["a", "<kind integer>:<32-bytes lowercase hex of a pubkey>:<d tag value>", <recommended relay URL, optional>]
 - for a normal replaceable event: ["a", "<kind integer>:<32-bytes lowercase hex of a pubkey>:", <recommended relay URL, optional>] (note: include the trailing colon)

As a convention, all single-letter (only english alphabet letters: a-z, A-Z) key tags are expected to be indexed by relays, such that it is possible, for example, to query or subscribe to events that reference the event "5c83da77af1dec6d7289834998ad7aafbd9e2191396d75ec3cc27f5a77226f36" by using the {"#e": ["5c83da77af1dec6d7289834998ad7aafbd9e2191396d75ec3cc27f5a77226f36"]} filter. Only the first value in any given tag is indexed.

Kinds

Kinds specify how clients should interpret the meaning of each event and the other fields of each event (e.g. an "r" tag may have a meaning in an event of kind 1 and an entirely different meaning in an event of kind 10002). Each NIP may define the meaning of a set of kinds that weren't defined elsewhere. NIP-10, for instance, specifies the kind:1 text note for social media applications.

This NIP defines one basic kind:

- **0: user metadata:** the content is set to a stringified JSON object {name: <nickname or full name>, about: <short bio>, picture: <url of the image>} describing the user who created the event. Extra metadata fields may be set. A relay may delete older events once it gets a new one for the same pubkey.

And also a convention for kind ranges that allow for easier experimentation and flexibility of relay implementation:

- for kind *n* such that $1000 \leq n < 10000$ || $4 \leq n < 45$ || $n == 1$ || $n == 2$, events are **regular**, which means they're all expected to be stored by relays.
- for kind *n* such that $10000 \leq n < 20000$ || $n == 0$ || $n == 3$, events are **replaceable**, which means that, for each combination of pubkey and kind, only the latest event **MUST** be stored by relays, older versions **MAY** be discarded.
- for kind *n* such that $20000 \leq n < 30000$, events are **ephemeral**, which means they are not expected to be stored by relays.
- for kind *n* such that $30000 \leq n < 40000$, events are **addressable** by their kind, pubkey and *d* tag value – which means that, for each combination of kind, pubkey and the *d* tag value, only the latest event **MUST** be stored by relays, older versions **MAY** be discarded.

In case of replaceable events with the same timestamp, the event with the lowest id (first in lexical order) should be retained, and the other discarded.

When answering to REQ messages for replaceable events such as {"kinds": [0], "authors": [<hex-key>]}, even if the relay has more than one version stored, it **SHOULD** return just the latest one.

These are just conventions and relay implementations may differ.

Communication between clients and relays

Relays expose a websocket endpoint to which clients can connect. Clients SHOULD open a single websocket connection to each relay and use it for all their subscriptions. Relays MAY limit number of connections from specific IP/client/etc.

From client to relay: sending events and creating subscriptions

Clients can send 3 types of messages, which must be JSON arrays, according to the following patterns:

- ["EVENT", <event JSON as defined above>], used to publish events.
- ["REQ", <subscription_id>, <filters1>, <filters2>, ...], used to request events and subscribe to new updates.
- ["CLOSE", <subscription_id>], used to stop previous subscriptions.

<subscription_id> is an arbitrary, non-empty string of max length 64 chars. It represents a subscription per connection. Relays MUST manage <subscription_id>s independently for each WebSocket connection. <subscription_id>s are not guaranteed to be globally unique.

<filtersX> is a JSON object that determines what events will be sent in that subscription, it can have the following attributes:

```
{
  "ids": <a list of event ids>,
  "authors": <a list of lowercase pubkeys, the pubkey of an event must be one of these>,
  "kinds": <a list of a kind numbers>,
  "#<single-letter (a-zA-Z)>": <a list of tag values, for #e – a list of event ids, for #p – a list of pubkeys,
    etc.>,
  "since": <an integer unix timestamp in seconds. Events must have a created_at >= to this to pass>,
  "until": <an integer unix timestamp in seconds. Events must have a created_at <= to this to pass>,
  "limit": <maximum number of events relays SHOULD return in the initial query>
}
```

Upon receiving a REQ message, the relay SHOULD return events that match the filter. Any new events it receives SHOULD be sent to that same websocket until the connection is closed, a CLOSE event is received with the same <subscription_id>, or a new REQ is sent using the same <subscription_id> (in which case a new subscription is created, replacing the old one).

Filter attributes containing lists (ids, authors, kinds and tag filters like #e) are JSON arrays with one or more values. At least one of the arrays' values must match the relevant field in an event for the condition to be considered a match. For scalar event attributes such as authors and kind, the attribute from the event must be contained in the filter list. In the case of tag attributes such as #e, for which an event may have multiple values, the event and filter condition values must have at least one item in common.

The ids, authors, #e and #p filter lists MUST contain exact 64-character lowercase hex values.

The since and until properties can be used to specify the time range of events returned in the subscription. If a filter includes the since property, events with created_at greater than or equal to since are considered to match the filter. The until property is similar except that created_at must be less than or equal to until. In short, an event matches a filter if since <= created_at <= until holds.

All conditions of a filter that are specified must match for an event for it to pass the filter, i.e., multiple conditions are interpreted as && conditions.

A REQ message may contain multiple filters. In this case, events that match any of the filters are to be returned, i.e., multiple filters are to be interpreted as || conditions.

The limit property of a filter is only valid for the initial query and MUST be ignored afterwards. When limit: n is present it is assumed that the events returned in the initial query will be the last n events ordered by the created_at. Newer events should appear first, and in the case of ties the event with the lowest id (first in lexical order) should be first. Relays SHOULD use the limit value to guide how many events are returned in the initial response. Returning fewer events is acceptable, but returning (much) more should be avoided to prevent overwhelming clients.

From relay to client: sending events and notices

Relays can send 5 types of messages, which must also be JSON arrays, according to the following patterns:

- ["EVENT", <subscription_id>, <event JSON as defined above>], used to send events requested by clients.
- ["OK", <event_id>, <true|false>, <message>], used to indicate acceptance or denial of an EVENT message.
- ["EOSE", <subscription_id>], used to indicate the *end of stored events* and the beginning of events newly received in real-time.
- ["CLOSED", <subscription_id>, <message>], used to indicate that a subscription was ended on the server side.
- ["NOTICE", <message>], used to send human-readable error messages or other things to clients.

This NIP defines no rules for how NOTICE messages should be sent or treated.

- EVENT messages MUST be sent only with a subscription ID related to a subscription previously initiated by the client (using the REQ message above).
- OK messages MUST be sent in response to EVENT messages received from clients, they must have the 3rd parameter set to true when an event has been accepted by the relay, false otherwise. The 4th parameter MUST always be present, but MAY be an empty string when the 3rd is true, otherwise it MUST be a string formed by a machine-readable single-word prefix followed by a : and then a human-readable message. Some examples:
 - ["OK", "b1a649ebe8...", true, ""]
 - ["OK", "b1a649ebe8...", true, "pow: difficulty 25>=24"]
 - ["OK", "b1a649ebe8...", true, "duplicate: already have this event"]
 - ["OK", "b1a649ebe8...", false, "blocked: you are banned from posting here"]
 - ["OK", "b1a649ebe8...", false, "blocked: please register your pubkey at <https://my-expensive-relay>"]
 - ["OK", "b1a649ebe8...", false, "rate-limited: slow down there chief"]
 - ["OK", "b1a649ebe8...", false, "invalid: event creation date is too far off from the current time"]
 - ["OK", "b1a649ebe8...", false, "pow: difficulty 26 is less than 30"]
 - ["OK", "b1a649ebe8...", false, "restricted: not allowed to write."]
 - ["OK", "b1a649ebe8...", false, "error: could not connect to the database"]
 - ["OK", "b1a649ebe8...", false, "mute: no one was listening to your ephemeral event and it wasn't handled in any way, it was ignored"]
- CLOSED messages MUST be sent in response to a REQ when the relay refuses to fulfill it. It can also be sent when a relay decides to kill a subscription on its side before a client has disconnected or sent a CLOSE. This message uses the same pattern of OK messages with the machine-readable prefix and human-readable message. Some examples:
 - ["CLOSED", "sub1", "unsupported: filter contains unknown elements"]
 - ["CLOSED", "sub1", "error: could not connect to the database"]
 - ["CLOSED", "sub1", "error: shutting down idle subscription"]
- The standardized machine-readable prefixes for OK and CLOSED are: duplicate, pow, blocked, rate-limited, invalid, restricted, mute and error for when none of that fits.

Communication

NIP-10

Text Notes and Threads

draft optional

This NIP defines kind:1 as a simple plaintext note.

Abstract

The .content property contains some human-readable text.

e tags can be used to define note thread roots and replies. They SHOULD be sorted by the reply stack from root to the direct parent.

q tags MAY be used when citing events in the .content with NIP-21.

```
["q", "<event-id> or <event-address>", "<relay-url>", "<pubkey-if-a-regular-event>"]
```

Authors of the e and q tags SHOULD be added as p tags to notify of a new reply or quote.

Markup languages such as markdown and HTML SHOULD NOT be used.

Marked “e” tags (PREFERRED)

Kind 1 events with e tags are replies to other kind 1 events. Kind 1 replies MUST NOT be used to reply to other kinds, use NIP-22 instead.

```
["e", <event-id>, <relay-url>, <marker>, <pubkey>]
```

Where:

- <event-id> is the id of the event being referenced.
- <relay-url> is the URL of a recommended relay associated with the reference. Clients SHOULD add a valid <relay-url> field, but may instead leave it as "".
- <marker> is optional and if present is one of "reply", "root".
- <pubkey> is optional, SHOULD be the pubkey of the author of the referenced event

Those marked with "reply" denote the id of the reply event being responded to. Those marked with "root" denote the root id of the reply thread being responded to. For top level replies (those replying directly to the root event), only the "root" marker should be used.

A direct reply to the root of a thread should have a single marked “e” tag of type “root”.

This scheme is preferred because it allows events to mention others without confusing them with <reply-id> or <root-id>.

<pubkey> SHOULD be the pubkey of the author of the e tagged event, this is used in the outbox model to search for that event from the authors write relays where relay hints did not resolve the event.

The “p” tag

Used in a text event contains a list of pubkeys used to record who is involved in a reply thread.

When replying to a text event E the reply event’s “p” tags should contain all of E’s “p” tags as well as the "pubkey" of the event being replied to.

Example: Given a text event authored by a1 with “p” tags [p1, p2, p3] then the “p” tags of the reply should be [a1, p1, p2, p3] in no particular order.

Deprecated Positional “e” tags

This scheme is not in common use anymore and is here just to keep backward compatibility with older events on the network.

Positional e tags are deprecated because they create ambiguities that are difficult, or impossible to resolve when an event references another but is not a reply.

They use simple e tags without any marker.

[“e”, <event-id>, <relay-url>] as per NIP-01.

Where:

- <event-id> is the id of the event being referenced.
- <relay-url> is the URL of a recommended relay associated with the reference. Many clients treat this field as optional.

The positions of the “e” tags within the event denote specific meanings as follows:

- No “e” tag: This event is not a reply to, nor does it refer to, any other event.
- One “e” tag: [“e”, <id>]: The id of the event to which this event is a reply.
- Two “e” tags: [“e”, <root-id>], [“e”, <reply-id>] <root-id> is the id of the event at the root of the reply chain. <reply-id> is the id of the article to which this event is a reply.
- Many “e” tags: [“e”, <root-id>] [“e”, <mention-id>], ..., [“e”, <reply-id>] There may be any number of <mention-ids>. These are the ids of events which may, or may not be in the reply chain. They are citing from this event. root-id and reply-id are as above.

NIP-14

Subject tag in Text events

draft optional

This NIP defines the use of the “subject” tag in text (kind: 1) events. (implemented in more-speech)

```
[“subject”: <string>]
```

Browsers often display threaded lists of messages. The contents of the subject tag can be used in such lists, instead of the more ad hoc approach of using the first few words of the message. This is very similar to the way email browsers display lists of incoming emails by subject rather than by contents.

When replying to a message with a subject, clients SHOULD replicate the subject tag. Clients MAY adorn the subject to denote that it is a reply. e.g. by prepending “Re:”.

Subjects should generally be shorter than 80 chars. Long subjects will likely be trimmed by clients.

NIP-23

Long-form Content

draft optional

This NIP defines kind:30023 (an *addressable event*) for long-form text content, generally referred to as “articles” or “blog posts”. kind:30024 has the same structure as kind:30023 and is used to save long form drafts.

“Social” clients that deal primarily with kind:1 notes should not be expected to implement this NIP.

Format

The .content of these events should be a string text in Markdown syntax. To maximize compatibility and readability between different clients and devices, any client that is creating long form notes:

- MUST NOT hard line-break paragraphs of text, such as arbitrary line breaks at 80 column boundaries.
- MUST NOT support adding HTML to Markdown.

Metadata

For the date of the last update the .created_at field should be used, for “tags” / “hashtags” (i.e. topics about which the event might be of relevance) the t tag should be used.

Other metadata fields can be added as tags to the event as necessary. Here we standardize 4 that may be useful, although they remain strictly optional:

- "title", for the article title
- "image", for a URL pointing to an image to be shown along with the title
- "summary", for the article summary
- "published_at", for the timestamp in unix seconds (stringified) of the first time the article was published

Editability

These articles are meant to be editable, so they should include a d tag with an identifier for the article. Clients should take care to only publish and read these events from relays that implement that. If they don't do that they should also take care to hide old versions of the same article they may receive.

Linking

The article may be linked to using the NIP-19 naddr code along with the a tag.

References

References to other Nostr notes, articles or profiles must be made according to NIP-27, i.e. by using NIP-21 nostr: . . . links and optionally adding tags for these (see example below).

Example Event

```
{
  "kind": 30023,
  "created_at": 1675642635,
  "content": "Lorem
    [ipsum][nostr:nevent1qqst8cuiky046negxgwm5ynqwn53t8aqjr6afd8g59nfqwxpdhylpcpzamhxue69uhhyetw9ujuetcv9khqmr99e3k7mg8arnc9]
    dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna
    aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo
    consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
    pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id
    est laborum.\n\nRead more at
    nostr:naddr1qqzkjurnw4ksz9thwden5te0wfkccete9ehx7un8qgs2d90kkcq3nk2jry62dyf50k0h36rhpdt594my40w9pkal876jxgrqsqqqa2
```

```

"tags": [
  ["d", "lorem-ipsum"],
  ["title", "Lorem Ipsum"],
  ["published_at", "1296962229"],
  ["t", "placeholder"],
  ["e", "b3e392b11f5d4f28321cedd09303a748acfd0487aea5a7450b3481c60b6e4f87", "wss://relay.example.com"],
  ["a", "30023:a695f6b60119d9521934a691347d9f78e8770b56da16bb255ee286ddf9fda919:ipsum", "wss://relay.nostr.org"]
],
"pubkey": "...",
"id": "..."
}

```

Replies & Comments

Replies to kind 30023 MUST use NIP-22 kind 1111 comments.

NIP-24

Extra metadata fields and tags

draft optional

This NIP keeps track of extra optional fields that can be added to events which are not defined anywhere else but have become *de facto* standards and other minor implementation possibilities that do not deserve their own NIP and do not have a place in other NIPs.

kind 0

These are extra fields not specified in NIP-01 that may be present in the stringified JSON of metadata events:

- **display_name**: an alternative, bigger name with richer characters than **name**. **name** should always be set regardless of the presence of **display_name** in the metadata.
- **website**: a web URL related in any way to the event author.
- **banner**: an URL to a wide (~1024x768) picture to be optionally displayed in the background of a profile screen.
- **bot**: a boolean to clarify that the content is entirely or partially the result of automation, such as with chatbots or newsfeeds.
- **birthday**: an object representing the author's birth date. The format is { "year": number, "month": number, "day": number }. Each field MAY be omitted.

Deprecated fields These are fields that should be ignored or removed when found in the wild:

- **displayName**: use **display_name** instead.
- **username**: use **name** instead.

kind 3

These are extra fields not specified in NIP-02 that may be present in the stringified JSON of follow events:

Deprecated fields

- {<relay-url>: {"read": <true|false>, "write": <true|false>}, ...}: an object of relays used by a user to read/write. NIP-65 should be used instead.

tags

These tags may be present in multiple event kinds. Whenever a different meaning is not specified by some more specific NIP, they have the following meanings:

- **r**: a web URL the event is referring to in some way.
- **i**: an external id the event is referring to in some way - see NIP-73.
- **title**: name of NIP-51 sets, NIP-52 calendar event, NIP-53 live event or NIP-99 listing.
- **t**: a hashtag. The value MUST be a lowercase string.

Warning unrecommended: superseded by the Marmot Protocol

NIP-EE

E2EE Messaging using the Messaging Layer Security (MLS) Protocol

final unrecommended optional

This NIP standardizes how to use the MLS Protocol with Nostr for efficient and E2EE (end-to-end encrypted) direct and group messaging.

Context

Originally, one-to-one direct messages (DMs) in Nostr happened via the scheme defined in NIP-04. This NIP is not recommended because, while it encrypts the content of the message (provides decent confidentiality), it leaks significant amounts of metadata about the parties involved in the conversation (completely lacks privacy).

With the addition of NIP-44, we have an updated encryption scheme that improves confidentiality guarantees but stops short of defining a new scheme for doing direct messages using this encryption scheme. Hence, makes little to no difference to privacy.

Most recently, NIP-17 combines NIP-44 encryption with NIP-59 gift-wrapping to hide the encrypted direct message inside another set of events to ensure that it's impossible to see who is talking to who and when messages passed between the users. This largely solves the metadata leakage problem; while it's still possible to see that a user is receiving gift-wrapped events, you can't tell from whom and what kind of events are within the gift-wrap outer event. This gives some degree of deniability/repudiation but doesn't solve forward secrecy or post compromise security. That is to say, if a user's private key (or the calculated conversation key shared between two users used to encrypt messages) is compromised, the attacker will have full access to all past and future DMs sent between those users.

In addition, neither NIP-04 or NIP-17 attempt to solve the problem of group messages.

Why is this important?

Without proper E2EE, Nostr cannot be used as the protocol for secure messaging clients. While clients like Signal do a fantastic job with E2EE, they still rely on centralized servers and as a result can be shut down by a powerful (i.e. state-level) actor. The goal of Nostr is not only to protect against centralized entities censoring you and your communications, but also protect against the ability of a state-level actor to stop these sorts of services from existing in the first place. By replacing centralized servers with decentralized relays, we make it nearly impossible for a centralized actor to completely stop communications between individual users.

Goals of this NIP

1. Private *and* Confidential DMs and Group messages
 1. **Private** means that an observer cannot tell that Alice and Bob are talking to one another, or that Alice is part of a specific group. This necessarily requires protecting metadata.
 2. **Confidential** means that the contents of conversations can only be viewed by the intended recipients.
2. Forward secrecy and Post-compromise security
 1. **Forward secrecy** means that encrypted content in the past remains encrypted even if a key material is leaked.
 2. **Post compromise security** means that leaking key material doesn't allow an attacker to continue to read messages indefinitely into the future.
3. Scales efficiently for large groups
4. Allows for the use of multiple device/clients in a single conversation/group.

Why MLS?

This scheme adapts the Message Layer Security (MLS) protocol for use with Nostr. You can think of MLS as an evolution of the Signal Protocol. However, it significantly improves the scalability of encryption operations for

large group messaging significantly (linear -> log), is built to accommodate federated environments, and also allows for graceful updating of ciphersuites and versions over time. In addition, it's very flexible and agnostic about the message content that is sent.

It's beyond the scope of this NIP to explain the MLS protocol but you can read more about it in its Architectural Overview or the RFC. MLS is on track to become an internet standard under the IETF so the protocol itself is extremely well vetted and researched. This also means there is the potential for cross network messaging interoperability in the future as MLS gains more adoption.

Core MLS Concepts

From the MLS Architectural Overview:

MLS provides a way for clients to form groups within which they can communicate securely. For example, a set of users might use clients on their phones or laptops to join a group and communicate with each other. A group may be as small as two clients (e.g., for simple person to person messaging) or as large as hundreds of thousands. A client that is part of a group is a member of that group. As groups change membership and group or member properties, they advance from one epoch to another and the cryptographic state of the group evolves.

The group is represented as a tree, which represents the members as the leaves of a tree. It is used to efficiently encrypt to subsets of the members. Each member has a state called a LeafNode object holding the client's identity, credentials, and capabilities.

The MLS protocol's job is to manage and evolve the cryptographic state of a group. This includes managing the membership of a group, the cryptographic state of a group (ratchet tree, keys, and encryption/decryption/authentication of messages), and managing the evolution of the group over time.

Groups

Groups are created by their first member, who then invites one or more other members. Groups evolve over time in blocks called Epochs. New epochs are proposed via one or more Proposal messages and then committed to via a Commit message.

Clients

The device/client pair (e.g. Primal on iOS or Coracle on web) with which a user joins the group is represented as a LeafNode in the tree. The terms Client and Member are interchangeable in this regard. It is not possible to share group state across multiple Clients. If a user joins a group from 2 separate devices, their state is separate and they will be tracked as 2 separate members of the group.

Messages

There are several different types of messages sent within a group. Some of these are control messages that are used to update the group state over time. These include Welcome, Proposal, and Commit messages. Others are the actual messages that are sent between members in a group. These include Application messages.

Messages in MLS are "framed". Meaning that they are wrapped in a data structure that includes information about the sender, the epoch, the message index within the epoch and the message content. This framing makes it possible to authenticate and decrypt messages correctly, even if they arrive out of order.

MLS is agnostic to the "content" of the messages that are sent. This is a key feature of MLS that allows for the use of MLS for a wide variety of applications.

MLS is also agnostic to the transport protocol that is used to send messages. Obviously for us, we'll be using websockets, Nostr events and relays.

The focus of this NIP

This NIP focuses on how to use Nostr to perform the Authentication Service and Delivery Service functions required by the MLS protocol. Most clients will choose to use an MLS implementation to handle keys, ratcheting, group state

management, and other aspects of the MLS protocol itself. OpenMLS is the most actively developed library that implements MLS.

This NIP specifies the following:

1. A standardized way that Nostr clients should create MLS groups.
2. The required format of the `MLS Credential` that Nostr clients should use to represent a Nostr user in a group.
3. The structure of `KeyPackage Events` published to relays that allow Nostr users to be added to a group asynchronously.
4. The structure of `Group Events` published to relays that represent the evolution of a group's state and the contents of the messages sent in the group.

Security Considerations

This is a concise overview of the security trade-offs and considerations of this NIP in various scenarios. The NIP strives to fully maintain MLS security guarantees.

Forward Secrecy and Post-compromise Security

- As per the MLS spec, keys are deleted as soon as they are used to encrypt or decrypt a message. This is usually handled by the MLS implementation library itself but attention needs to be paid by clients to ensure they're not storing secrets (especially the exporter secret) for longer than absolutely necessary.
- This NIP maintains MLS forward secrecy and post-compromise security guarantees. You can read more about those in the [MLS Architectural Overview](#) section on Forward Secrecy and Post-compromise Security.

Leakage of various keys

- This NIP does not depend on a user's Nostr identity key for any aspect of the MLS messaging protocol. Compromise of a user's Nostr identity key does not give access to past or future messages in any MLS-based group.
- For a complete discussion of MLS key leakage, please see the [Endpoint Compromise](#) section of the [MLS Architectural Overview](#).

Metadata

- The only group specific metadata published to relays is the Nostr group ID value. This value is used to identify the group in the `h` tag of the `Group Message Event` (`kind: 445`). These events are published ephemerally and this Nostr group ID value can be updated over the lifetime of the group by group admins. This is a tradeoff to ensure that group participants and group size are obfuscated but still makes it possible to efficiently fan out group messages to all participants. The content field of this event is a value encrypted in two separate ways (using NIP-44 and MLS) with MLS group state/keys. Only group members with up-to-date group state can decrypt and read these messages.
- A user's key package events can be used one or more times to be added to groups. There is a tradeoff inherent here: Reusing key packages (initial signing keys) carries some degree of risk but this risk is mitigated as long as a user rotates their signing key immediately upon joining a group. This step also improves the forward secrecy of the entire group.

Device Compromise

Clients implementing this NIP should take every precaution to ensure that data is stored in a secure way on the device and is protected against unwanted access in the case that a device is compromised (e.g. encryption at rest, biometric authentication, etc.). That said, full device compromise should be viewed as a catastrophic event and any group the compromised device was a part of should be considered compromised until they can remove that member and update their group's state. Some suggestions:

- Clients should support and encourage self-destructing messages (ensuring that full transcript history isn't available on a device forever).
- Clients should regularly suggest to group admins that inactive users be removed.
- Clients should regularly suggest (or automatically) rotate a user's signing key in each of their groups.

- Clients should encrypt group state and keys on the device using a secret value that isn't part of the group state or the user's Nostr identity key.
- Clients should use secure enclave storage where possible.

For a full discussion of the security considerations of MLS, please see the Security Considerations section of the MLS RFC.

Creating groups

MLS Groups are created with a random 32-byte ID value that is effectively permanent. This ID should be treated as private to the group and **MUST** not be published to relays in any form.

Clients must also ensure that the ciphersuite, capabilities, and extensions they use when creating the group are compatible with those advertised by the users they'd like to invite to the group. They can check this info via the user's published KeyPackage Events.

When creating a new group, the following MLS extensions **MUST** be used.

- `required_capabilities`
- `ratchet_tree`
- `nostr_group_data`

And the following MLS extension is highly recommended (more here): - `last_resort`

Changes to an MLS group are affected by first creating one or more `Proposal` events and then committing to a set of proposals in a `Commit` event. These are MLS events, not Nostr events. However, for the group state to properly evolve the `Commit` events (which represent a specific set of proposals - like adding a new user to the group) must be published to relays for the other group members to see. See Group Messages for more information.

MLS Credentials

A `Credential` in MLS is an assertion of who the user is coupled with a signing key. When constructing `Credentials` for MLS, clients **MUST** use the `BasicCredential` type and set the `identity` value as the 32-byte hex-encoded public key of the user's Nostr identity key. Clients **MUST** not allow users to change the `identity` field and **MUST** validate that all `Proposal` messages do not attempt to change the `identity` field on any credential in the group.

A `Credential` also has an associated signing key. The initial signing key for a user is included in the `KeyPackage` event. The signing key **MUST** be different from the user's Nostr identity key. This signing key **SHOULD** be rotated over time to provide improved post-compromise security.

Nostr Group Data Extension

As mentioned above, the `nostr_group_data` extension is a required MLS extension used to associate Nostr-specific data with an MLS group in a cryptographically secure and proveable way. This extension **MUST** be included as a required capability when creating a new group.

The extension stores the following data about the group:

- `nostr_group_id`: A 32-byte ID for the group. This is a different value from the group ID used by MLS and **CAN** be changed over time. This value is the group ID value used in the `h` tags when sending group message events.
- `name`: The name of the group.
- `description`: A short description of the group.
- `admin_pubkeys`: An array of the hex-encoded public keys of the group admins. The MLS protocol itself does not have a concept of group admins. Clients **MUST** check the list of `admin_pubkeys` before making any change to the group data (anything in this extension), or before changing group membership (add/remove members), or updating any other aspect of the group itself (e.g. ciphersuite, etc.). Note, all members of the group can send `Proposal` and `Commits` messages for changes to their own credentials (e.g. updating their signing key).
- `relays`: An array of the Nostr relay URLs that the group uses to publish and receive messages.

All of these values can be updated over time using MLS `Proposal` and `Commit` events (by group admins).

KeyPackage Event and Signing Keys

Each user that wishes to be reachable via MLS-based messaging MUST first publish at least one KeyPackage event. The KeyPackage Event is used to authenticate users and create the necessary `Credential` to add members to groups in an asynchronous way. Users can publish multiple KeyPackage Events with different parameters (supporting different ciphersuites or MLS extensions, for example). KeyPackages include a signing key that is used for signing MLS messages within a group. This signing key MUST not be the same as the user's Nostr identity key.

KeyPackage reuse SHOULD be minimized. However, in normal MLS use, KeyPackages are consumed when joining a group. In order to reduce race conditions between invites for multiple groups using the same Key Package, Nostr clients SHOULD use "Last resort" KeyPackages. This requires the inclusion of the `last_resort` extension on the KeyPackage's capabilities (same as with the Group).

It's important that clients immediately rotate a user's signing key after joining a group via a last resort key package to improve post-compromise security. The signing key (the public key included in the KeyPackage Event) is used for signing within the group. Therefore, clients implementing this NIP MUST ensure that they retain access to the private key material of the signing key for each group they are a member of.

In most cases, it's assumed that clients implementing this NIP will manage the creation and rotation of KeyPackage Events.

Example KeyPackage Event

```
{
  "id": <id>,
  "kind": 443,
  "created_at": <unix timestamp in seconds>,
  "pubkey": <main identity pubkey>,
  "content": "",
  "tags": [
    ["mls_protocol_version", "1.0"],
    ["ciphersuite", <MLS CipherSuite ID value e.g. "0x0001">],
    ["extensions", <An array of MLS Extension ID values e.g. "0x0001, 0x0002">],
    ["client", <client name>, <handler event id>, <optional relay url>],
    ["relays", <array of relay urls>],
    ["-"]
  ],
  "sig": <signed with main identity key>
}
```

- The content hex encoded serialized `KeyPackageBundle` from MLS.
- The `mls_protocol_version` tag is required and MUST be the version number of the MLS protocol version being used. For now, this is 1.0.
- The `ciphersuite` tag is the value of the MLS ciphersuite that this KeyPackage Event supports. Read more about ciphersuites in MLS.
- The `extensions` tag is an array of MLS extension IDs that this KeyPackage Event supports. Read more about MLS extensions.
- (optional) The `client` tag helps other clients manage the user experience when they receive group invites but don't have access to the signing key.
- The `relays` tag identifies each of the relays that the client will attempt to publish this KeyPackage event. This allows for deletion of KeyPackage Events at a later date.
- (optional) The `-` tag can be used to ensure that KeyPackage Events are only published by their authenticated author. Read more in NIP-70

Deleting KeyPackage Events

Clients SHOULD delete the KeyPackage Event on all the listed relays any time they successfully process a group request event for a given KeyPackage Event. Clients MAY also create a new KeyPackage Event at the same time.

If clients cannot process a Welcome message (e.g. because the signing key was generated on another client), clients **MUST** not delete the KeyPackage Event and **SHOULD** show a human-understandable error to the user.

Rotating Signing Keys

Clients **MUST** regularly rotate the user's signing key in each group that they are a part of. The more often the signing key is rotated the stronger the post-compromise security. This rotation is done via `Proposal` and `Commit` events and broadcast to the group via a Group Event. Read more about forward secrecy and post-compromise security inherent in MLS.

KeyPackage Relays List Event

A kind: 10051 event indicates the relays that a user will publish their KeyPackage Events to. The event **MUST** include a list of relay tags with relay URLs. These relays **SHOULD** be readable by anyone the user wants to be able to contact them.

```
{
  "kind": 10051,
  "tags": [
    ["relay", "wss://inbox.nostr.wine"],
    ["relay", "wss://myrelay.nostr1.com"],
  ],
  "content": "",
  //...other fields
}
```

Welcome Event

When a new user is added to a group via an MLS `Commit` message. The member who sends the `Commit` message to the group is responsible for sending the user being added to the group a Welcome Event. This Welcome Event is sent to the user as a NIP-59 gift-wrapped event. The Welcome Event gives the new member the context they need to join the group and start sending messages.

Clients creating the Welcome Event **SHOULD** wait until they have received acknowledgement from relays that their Group Event with the `Commit` has been received before publishing the Welcome Event.

```
{
  "id": <id>,
  "kind": 444,
  "created_at": <unix timestamp in seconds>,
  "pubkey": <nostr identity pubkey of sender>,
  "content": <serialized Welcome object>,
  "tags": [
    ["e", <ID of the KeyPackage Event used to add the user to the group>],
    ["relays", <array of relay urls>],
  ],
  "sig": <NOT SIGNED>
}
```

- The content field is required and is a serialized MLSMessage object containing the MLS Welcome object.
- The e tag is required and is the ID of the KeyPackage Event used to add the user to the group.
- The relays tag is required and is a list of relays clients should query for Group Events.

Welcome Events are then sealed and gift-wrapped as detailed in NIP-59 before being published. Like all events that are sealed and gift-wrapped, kind: 444 events **MUST** never be signed. This ensures that if they were ever leaked they would not be publishable to relays.

Large Groups For groups above ~150 participants, welcome messages will become larger than the maximum event size allowed by Nostr. There is currently work underway on the MLS protocol to support “light” client welcomes that don’t require the full Ratchet Tree state to be sent to the new member. This section will be updated with recommendations for how to handle large groups.

Group Events

Group Events are all the messages that are sent within a group. This includes all “control” events that update the shared group state over time (Proposal, Commit) and messages sent between members of the group (Application messages).

Group Events are published using an ephemeral Nostr keypair to obfuscate the number and identity of group participants. Clients **MUST** use a new Nostr keypair for each Group Event they publish.

```
{
  "id": <id>,
  "kind": 445,
  "created_at": <unix timestamp in seconds>,
  "pubkey": <ephemeral sender pubkey>,
  "content": <NIP-44 encrypted serialized MLSMessage object>,
  "tags": [
    ["h", <group id>]
  ],
  "sig": <signed with ephemeral sender key>
}
```

- The content field is a tls-style serialized MLSMessage object which is then encrypted according to NIP-44. However, instead of using the sender and receivers keys to derive a conversation_key, the NIP-44 encryption is done using a Nostr keypair generated from the MLS exporter_secret to calculate the conversation_key value. Essentially, you use the hex-encoded exporter_secret value as the private key (used as the sender key), calculate the public key for that private key (used as the receiver key), and then proceed with the standard NIP-44 scheme to encrypt and decrypt messages.
- The exporter_secret value should be generated with a 32-byte length and labeled nostr. This exporter_secret value is rotated on each new epoch in the group. Clients should generate a new 32-byte value each time they process a valid Commit message.
- The pubkey is the hex-encoded public key of the ephemeral sender.
- The h tag is the nostr group ID value (from the Nostr Group Data Extension).

Application Messages

Application messages are the messages that are sent within the group by members. These are contained within the MLSMessage object. The format of these messages should be unsigned Nostr events of the appropriate kind. For normal DM or group messages, clients **SHOULD** use kind: 9 chat message events. If the user reacts to a message, it would be a kind: 7 event, and so on.

This means that once the application message has been decrypted and deserialized, clients can store those events and treat them as any other Nostr event, effectively creating a private Nostr feed of the group’s activity and taking advantage of all the features of Nostr.

These inner unsigned Nostr events **MUST** use the member’s Nostr identity key for the pubkey field and clients **MUST** check that the identity of the member who sent the message matches the pubkey of the inner Nostr event.

These Nostr events **MUST** remain **unsigned** to ensure that if they were to leak to relays they would not be published publicly. These Nostr events **MUST** not include any “h” tags or other tags that would identify the group that they belong to.

Commit Message race conditions

The MLS protocol is resilient to almost all messages arriving out of order. However, the order of Commit messages is important for the group state to move forward from one epoch to the next correctly. Given Nostr’s nature as a

decentralized network, it is possible for a client to receive 2 or more `Commit` messages all attempting to update to a new epoch at the same time.

Clients sending commit messages **MUST** wait until they receive acknowledgement from at least one relay that their Group Message Event with the `Commit` has been received before applying the commit to their own group state.

If a client receives 2 or more `Commit` messages attempting to change same epoch, they **MUST** apply only one of the `Commit` messages they receive, determined by the following:

1. Using the `created_at` timestamp on the kind 445 event. The `Commit` with the lowest value for `created_at` is the message to be applied. The other `Commit` message is discarded.
2. If the `created_at` timestamp is the same for two or more `Commit` messages, the `Commit` message with the lowest value for `id` field is the message to be applied.

Clients **SHOULD** retain previous group state for a short period of time in order to recover from forked group state.

NIP-17

Private Direct Messages

draft optional

This NIP defines an encrypted chat scheme which uses NIP-44 encryption and NIP-59 seals and gift wraps.

Any event sent to an encrypted chat MUST NOT be signed, and MUST be encrypted as described in NIP-59 and illustrated below. Omitting signatures makes messages deniable in case they are accidentally or maliciously leaked, while still allowing the recipient to authenticate them.

By convention, kind 14 direct messages, kind 15 file messages, and kind 7 reactions may be sent to an encrypted chat.

Kind Definitions

Chat Message

Kind 14 is a chat message. p tags identify one or more receivers of the message.

```
{
  "id": "<usual hash>",
  "pubkey": "<sender-pubkey>",
  "created_at": "<current-time>",
  "kind": 14,
  "tags": [
    ["p", "<receiver-1-pubkey>", "<relay-url>"],
    ["p", "<receiver-2-pubkey>", "<relay-url>"],
    ["e", "<kind-14-id>", "<relay-url>"] // if this is a reply
    ["subject", "<conversation-title>"],
    // rest of tags...
  ],
  "content": "<message-in-plain-text>",
}
```

.content MUST be plain text. Fields id and created_at are required.

An e tag denotes the direct parent message this post is replying to.

q tags MAY be used when citing events in the .content with NIP-21.

```
["q", "<event-id> or <event-address>", "<relay-url>", "<pubkey-if-a-regular-event>"]
```

File Message

```
{
  "id": "<usual hash>",
  "pubkey": "<sender-pubkey>",
  "created_at": "<current-time>",
  "kind": 15,
  "tags": [
    ["p", "<receiver-1-pubkey>", "<relay-url>"],
    ["p", "<receiver-2-pubkey>", "<relay-url>"],
    ["e", "<kind-14-id>", "<relay-url>", "reply"], // if this is a reply
    ["subject", "<conversation-title>"],
    ["file-type", "<file-mime-type>"],
    ["encryption-algorithm", "<encryption-algorithm>"],
    ["decryption-key", "<decryption-key>"],
    ["decryption-nonce", "<decryption-nonce>"],
    ["x", "<the SHA-256 hexencoded string of the file>"],
  ],
}
```

```

    // rest of tags...
  ],
  "content": "<file-url>"
}

```

Kind 15 is used for sending encrypted file event messages:

- **file-type**: Specifies the MIME type of the attached file (e.g., `image/jpeg`, `audio/mpeg`, etc.) before encryption.
- **encryption-algorithm**: Indicates the encryption algorithm used for encrypting the file. Supported algorithms: `aes-gcm`.
- **decryption-key**: The decryption key that will be used by the recipient to decrypt the file.
- **decryption-nonce**: The decryption nonce that will be used by the recipient to decrypt the file.
- **content**: The URL of the file (`<file-url>`).
- **x** containing the SHA-256 hexencoded string of the encrypted file.
- **ox** containing the SHA-256 hexencoded string of the file before encryption.
- **size** (optional) size of the encrypted file in bytes
- **dim** (optional) size in pixels in the form `<width>x<height>`
- **blurhash** (optional) the blurhash to show while the client is loading the file
- **thumb** (optional) URL of thumbnail with same aspect ratio (encrypted with the same key, nonce)
- **fallback** (optional) zero or more fallback file sources in case url fails (encrypted with the same key, nonce)

Chat Rooms

The set of `pubkey` + `p` tags defines a chat room. If a new `p` tag is added or a current one is removed, a new room is created with a clean message history.

Clients SHOULD render messages of the same room in a continuous thread.

An optional `subject` tag defines the current name/topic of the conversation. Any member can change the topic by simply submitting a new `subject` to an existing `pubkey` + `p` tags room. There is no need to send `subject` in every message. The newest `subject` in the chat room is the subject of the conversation.

Encrypting

Following NIP-59, the **unsigned** chat messages must be sealed (`kind:13`) and then gift-wrapped (`kind:1059`) to each receiver and the sender individually.

```

{
  "id": "<usual hash>",
  "pubkey": randomPublicKey,
  "created_at": randomTimeUpTo2DaysInThePast(),
  "kind": 1059, // gift wrap
  "tags": [
    ["p", receiverPublicKey, "<relay-url>"] // receiver
  ],
  "content": nip44Encrypt(
    {
      "id": "<usual hash>",
      "pubkey": senderPublicKey,
      "created_at": randomTimeUpTo2DaysInThePast(),
      "kind": 13, // seal
      "tags": [], // no tags
      "content": nip44Encrypt(unsignedKind14, senderPrivateKey, receiverPublicKey),
      "sig": "<signed by senderPrivateKey>"
    },
    randomPrivateKey, receiverPublicKey
  ),
  "sig": "<signed by randomPrivateKey>"
}

```


The encryption algorithm MUST use the latest version of NIP-44.

Clients MUST verify if pubkey of the kind:13 is the same pubkey on the kind:14, otherwise any sender can impersonate others by simply changing the pubkey on kind:14.

Clients SHOULD randomize created_at in up to two days in the past in both the seal and the gift wrap to make sure grouping by created_at doesn't reveal any metadata.

The gift wrap's p tag can be the receiver's main pubkey or an alias key created to receive DMs without exposing the receiver's identity.

Clients MAY offer disappearing messages by setting an expiration tag in the gift wrap of each receiver or by not generating a gift wrap to the sender's public key. This tag SHOULD be included on the kind 13 seal as well, in case it leaks.

Publishing

Kind 10050 indicates the user's preferred relays to receive DMs. The event MUST include a list of relay tags with relay URIs.

```
{
  "kind": 10050,
  "tags": [
    ["relay", "wss://inbox.nostr.wine"],
    ["relay", "wss://myrelay.nostr1.com"],
  ],
  "content": "",
  // other fields...
}
```

Clients SHOULD publish the gift-wrapped kind 1059 events that contain the sealed rumors to the relays listed in the recipient's kind 10050 event. If that is not found that indicates the user is not ready to receive messages under this NIP and clients shouldn't try.

Relays

Relays MAY protect message metadata by only serving kind:1059 events to users p-tagged on the event (enforced using NIP 42 AUTH).

Clients SHOULD guide users to keep kind:10050 lists small (1-3 relays) and SHOULD spread them to as many relays as viable.

Benefits & Limitations

This NIP offers the following privacy and security features:

1. **No Metadata Leak:** Participant identities, each message's real date and time, event kinds, and other event tags are all hidden from the public. Senders and receivers cannot be linked with public information alone.
2. **No Public Group Identifiers:** There is no public central queue, channel or otherwise converging identifier to correlate or count all messages in the same group.
3. **No Moderation:** There are no group admins: no invitations or bans.
4. **No Shared Secrets:** No secret must be known to all members that can leak or be mistakenly shared
5. **Fully Recoverable:** Messages can be fully recoverable by any client with the user's private key
6. **Optional Forward Secrecy:** Users and clients can opt-in for "disappearing messages".
7. **Uses Public Relays:** Messages can flow through public relays without loss of privacy. Private relays can increase privacy further, but they are not required.
8. **Cold Storage:** Users can unilaterally opt-in to sharing their messages with a separate key that is exclusive for DM backup and recovery.

The main limitation of this approach is having to send a separate encrypted event to each receiver. Group chats with more than 100 participants should find a more suitable messaging scheme.

Examples

This example sends the message `Hola, que tal?` from `nsec1w8udu59yjdjvedgs3yv5qccshcj8k05fh3l60k9x57asjrqdpa00qkmr89` to `nsec12ywtkplvyq5t6twdqwwygavp5lm4fhuang89c943nf2z92eez43szvn4dt`.

The two final GiftWraps, one to the receiver and the other to the sender, respectively, are:

```
{
  "id": "2886780f7349afc1344047524540ee716f7bdc1b64191699855662330bf235d8",
  "pubkey": "8f8a7ec43b77d25799281207e1a47f7a654755055788f7482653f9c9661c6d51",
  "created_at": 1703128320,
  "kind": 1059,
  "tags": [
    ["p", "918e2da906df4ccd12c8ac672d8335add131a4cf9d27ce42b3bb3625755f0788"]
  ],
  "content": "Asqzd1MsG304G8h08bE67dhAR1gFTzTckUuyvndZ8LrGCwI4pgC3d6hyAK0Wo9gtkLq5r2rT2RyHLE5wRqbC0lQ8WwJEKwqwIJwT5P03l2RxxGCHD",
  "sig": "a3c6ce632b145c0869423c1afaff4a6d764a9b64dedaf15f170b944ead67227518a72e455567ca1c2a0d187832cecbde7ed478395ec4c95dd3e71749"
}
```

```
{
  "id": "162b0611a1911cfcb30f8a5502792b346e535a45658b3a31ae5c178465509721",
  "pubkey": "626be2af274b29ea4816ad672ee452b7cf96bbb4836815a55699ae402183f512",
  "created_at": 1702711587,
  "kind": 1059,
  "tags": [
    ["p", "44900586091b284416a0c001f677f9c49f7639a55c3f1e2ec130a8e1a7998e1b"]
  ],
  "content": "AsTCLTzr0gzXXji7uyeSUB6LYrx3HDjWGdkNaBS6BAX9CpHa+Vtt5oI2xJmWLen+Fo2NB0Fazv1285Gb3HSM82gVycrx1HUAaQDUG6HI7XBEGqBhC",
  "sig": "c94e74533b482aa8eeeb54ae72a5303e0b21f62909ca43c8ef06b0357412d6f8a92f96e1a205102753777fd25321a58fba3fb384eee114bd53ce6c00"
}
```

Warning unrecommended: deprecated in favor of NIP-17

NIP-04

Encrypted Direct Message

final unrecommended optional

A special event with kind 4, meaning “encrypted direct message”. It is supposed to have the following attributes:

content **MUST** be equal to the base64-encoded, aes-256-cbc encrypted string of anything a user wants to write, encrypted using a shared cipher generated by combining the recipient’s public-key with the sender’s private-key; this appended by the base64-encoded initialization vector as if it was a querystring parameter named “iv”. The format is the following: "content": "<encrypted_text>?iv=<initialization_vector>".

tags **MUST** contain an entry identifying the receiver of the message (such that relays may naturally forward this event to them), in the form ["p", "<pubkey, as a hex string>"].

tags **MAY** contain an entry identifying the previous message in a conversation or a message we are explicitly replying to (such that contextual, more organized conversations may happen), in the form ["e", "<event_id>"].

Note: By default in the libsecp256k1 ECDH implementation, the secret is the SHA256 hash of the shared point (both X and Y coordinates). In Nostr, only the X coordinate of the shared point is used as the secret and it is NOT hashed. If using libsecp256k1, a custom function that copies the X coordinate must be passed as the hashfp argument in secp256k1_ecdh. See [here](#).

Code sample for generating such an event in JavaScript:

```
import crypto from 'crypto'
import * as secp from '@noble/secp256k1'

let sharedPoint = secp.getSharedSecret(ourPrivateKey, '02' + theirPublicKey)
let sharedX = sharedPoint.slice(1, 33)

let iv = crypto.randomFillSync(new Uint8Array(16))
var cipher = crypto.createCipheriv(
  'aes-256-cbc',
  Buffer.from(sharedX),
  iv
)
let encryptedMessage = cipher.update(text, 'utf8', 'base64')
encryptedMessage += cipher.final('base64')
let ivBase64 = Buffer.from(iv.buffer).toString('base64')

let event = {
  pubkey: ourPubKey,
  created_at: Math.floor(Date.now() / 1000),
  kind: 4,
  tags: [['p', theirPublicKey]],
  content: encryptedMessage + '?iv=' + ivBase64
}
```

Security Warning

This standard does not go anywhere near what is considered the state-of-the-art in encrypted communication between peers, and it leaks metadata in the events, therefore it must not be used for anything you really need to keep secret, and only with relays that use AUTH to restrict who can fetch your kind:4 events.

Client Implementation Warning

Clients *should not* search and replace public key or note references from the .content. If processed like a regular text note (where @npub... is replaced with #[0] with a ["p", "..."] tag) the tags are leaked and the mentioned user will receive the message in their inbox.

NIP-40

Expiration Timestamp

draft optional

The expiration tag enables users to specify a unix timestamp at which the message SHOULD be considered expired (by relays and clients) and SHOULD be deleted by relays.

```
tag: expiration
values:
- [UNIX timestamp in seconds]: required
```

```
{
  "pubkey": "<pub-key>",
  "created_at": 1000000000,
  "kind": 1,
  "tags": [
    ["expiration", "1600000000"]
  ],
  "content": "This message will expire at the specified timestamp and be deleted by relays.\n",
  "id": "<event-id>"
}
```

Note: The timestamp should be in the same format as the created_at timestamp and should be interpreted as the time at which the message should be deleted by relays.

Client Behavior

Clients SHOULD use the supported_nips field to learn if a relay supports this NIP. Clients SHOULD NOT send expiration events to relays that do not support this NIP.

Clients SHOULD ignore events that have expired.

Relay Behavior

Relays MAY NOT delete expired messages immediately on expiration and MAY persist them indefinitely. Relays SHOULD NOT send expired events to clients, even if they are stored. Relays SHOULD drop any events that are published to them if they are expired. An expiration timestamp does not affect storage of ephemeral events.

Suggested Use Cases

- Temporary announcements - This tag can be used to make temporary announcements. For example, an event organizer could use this tag to post announcements about an upcoming event.
- Limited-time offers - This tag can be used by businesses to make limited-time offers that expire after a certain amount of time. For example, a business could use this tag to make a special offer that is only available for a limited time.

Warning The events could be downloaded by third parties as they are publicly accessible all the time on the relays. So don't consider expiring messages as a security feature for your conversations or other uses.

NIP-09

Event Deletion Request

draft optional

A special event with kind 5, meaning “deletion request” is defined as having a list of one or more `e` or `a` tags, each referencing an event the author is requesting to be deleted. Deletion requests **SHOULD** include a `k` tag for the kind of each event being requested for deletion.

The event’s content field **MAY** contain a text note describing the reason for the deletion request.

For example:

```
{
  "kind": 5,
  "pubkey": <32-bytes hex-encoded public key of the event creator>,
  "tags": [
    ["e", "dcd59..464a2"],
    ["e", "968c5..ad7a4"],
    ["a", "<kind>:<pubkey>:<d-identifier>"],
    ["k", "1"],
    ["k", "30023"]
  ],
  "content": "these posts were published by accident",
  // other fields...
}
```

Relays **SHOULD** delete or stop publishing any referenced events that have an identical `pubkey` as the deletion request. Clients **SHOULD** hide or otherwise indicate a deletion request status for referenced events.

Relays **SHOULD** continue to publish/share the deletion request events indefinitely, as clients may already have the event that’s intended to be deleted. Additionally, clients **SHOULD** broadcast deletion request events to other relays which don’t have it.

When an `a` tag is used, relays **SHOULD** delete all versions of the replaceable event up to the `created_at` timestamp of the deletion request event.

Client Usage

Clients **MAY** choose to fully hide any events that are referenced by valid deletion request events. This includes text notes, direct messages, or other yet-to-be defined event kinds. Alternatively, they **MAY** show the event along with an icon or other indication that the author has “disowned” the event. The `content` field **MAY** also be used to replace the deleted events’ own content, although a user interface should clearly indicate that this is a deletion request reason, not the original content.

A client **MUST** validate that each event `pubkey` referenced in the `e` tag of the deletion request is identical to the deletion request `pubkey`, before hiding or deleting any event. Relays can not, in general, perform this validation and should not be treated as authoritative.

Clients display the deletion request event itself in any way they choose, e.g., not at all, or with a prominent notice.

Clients **MAY** choose to inform the user that their request for deletion does not guarantee deletion because it is impossible to delete events from all relays and clients.

Relay Usage

Relays **MAY** validate that a deletion request event only references events that have the same `pubkey` as the deletion request itself, however this is not required since relays may not have knowledge of all referenced events.

Deletion Request of a Deletion Request

Publishing a deletion request event against a deletion request has no effect. Clients and relays are not obliged to support “unrequest deletion” functionality.

NIP-92

Media Attachments

Media attachments (images, videos, and other files) may be added to events by including a URL in the event content, along with a matching `imeta` tag.

`imeta` ("inline metadata") tags MAY add information about media URLs in the event's content. Each `imeta` tag SHOULD match a URL in the event content. Clients MAY replace `imeta` URLs with rich previews.

The `imeta` tag is variadic, and each entry is a space-delimited key / value pair. Each `imeta` tag MUST have a `url`, and at least one other field. `imeta` MAY include any field specified by NIP 94. There SHOULD be only one `imeta` tag per URL.

Example

```
{
  "content": "More image metadata tests 'dont mind me https://nostr.build/i/my-image.jpg",
  "kind": 1,
  "tags": [
    [
      "imeta",
      "url https://nostr.build/i/my-image.jpg",
      "m image/jpeg",
      "blurhash eVF$^OI:{$M{o#*0-nNFxakD-?xVMjWEWB%inkxvR-oetmo#R-aen$",
      "dim 3024x4032",
      "alt A scenic photo overlooking the coast of Costa Rica",
      "x <sha256 hash as specified in NIP 94>",
      "fallback https://nostrcheck.me/alt1.jpg",
      "fallback https://void.cat/alt1.jpg"
    ]
  ]
}
```

Recommended client behavior

When uploading files during a new post, clients MAY include this metadata after the file is uploaded and included in the post.

When pasting URLs during post composition, the client MAY download the file and add this metadata before the post is sent.

The client MAY ignore `imeta` tags that do not match the URL in the event content.

Social

NIP-02

Follow List

final optional

A special event with kind 3, meaning “follow list” is defined as having a list of `p` tags, one for each of the followed /-known profiles one is following.

Each tag entry should contain the key for the profile, a relay URL where events from that key can be found (can be set to an empty string if not needed), and a local name (or “petname”) for that profile (can also be set to an empty string or not provided), i.e., `[“p”, <32-bytes hex key>, <main relay URL>, <petname>]`.

The `.content` is not used.

For example:

```
{
  "kind": 3,
  "tags": [
    ["p", "91cf9..4e5ca", "wss://alicerelay.com/", "alice"],
    ["p", "14aeb..8dad4", "wss://bobrelay.com/nostr", "bob"],
    ["p", "612ae..e610f", "ws://carolrelay.com/ws", "carol"]
  ],
  "content": "",
  // other fields...
}
```

Every new following list that gets published overwrites the past ones, so it should contain all entries. Relays and clients SHOULD delete past following lists as soon as they receive a new one.

Whenever new follows are added to an existing list, clients SHOULD append them to the end of the list, so they are stored in chronological order.

Uses

Follow list backup

If one believes a relay will store their events for sufficient time, they can use this kind-3 event to backup their following list and recover on a different device.

Profile discovery and context augmentation

A client may rely on the kind-3 event to display a list of followed people by profiles one is browsing; make lists of suggestions on who to follow based on the follow lists of other people one might be following or browsing; or show the data in other contexts.

Relay sharing

A client may publish a follow list with good relays for each of their follows so other clients may use these to update their internal relay lists if needed, increasing censorship-resistance.

Petname scheme

The data from these follow lists can be used by clients to construct local “petname” tables derived from other people’s follow lists. This alleviates the need for global human-readable names. For example:

A user has an internal follow list that says

```
[
  ["p", "21df6d143fb96c2ec9d63726bf9edc71", "", "erin"]
]
```

And receives two follow lists, one from 21df6d143fb96c2ec9d63726bf9edc71 that says

```
[  
  ["p", "a8bb3d884d5d90b413d9891fe4c4e46d", "", "david"]  
]
```

and another from a8bb3d884d5d90b413d9891fe4c4e46d that says

```
[  
  ["p", "f57f54057d2a7af0efecc8b0b66f5708", "", "frank"]  
]
```

When the user sees 21df6d143fb96c2ec9d63726bf9edc71 the client can show *erin* instead; When the user sees a8bb3d884d5d90b413d9891fe4c4e46d the client can show *david.erin* instead; When the user sees f57f54057d2a7af0efecc8b0b66f the client can show *frank.david.erin* instead.

NIP-05

Mapping Nostr keys to DNS-based internet identifiers

final optional

On events of kind 0 (user metadata) one can specify the key "nip05" with an internet identifier (an email-like address) as the value. Although there is a link to a very liberal "internet identifier" specification above, NIP-05 assumes the <local-part> part will be restricted to the characters a-z0-9-_. , case-insensitive.

Upon seeing that, the client splits the identifier into <local-part> and <domain> and use these values to make a GET request to `https://<domain>/.well-known/nostr.json?name=<local-part>`.

The result should be a JSON document object with a key "names" that should then be a mapping of names to hex formatted public keys. If the public key for the given <name> matches the pubkey from the user metadata event, the client then concludes that the given pubkey can indeed be referenced by its identifier.

Example

If a client sees an event like this:

```
{
  "pubkey": "b0635d6a9851d3aed0cd6c495b282167acf761729078d975fc341b22650b07b9",
  "kind": 0,
  "content": "{ \"name\": \"bob\", \"nip05\": \"bob@example.com\" }"
  // other fields...
}
```

It will make a GET request to `https://example.com/.well-known/nostr.json?name=bob` and get back a response that will look like

```
{
  "names": {
    "bob": "b0635d6a9851d3aed0cd6c495b282167acf761729078d975fc341b22650b07b9"
  }
}
```

or with the **recommended** "relays" attribute:

```
{
  "names": {
    "bob": "b0635d6a9851d3aed0cd6c495b282167acf761729078d975fc341b22650b07b9"
  },
  "relays": {
    "b0635d6a9851d3aed0cd6c495b282167acf761729078d975fc341b22650b07b9": [ "wss://relay.example.com",
    "wss://relay2.example.com" ]
  }
}
```

If the pubkey matches the one given in "names" (as in the example above) that means the association is right and the "nip05" identifier is valid and can be displayed.

The recommended "relays" attribute may contain an object with public keys as properties and arrays of relay URLs as values. When present, that can be used to help clients learn in which relays the specific user may be found. Web servers which serve `/.well-known/nostr.json` files dynamically based on the query string SHOULD also serve the relays data for any name they serve in the same reply when that is available.

Finding users from their NIP-05 identifier

A client may implement support for finding users' public keys from *internet identifiers*, the flow is the same as above, but reversed: first the client fetches the *well-known* URL and from there it gets the public key of the user, then it tries to fetch the kind 0 event for that user and check if it has a matching "nip05".

Notes

Identification, not verification

The NIP-05 is not intended to *verify* a user, but only to *identify* them, for the purpose of facilitating the exchange of a contact or their search.

Exceptions are people who own (e.g., a company) or are connected (e.g., a project) to a well-known domain, who can exploit NIP-05 as an attestation of their relationship with it, and thus to the organization behind it, thereby gaining an element of trust.

User discovery implementation suggestion

A client can use this to allow users to search other profiles. If a client has a search box or something like that, a user may be able to type “bob@example.com” there and the client would recognize that and do the proper queries to obtain a pubkey and suggest that to the user.

Clients must always follow public keys, not NIP-05 addresses

For example, if after finding that bob@bob.com has the public key abc...def, the user clicks a button to follow that profile, the client must keep a primary reference to abc...def, not bob@bob.com. If, for any reason, the address https://bob.com/.well-known/nostr.json?name=bob starts returning the public key 1d2...e3f at any time in the future, the client must not replace abc...def in his list of followed profiles for the user (but it should stop displaying “bob@bob.com” for that user, as that will have become an invalid “nip05” property).

Public keys must be in hex format

Keys must be returned in hex format. Keys in NIP-19 npub format are only meant to be used for display in client UIs, not in this NIP.

Showing just the domain as an identifier

Clients may treat the identifier _@domain as the “root” identifier, and choose to display it as just the <domain>. For example, if Bob owns bob.com, he may not want an identifier like bob@bob.com as that is redundant. Instead, Bob can use the identifier _@bob.com and expect Nostr clients to show and treat that as just bob.com for all purposes.

Reasoning for the /.well-known/nostr.json?name=<local-part> format

By adding the <local-part> as a query string instead of as part of the path, the protocol can support both dynamic servers that can generate JSON on-demand and static servers with a JSON file in it that may contain multiple names.

Allowing access from JavaScript apps

JavaScript Nostr apps may be restricted by browser CORS policies that prevent them from accessing /.well-known/nostr.json on the user’s domain. When CORS prevents JS from loading a resource, the JS program sees it as a network failure identical to the resource not existing, so it is not possible for a pure-JS app to tell the user for certain that the failure was caused by a CORS issue. JS Nostr apps that see network failures requesting /.well-known/nostr.json files may want to recommend to users that they check the CORS policy of their servers, e.g.:

```
$ curl -sI https://example.com/.well-known/nostr.json?name=bob | grep -i ^Access-Control
Access-Control-Allow-Origin: *
```

Users should ensure that their /.well-known/nostr.json is served with the HTTP header Access-Control-Allow-Origin: * to ensure it can be validated by pure JS apps running in modern browsers.

Security Constraints

The /.well-known/nostr.json endpoint MUST NOT return any HTTP redirects.

Fetchers MUST ignore any HTTP redirects given by the /.well-known/nostr.json endpoint.

NIP-25

Reactions

draft optional

A reaction is a kind 7 event that is used to indicate user reactions to other events. A reaction's content field MUST include user-generated-content indicating the value of the reaction (conventionally +, -, or an emoji).

A reaction with content set to + or an empty string MUST be interpreted as a "like" or "upvote". A reaction with content set to - MUST be interpreted as a "dislike" or "downvote".

A reaction with content set to an emoji or NIP-30 custom emoji SHOULD NOT be interpreted as a "like" or "dislike". Clients MAY instead display this emoji reaction on the post.

Tags

There MUST be always an e tag set to the id of the event that is being reacted to. The e tag SHOULD include a relay hint pointing to a relay where the event being reacted to can be found. If a client decides to include other e, which not recommended, the target event id should be last of the e tags.

There SHOULD be a p tag set to the pubkey of the event being reacted to. If a client decides to include other p tags, which not recommended, the target event pubkey should be last the p tags.

If the event being reacted to is an addressable event, an a SHOULD be included together with the e tag, it must be set to the coordinates (kind:pubkey:d-tag) of the event being reacted to.

The e and a tags SHOULD include relay and pubkey hints. The p tags SHOULD include relay hints.

The reaction event MAY include a k tag with the stringified kind number of the reacted event as its value.

Example code

```
func make_like_event(pubkey: String, privkey: String, liked: NostrEvent, hint: String) -> NostrEvent {
    var tags: [[String]] = []
    tags.append(["e", liked.id, hint, liked.pubkey])
    tags.append(["p", liked.pubkey, hint])
    tags.append(["k", String(liked.kind)])
    let ev = NostrEvent(content: "+", pubkey: pubkey, kind: 7, tags: tags)
    ev.calculate_id()
    ev.sign(privkey: privkey)
    return ev
}
```

External Content Reactions

If the target of a reaction is not a native nostr event, the reaction MUST be a kind 17 event and MUST include NIP-73 external content k + i tags to properly reference the content.

Reacting to a website:

```
{
  "kind": 17,
  "content": " ",
  "tags": [
    ["k", "web"],
    ["i", "https://example.com"]
  ],
}
```

Reacting to a podcast episode:

```
{
  "kind": 17,
  "content": "+",
  "tags": [
    ["k", "podcast:guid"],
    ["i", "podcast:guid:917393e3-1b1e-5cef-ace4-edaa54e1f810", "https://fountain.fm/show/QRt0L2EfrKXNGDlRmjL"],
    ["k", "podcast:item:guid"],
    ["i", "podcast:item:guid:PC20-229", "https://fountain.fm/episode/DQqBg5sD3qFGMCZoSuLF"]
  ],
}
```

Custom Emoji Reaction

The client may specify a custom emoji (NIP-30) `:shortcode:` in the reaction content. The client should refer to the emoji tag and render the content as an emoji if shortcode is specified.

```
{
  "kind": 7,
  "content": ":soapbox:",
  "tags": [
    ["emoji", "soapbox", "https://gleasonator.com/emoji/Gleasonator/soapbox.png"]
  ],
  // other fields...
}
```

The content can be set only one `:shortcode:`. And emoji tag should be one.

NIP-30

Custom Emoji

draft optional

Custom emoji may be added to **kind 0**, **kind 1**, **kind 7** (NIP-25) and **kind 30315** (NIP-38) events by including one or more "emoji" tags, in the form:

```
["emoji", <shortcode>, <image-url>]
```

Where:

- <shortcode> is a name given for the emoji, which MUST be comprised of only alphanumeric characters and underscores.
- <image-url> is a URL to the corresponding image file of the emoji.

For each emoji tag, clients should parse emoji shortcodes (aka "emojify") like :shortcode: in the event to display custom emoji.

Clients may allow users to add custom emoji to an event by including :shortcode: identifier in the event, and adding the relevant "emoji" tags.

Kind 0 events

In kind 0 events, the name and about fields should be emojified.

```
{
  "kind": 0,
  "content": "{ \"name\": \"Alex Gleason :soapbox:\"",
  "tags": [
    ["emoji", "soapbox", "https://gleasonator.com/emoji/Gleasonator/soapbox.png"]
  ],
  "pubkey": "79c2cae114ea28a981e7559b4fe7854a473521a8d22a66bbab9fa248eb820ff6",
  "created_at": 1682790000
}
```

Kind 1 events

In kind 1 events, the content should be emojified.

```
{
  "kind": 1,
  "content": "Hello :gleasonator:  :ablobcatrainbow: :disputed: yolo",
  "tags": [
    ["emoji", "ablobcatrainbow", "https://gleasonator.com/emoji/blobcat/ablobcatrainbow.png"],
    ["emoji", "disputed", "https://gleasonator.com/emoji/Fun/disputed.png"],
    ["emoji", "gleasonator", "https://gleasonator.com/emoji/Gleasonator/gleasonator.png"]
  ],
  "pubkey": "79c2cae114ea28a981e7559b4fe7854a473521a8d22a66bbab9fa248eb820ff6",
  "created_at": 1682630000
}
```

Kind 7 events

In kind 7 events, the content should be emojified.

```
{
  "kind": 7,
  "content": ":dezh:",
  "tags": [
```



```
["emoji", "dezh", "https://raw.githubusercontent.com/dezh-tech/brand-assets/main/dezh/logo/black-normal.svg"]
],
"pubkey": "79c2cae114ea28a981e7559b4fe7854a473521a8d22a66bbab9fa248eb820ff6",
"created_at": 1682630000
}
```

NIP-18

Reposts

draft optional

A repost is a kind 6 event that is used to signal to followers that a kind 1 text note is worth reading.

The content of a repost event is *the stringified JSON of the reposted note*. It MAY also be empty, but that is not recommended. Reposts of NIP-70-protected events SHOULD always have an empty content.

The repost event MUST include an e tag with the id of the note that is being reposted. That tag MUST include a relay URL as its third entry to indicate where it can be fetched.

The repost SHOULD include a p tag with the pubkey of the event being reposted.

Quote Reposts

Mentions to NIP-21 entities like nevent, note and naddr on any event must be converted into q tags. The q tag ensures quote reposts are not pulled and included as replies in threads. It also allows you to easily pull and count all of the quotes for a post. The syntax follows

```
["q", "<event-id> or <event-address>", "<relay-url>", "<pubkey-if-a-regular-event>"]
```

Generic Reposts

Since kind 6 reposts are reserved for kind 1 contents, we use kind 16 as a “generic repost”, that can include any kind of event inside other than kind 1.

kind 16 reposts SHOULD contain a "k" tag with the stringified kind number of the reposted event as its value.

When reposting a replaceable event, the repost SHOULD include an "a" tag with the event coordinate (kind:pubkey:d-tag) of the reposted event.

If the "a" tag is not present, it indicates that a specific version of a replaceable event is being reposted, in which case the content field must contain the full JSON string of the reposted event.

NIP-27

Text Note References

draft optional

This document standardizes the treatment given by clients of inline references of other events and profiles inside the `.content` of any event that has readable text in its `.content` (such as kinds 1 and 30023).

When creating an event, clients should include mentions to other profiles and to other events in the middle of the `.content` using NIP-21 codes, such as `nostr:nprofile1qqsw3dy8cpu...6x2argwghx6egsqstvg`.

Including NIP-18's quote tags (`["q", "<event-id> or <event-address>", "<relay-url>", "<pubkey-if-a-regular-event>"]`) for each reference is optional, clients should do it whenever they want the profile being mentioned to be notified of the mention, or when they want the referenced event to recognize their mention as a reply.

A reader client that receives an event with such `nostr:... mentions` in its `.content` can do any desired context augmentation (for example, linking to the profile or showing a preview of the mentioned event contents) it wants in the process. If turning such mentions into links, they could become internal links, NIP-21 links or direct links to web clients that will handle these references.

Example of a profile mention process

Suppose Bob is writing a note in a client that has search-and-autocomplete functionality for users that is triggered when they write the character `@`.

As Bob types `"hello @mat"` the client will prompt him to autocomplete with `mattn's` profile, showing a picture and name.

Bob presses "enter" and now he sees his typed note as `"hello @mattn"`, `@mattn` is highlighted, indicating that it is a mention. Internally, however, the event looks like this:

```
{
  "content": "hello nostr:nprofile1qqszclxx9f5haga8sfjjrulaxncvkfekj097t6f3pu65f86rvvg49ehqj6f9dh",
  "created_at": 1679790774,
  "id": "f39e9b451a73d62abc5016cffdd294b1a904e2f34536a208874fe5e22bbd47cf",
  "kind": 1,
  "pubkey": "79be667ef9dcbac55a06295ce870b07029bfcdb2dce28d959f2815b16f81798",
  "sig":
    "f8c8bab1b90cc3d2ae1ad999e6af8af449ad8bb4edf64807386493163e29162b5852a796a8f474d6b1001cddbaac0de4392838574f5366f03cc94cf5df",
  "tags": [
    [
      "p",
      "2c7cc62a697ea3a7826521f3fd34f0cb273693cbe5e9310f35449f43622a5cdc"
    ]
  ]
}
```

(Alternatively, the mention could have been a `nostr:npub1... URL`.)

After Bob publishes this event and Carol sees it, her client will initially display the `.content` as it is, but later it will parse the `.content` and see that there is a `nostr: URL` in there, decode it, extract the public key from it (and possibly relay hints), fetch that profile from its internal database or relays, then replace the full URL with the name `@mattn`, with a link to the internal page view for that profile.

Verbose and probably unnecessary considerations

- The example above was very concrete, but it doesn't mean all clients have to implement the same flow. There could be clients that do not support autocomplete at all, so they just allow users to paste raw NIP-19 codes into the body of text, then prefix these with `nostr:` before publishing the event.

- The flow for referencing other events is similar: a user could paste a `note1...` or `nevent1...` code and the client will turn that into a `nostr:note1...` or `nostr:nevent1...` URL. Then upon reading such references the client may show the referenced note in a preview box or something like that – or nothing at all.
- Other display procedures can be employed: for example, if a client that is designed for dealing with only `kind:1` text notes sees, for example, a `kind:30023 nostr:naddr1...` URL reference in the `.content`, it can, for example, decide to turn that into a link to some hardcoded webapp capable of displaying such events.
- Clients may give the user the option to include or not include tags for mentioned events or profiles. If someone wants to mention `mattn` without notifying them, but still have a nice augmentable / clickable link to their profile inside their note, they can instruct their client to *not* create a `["p", ...]` tag for that specific mention.
- In the same way, if someone wants to reference another note but their reference is not meant to show up along other replies to that same note, their client can choose to not include a corresponding `["e", ...]` tag for any given `nostr:nevent1...` URL inside `.content`. Clients may decide to expose these advanced functionalities to users or be more opinionated about things.

Warning unrecommended: deprecated in favor of NIP-27

NIP-08

Handling Mentions

final unrecommended optional

This document standardizes the treatment given by clients of inline mentions of other events and pubkeys inside the content of `text_notes`.

Clients that want to allow tagged mentions they **MUST** show an autocomplete component or something analogous to that whenever the user starts typing a special key (for example, “@”) or presses some button to include a mention etc – or these clients can come up with other ways to unambiguously differentiate between mentions and normal text.

Once a mention is identified, for example, the pubkey `27866e9d854c78ae625b867eefdfa9580434bc3e675be08d2acb526610d96fb` the client **MUST** add that pubkey to the `.tags` with the tag `p`, then replace its textual reference (inside `.content`) with the notation `#[index]` in which “index” is equal to the 0-based index of the related tag in the tags array.

The same process applies for mentioning event IDs.

A client that receives a `text_note` event with such `#[index]` mentions in its `.content` **CAN** do a search-and-replace using the actual contents from the `.tags` array with the actual pubkey or event ID that is mentioned, doing any desired context augmentation (for example, linking to the pubkey or showing a preview of the mentioned event contents) it wants in the process.

Where `#[index]` has an `index` that is outside the range of the tags array or points to a tag that is not an `e` or `p` tag or a tag otherwise declared to support this notation, the client **MUST NOT** perform such replacement or augmentation, but instead display it as normal text.

NIP-38

User Statuses

draft optional

Abstract

This NIP enables a way for users to share live statuses such as what music they are listening to, as well as what they are currently doing: work, play, out of office, etc.

Live Statuses

A special event with kind:30315 “User Status” is defined as an *optionally expiring addressable event*, where the d tag represents the status type:

For example:

```
{
  "kind": 30315,
  "content": "Sign up for nostrasia!",
  "tags": [
    ["d", "general"],
    ["r", "https://nostr.world"]
  ],
}

{
  "kind": 30315,
  "content": "Intergalatic - Beastie Boys",
  "tags": [
    ["d", "music"],
    ["r", "spotify:search:Intergalatic%20-%20Beastie%20Boys"],
    ["expiration", "1692845589"]
  ],
}
```

Two common status types are defined: `general` and `music`. `general` represent general statuses: “Working”, “Hiking”, etc.

`music` status events are for live streaming what you are currently listening to. The expiry of the `music` status should be when the track will stop playing.

Any other status types can be used but they are not defined by this NIP.

The status MAY include an `r`, `p`, `e` or `a` tag linking to a URL, profile, note, or addressable event.

The content MAY include emoji(s), or NIP-30 custom emoji(s). If the content is an empty string then the client should clear the status.

Client behavior

Clients MAY display this next to the username on posts or profiles to provide live user status information.

Use Cases

- Calendar nostr apps that update your general status when you’re in a meeting
- Nostr Nests that update your general status with a link to the nest when you join
- Nostr music streaming services that update your music status when you’re listening
- Podcasting apps that update your music status when you’re listening to a podcast, with a link for others to listen as well

- Clients can use the system media player to update playing music status

NIP-58

Badges

draft optional

Three special events are used to define, award and display badges in user profiles:

1. A “Badge Definition” event is defined as an addressable event with kind `30009` having a `d` tag with a value that uniquely identifies the badge (e.g. `bravery`) published by the badge issuer. Badge definitions can be updated.
2. A “Badge Award” event is a kind `8` event with a single `a` tag referencing a “Badge Definition” event and one or more `p` tags, one for each pubkey the badge issuer wishes to award. Awarded badges are immutable and non-transferable.
3. A “Profile Badges” event is defined as an *addressable event* with kind `30008` with a `d` tag with the value `profile_badges`. Profile badges contain an ordered list of pairs of `a` and `e` tags referencing a Badge Definition and a Badge Award for each badge to be displayed.

Badge Definition event

The following tags **MUST** be present:

- `d` tag with the unique name of the badge.

The following tags **MAY** be present:

- A `name` tag with a short name for the badge.
- `image` tag whose value is the URL of a high-resolution image representing the badge. The second value optionally specifies the dimensions of the image as `widthxheight` in pixels. Badge recommended dimensions is 1024x1024 pixels.
- A `description` tag whose value MAY contain a textual representation of the image, the meaning behind the badge, or the reason of its issuance.
- One or more `thumb` tags whose first value is an URL pointing to a thumbnail version of the image referenced in the `image` tag. The second value optionally specifies the dimensions of the thumbnail as `widthxheight` in pixels.

Badge Award event

The following tags **MUST** be present:

- An `a` tag referencing a kind `30009` Badge Definition event.
- One or more `p` tags referencing each pubkey awarded.

Profile Badges Event

The number of badges a pubkey can be awarded is unbounded. The Profile Badge event allows individual users to accept or reject awarded badges, as well as choose the display order of badges on their profiles.

The following tags **MUST** be present:

- A `d` tag with the unique identifier `profile_badges`

The following tags **MAY** be present:

- Zero or more ordered consecutive pairs of `a` and `e` tags referencing a kind `30009` Badge Definition and kind `8` Badge Award, respectively. Clients **SHOULD** ignore `a` without corresponding `e` tag and viceversa. Badge Awards referenced by the `e` tags should contain the same `a` tag.

Motivation

Users **MAY** be awarded badges (but not limited to) in recognition, in gratitude, for participation, or in appreciation of a certain goal, task or cause.

Users MAY choose to decorate their profiles with badges for fame, notoriety, recognition, support, etc., from badge issuers they deem reputable.

Recommendations

Clients MAY whitelist badge issuers (pubkeys) for the purpose of ensuring they retain a valuable/special factor for their users.

Badge image recommended aspect ratio is 1:1 with a high-res size of 1024x1024 pixels.

Badge thumbnail image recommended dimensions are: 512x512 (xl), 256x256 (l), 64x64 (m), 32x32 (s) and 16x16 (xs).

Clients MAY choose to render less badges than those specified by users in the Profile Badges event or replace the badge image and thumbnails with ones that fits the theme of the client.

Clients SHOULD attempt to render the most appropriate badge thumbnail according to the number of badges chosen by the user and space available. Clients SHOULD attempt render the high-res version on user action (click, tap, hover).

Example of a Badge Definition event

```
{
  "pubkey": "alice",
  "kind": 30009,
  "tags": [
    ["d", "bravery"],
    ["name", "Medal of Bravery"],
    ["description", "Awarded to users demonstrating bravery"],
    ["image", "https://nostr.academy/awards/bravery.png", "1024x1024"],
    ["thumb", "https://nostr.academy/awards/bravery_256x256.png", "256x256"]
  ],
  // other fields...
}
```

Example of Badge Award event

```
{
  "id": "<badge award event id>",
  "kind": 8,
  "pubkey": "alice",
  "tags": [
    ["a", "30009:alice:bravery"],
    ["p", "bob", "wss://relay"],
    ["p", "charlie", "wss://relay"]
  ],
  // other fields...
}
```

Example of a Profile Badges event

Honorable Bob The Brave:

```
{
  "kind": 30008,
  "pubkey": "bob",
  "tags": [
    ["d", "profile_badges"],
    ["a", "30009:alice:bravery"],
    ["e", "<bravery badge award event id>", "wss://nostr.academy"],
  ]
}
```

```
["a", "30009:alice:honor"],  
["e", "<honor badge award event id>", "wss://nostr.academy"]  
],  
// other fields...  
}
```

NIP-39

External Identities in Profiles

draft optional

Abstract

Nostr protocol users may have other online identities such as usernames, profile pages, keypairs etc. they control and they may want to include this data in their profile metadata so clients can parse, validate and display this information.

i tag on a metadata event

A new optional i tag is introduced for kind 0 metadata event defined in NIP-01:

```
{
  "id": <id>,
  "pubkey": <pubkey>,
  "tags": [
    ["i", "github:semisol", "9721ce4ee4fceb91c9711ca2a6c9a5ab"],
    ["i", "twitter:semisol_public", "1619358434134196225"],
    ["i", "mastodon:bitcoinhackers.org@semisol", "109775066355589974"],
    ["i", "telegram:1087295469", "nostrdirectory/770"]
  ],
  // other fields...
}
```

An i tag will have two parameters, which are defined as the following: 1. platform:identity: This is the platform name (for example github) and the identity on that platform (for example semisol) joined together with :. 2. proof: String or object that points to the proof of owning this identity.

Clients SHOULD process any i tags with more than 2 values for future extensibility. Identity provider names SHOULD only include a-z, 0-9 and the characters ._-/ and MUST NOT include :. Identity names SHOULD be normalized if possible by replacing uppercase letters with lowercase letters, and if there are multiple aliases for an entity the primary one should be used.

Claim types

github

Identity: A GitHub username.

Proof: A GitHub Gist ID. This Gist should be created by <identity> with a single file that has the text Verifying that I control the following Nostr public key: <npub encoded public key>. This can be located at <https://gist.github.com/<identity>/<proof>>.

twitter

Identity: A Twitter username.

Proof: A Tweet ID. The tweet should be posted by <identity> and have the text Verifying my account on nostr My Public Key: "<npub encoded public key>". This can be located at <https://twitter.com/<identity>/status/<proof>>.

mastodon

Identity: A Mastodon instance and username in the format <instance>/@<username>.

Proof: A Mastodon post ID. This post should be published by <username>@<instance> and have the text Verifying that I control the following Nostr public key: "<npub encoded public key>". This can be located at <https://<identity>/<proof>>.

telegram

Identity: A Telegram user ID.

Proof: A string in the format <ref>/<id> which points to a message published in the public channel or group with name <ref> and message ID <id>. This message should be sent by user ID <identity> and have the text Verifying that I control the following Nostr public key: "<npub encoded public key>". This can be located at <https://t.me/<proof>>.

Groups

NIP-28

Public Chat

draft optional

This NIP defines new event kinds for public chat channels, channel messages, and basic client-side moderation.

It reserves five event kinds (40-44) for immediate use:

- 40 - channel create
- 41 - channel metadata
- 42 - channel message
- 43 - hide message
- 44 - mute user

Client-centric moderation gives client developers discretion over what types of content they want included in their apps, while imposing no additional requirements on relays.

Kind 40: Create channel

Create a public chat channel.

In the channel creation content field, Client SHOULD include basic channel metadata (name, about, picture and relays as specified in kind 41).

```
{
  "content": "{ \"name\": \"Demo Channel\", \"about\": \"A test channel.\", \"picture\":
    \"https://placekitten.com/200/200\", \"relays\": [\"wss://nos.lol\", \"wss://nostr.mom\"]}",
  // other fields...
}
```

Kind 41: Set channel metadata

Update a channel's public metadata.

Kind 41 is used to update the metadata without modifying the event id for the channel. Only the most recent kind 41 per e tag value MAY be available.

Clients SHOULD ignore kind 41s from pubkeys other than the kind 40 pubkey.

Clients SHOULD support basic metadata fields:

- name - string - Channel name
- about - string - Channel description
- picture - string - URL of channel picture
- relays - array - List of relays to download and broadcast events to

Clients MAY add additional metadata fields.

Clients SHOULD use NIP-10 marked "e" tags to recommend a relay.

It is also possible to set the category name using the "t" tag. This category name can be searched and filtered.

```
{
  "content": "{ \"name\": \"Updated Demo Channel\", \"about\": \"Updating a test channel.\", \"picture\":
    \"https://placekitten.com/201/201\", \"relays\": [\"wss://nos.lol\", \"wss://nostr.mom\"]}",
  "tags": [
    ["e", <channel_create_event_id>, <relay-url>, "root"],
    ["t", <category_name-1>],
    ["t", <category_name-2>],
    ["t", <category_name-3>],
  ],
  // other fields...
}
```

```
}
```

Kind 42: Create channel message

Send a text message to a channel.

Clients SHOULD use NIP-10 marked “e” tags to recommend a relay and specify whether it is a reply or root message.

Clients SHOULD append NIP-10 “p” tags to replies.

Root message:

```
{
  "content": <string>,
  "tags": [
    ["e", <kind_40_event_id>, <relay-url>, "root"],
    // other fields...
  ]
}
```

Reply to another message:

```
{
  "content": <string>,
  "tags": [
    ["e", <kind_40_event_id>, <relay-url>, "root"],
    ["e", <kind_42_event_id>, <relay-url>, "reply"],
    ["p", <pubkey>, <relay-url>],
    // rest of tags...
  ],
  // other fields...
}
```

Kind 43: Hide message

User no longer wants to see a certain message.

The content may optionally include metadata such as a reason.

Clients SHOULD hide event 42s shown to a given user, if there is an event 43 from that user matching the event 42 id.

Clients MAY hide event 42s for other users other than the user who sent the event 43.

(For example, if three users ‘hide’ an event giving a reason that includes the word ‘pornography’, a Nostr client that is an iOS app may choose to hide that message for all iOS clients.)

```
{
  "content": "{ \"reason\": \"Dick pic\" }",
  "tags": [
    ["e", <kind_42_event_id>],
    // other fields...
  ]
}
```

Kind 44: Mute user

User no longer wants to see messages from another user.

The content may optionally include metadata such as a reason.

Clients SHOULD hide event 42s shown to a given user, if there is an event 44 from that user matching the event 42 pubkey.

Clients MAY hide event 42s for users other than the user who sent the event 44.

```
{
  "content": "{ \"reason\": \"Posting dick pics\" }",
  "tags": [ "p", <pubkey> ],
  // other fields...
}
```

Relay recommendations

Clients SHOULD use the relay URLs of the metadata events.

Clients MAY use any relay URL. For example, if a relay hosting the original kind 40 event for a channel goes offline, clients could instead fetch channel data from a backup relay, or a relay that clients trust more than the original relay.

Motivation

If we're solving censorship-resistant communication for social media, we may as well solve it also for Telegram-style messaging.

We can bring the global conversation out from walled gardens into a true public square open to all.

Additional info

- Chat demo PR with fiatjaf+jb55 comments
- Conversation about NIP16

NIP-29

Relay-based Groups

draft optional

This NIP defines a standard for groups that are only writable by a closed set of users. They can be public for reading by external users or not.

Groups are identified by a random string of any length that serves as an *id*.

There is no way to create a group, what happens is just that relays (most likely when asked by users) will create rules around some specific ids so these ids can serve as an actual group, henceforth messages sent to that group will be subject to these rules.

Normally a group will originally belong to one specific relay, but the community may choose to move the group to other relays or even fork the group so it exists in different forms – still using the same *id* – across different relays.

Relay-generated events

Relays are supposed to generate the events that describe group metadata and group admins. These are *addressable* events signed by the relay keypair directly, with the group *id* as the *d* tag.

Group identifier

A group may be identified by a string in the format `<host>'<group-id>`. For example, a group with *id* `abcdef` hosted at the relay `wss://groups.nostr.com` would be identified by the string `groups.nostr.com'abcdef`.

Group identifiers must be strings restricted to the characters `a-z0-9-_-`, and SHOULD be random in order to avoid name collisions.

When encountering just the `<host>` without the `'<group-id>`, clients MAY infer `_` as the group id, which is a special top-level group dedicated to relay-local discussions.

The *h* tag

Events sent by users to groups (chat messages, text notes, moderation events etc) MUST have an *h* tag with the value set to the group *id*.

Timeline references

In order to not be used out of context, events sent to these groups may contain references to previous events seen from the same relay in the *previous* tag. The choice of which previous events to pick belongs to the clients. The references are to be made using the first 8 characters (4 bytes) of any event in the last 50 events seen by the user in the relay, excluding events by themselves. There can be any number of references (including zero), but it's recommended that clients include at least 3 and that relays enforce this.

This is a hack to prevent messages from being broadcasted to external relays that have forks of one group out of context. Relays are expected to reject any events that contain timeline references to events not found in their own database. Clients should also check these to keep relays honest about them.

Late publication

Relays should prevent late publication (messages published now with a timestamp from days or even hours ago) unless they are open to receive a group forked or moved from another relay.

Group management

Groups can have any number of users with elevated access. These users are identified by role labels which are arbitrarily defined by the relays (see also the description of *kind:39003*). What each role is capable of not defined in this NIP either, it's a relay policy that can vary. Roles can be assigned by other users (as long as they have the

capability to add roles) by publishing a `kind:9000` event with that user's pubkey in a `p` tag and the roles afterwards (even if the user is already a group member a `kind:9000` can be issued and the user roles must just be updated).

The roles supported by the group as to having some special privilege assigned to them should be accessible on the event `kind:39003`, but the relay may also accept other role names, arbitrarily defined by clients, and just not do anything with them.

Users with any roles that have any privilege can be considered *admins* in a broad sense and be returned in the `kind:39001` event for a group.

Unmanaged groups

Unmanaged groups are impromptu groups that can be used in any public relay unaware of NIP-29 specifics. They piggyback on relays' natural white/blacklists (or lack of) but aside from that are not actively managed and won't have any admins, group state or metadata events.

In *unmanaged* groups, everybody is considered to be a member.

Unmanaged groups can transition to managed groups, in that case the relay master key just has to publish moderation events setting the state of all groups and start enforcing the rules they choose to.

Event definitions

These are the events expected to be found in NIP-29 groups.

Normal user-created events

Groups may accept any event kind, including chats, threads, long-form articles, calendar, livestreams, market announcements and so on. These should be as defined in their respective NIPs, with the addition of the `h` tag.

User-related group management events

These are events that can be sent by users to manage their situation in a group, they also require the `h` tag.

- *join request* (`kind:9021`)

Any user can send a `kind 9021` event to the relay in order to request admission to the group. Relays **MUST** reject the request if the user has not been added to the group. The accompanying error message **SHOULD** explain whether the rejection is final, if the request is pending review, or if some other special handling is relevant (e.g. if payment is required). If a user is already a member, the event **MUST** be rejected with `duplicate:` as the error message prefix.

```
{
  "kind": 9021,
  "content": "optional reason",
  "tags": [
    ["h", "<group-id>"],
    ["code", "<optional-invite-code>"]
  ]
}
```

The optional code tag may be used by the relay to preauthorize acceptances in closed groups, together with the `kind:9009` create-invite moderation event.

- *leave request* (`kind:9022`)

Any user can send one of these events to the relay in order to be automatically removed from the group. The relay will automatically issue a `kind:9001` in response removing this user.

```
{
  "kind": 9022,
  "content": "optional reason",
  "tags": [
```

```

    ["h", "<group-id>"]
  ]
}

```

Group state – or moderation

These are events expected to be sent by the relay master key or by group admins – and relays should reject them if they don't come from an authorized admin. They also require the h tag.

- *moderation events* (kinds:9000-9020) (optional)

Clients can send these events to a relay in order to accomplish a moderation action. Relays must check if the pubkey sending the event is capable of performing the given action based on its role and the relay's internal policy (see also the description of kind:39003).

```

{
  "kind": 90xx,
  "content": "optional reason",
  "tags": [
    ["h", "<group-id>"],
    ["previous", /*...*/]
  ]
}

```

Each moderation action uses a different kind and requires different arguments, which are given as tags. These are defined in the following table:

kind	name	tags
9000	put-user	p with pubkey hex and optional roles
9001	remove-user	p with pubkey hex
9002	edit-metadata	fields from kind:39000 to be modified
9005	delete-event	e with event id hex
9007	create-group	
9008	delete-group	
9009	create-invite	

It's expected that the group state (of who is an allowed member or not, who is an admin and with which permission or not, what are the group name and picture etc) can be fully reconstructed from the canonical sequence of these events.

Group metadata events

These events contain the group id in a d tag instead of the h tag. They MUST be created by the relay master key only and a single instance of each (or none) should exist at all times for each group. They are merely informative but should reflect the latest group state (as it was changed by moderation events over time).

- *group metadata* (kind:39000) (optional)

This event defines the metadata for the group – basically how clients should display it. It must be generated and signed by the relay in which is found. Relays shouldn't accept these events if they're signed by anyone else.

If the group is forked and hosted in multiple relays, there will be multiple versions of this event in each different relay and so on.

When this event is not found, clients may still connect to the group, but treat it as having a different status, *unmanaged*,

```
{
  "kind": 39000,
  "content": "",
  "tags": [
    ["d", "<group-id>"],
    ["name", "Pizza Lovers"],
    ["picture", "https://pizza.com/pizza.png"],
    ["about", "a group for people who love pizza"],
    ["public"], // or ["private"]
    ["open"] // or ["closed"]
  ]
  // other fields...
}
```

name, picture and about are basic metadata for the group for display purposes. public signals the group can be read by anyone, while private signals that only AUTHed users can read. open signals that anyone can request to join and the request will be automatically granted, while closed signals that members must be pre-approved or that requests to join will be manually handled.

- *group admins* (kind:39001) (optional)

Each admin is listed along with one or more roles. These roles SHOULD have a correspondence with the roles supported by the relay, as advertised by the kind:39003 event.

```
{
  "kind": 39001,
  "content": "list of admins for the pizza lovers group",
  "tags": [
    ["d", "<group-id>"],
    ["p", "<pubkey1-as-hex>", "ceo"],
    ["p", "<pubkey2-as-hex>", "secretary", "gardener"],
    // other pubkeys...
  ],
  // other fields...
}
```

- *group members* (kind:39002) (optional)

It's a list of pubkeys that are members of the group. Relays might choose to not to publish this information, to restrict what pubkeys can fetch it or to only display a subset of the members in it.

Clients should not assume this will always be present or that it will contain a full list of members.

```
{
  "kind": 39002,
  "content": "list of members for the pizza lovers group",
  "tags": [
    ["d", "<group-id>"],
    ["p", "<admin1>"],
    ["p", "<member-pubkey1>"],
    ["p", "<member-pubkey2>"],
    // other pubkeys...
  ],
  // other fields...
}
```

- *group roles* (kind:39003) (optional)

This is an event that MAY be published by the relay informing users and clients about what are the roles supported by this relay according to its internal logic.

For example, a relay may choose to support the roles "admin" and "moderator", in which the "admin" will be allowed to edit the group metadata, delete messages and remove users from the group, while the "moderator" can only delete messages (or the relay may choose to call these roles "ceo" and "secretary" instead, the exact role name is not relevant).

The process through which the relay decides what roles to support and how to handle moderation events internally based on them is specific to each relay and not specified here.

```
{
  "kind": 39003,
  "content": "list of roles supported by this group",
  "tags": [
    ["d", "<group-id>"],
    ["role", "<role-name>", "<optional-description>"],
    ["role", "<role-name>", "<optional-description>"],
    // other roles...
  ],
  // other fields...
}
```

Implementation quirks

Checking your own membership in a group

The latest of either kind:9000 or kind:9001 events present in a group should tell a user that they are currently members of the group or if they were removed. In case none of these exist the user is assumed to not be a member of the group – unless the group is unmanaged, in which case the user is assumed to be a member.

Adding yourself to a group

When a group is open, anyone can send a kind:9021 event to it in order to be added, then expect a kind:9000 event to be emitted confirming that the user was added. The same happens with closed groups, except in that case a user may only send a kind:9021 if it has an invite code.

Storing your list of groups

A definition for kind:10009 was included in NIP-51 that allows clients to store the list of groups a user wants to remember being in.

Using unmanaged relays

To prevent event leakage, when using unmanaged relays, clients should include the NIP-70 - tag, as just the previous tag won't be checked by other unmanaged relays.

Groups MAY be named without relay support by adding a name to the corresponding tag in a user's kind 10009 group list.

Moderation

NIP-32

Labeling

draft optional

This NIP defines two new indexable tags to label events and a new event kind (`kind:1985`) to attach those labels to existing events. This supports several use cases, including distributed moderation, collection management, license assignment, and content classification.

New Tags:

- `L` denotes a label namespace
- `l` denotes a label

Label Namespace Tag

An `L` tag can be any string, but publishers **SHOULD** ensure they are unambiguous by using a well-defined namespace (such as an ISO standard) or reverse domain name notation.

`L` tags are **RECOMMENDED** in order to support searching by namespace rather than by a specific tag. The special `ugc` (“user generated content”) namespace **MAY** be used when the label content is provided by an end user.

`L` tags starting with `#` indicate that the label target should be associated with the label’s value. This is a way of attaching standard nostr tags to events, pubkeys, relays, urls, etc.

Label Tag

An `l` tag’s value can be any string. If using an `L` tag, `l` tags **MUST** include a mark matching an `L` tag value in the same event. If no `L` tag is included, a mark **SHOULD** still be included. If none is included, `ugc` is implied.

Label Target

The label event **MUST** include one or more tags representing the object or objects being labeled: `e`, `p`, `a`, `r`, or `t` tags. This allows for labeling of events, people, relays, or topics respectively. As with NIP-01, a relay hint **SHOULD** be included when using `e` and `p` tags.

Content

Labels should be short, meaningful strings. Longer discussions, such as for an explanation of why something was labeled the way it was, should go in the event’s `content` field.

Self-Reporting

`l` and `L` tags **MAY** be added to other event kinds to support self-reporting. For events with a kind other than 1985, labels refer to the event itself.

Example events

A suggestion that multiple pubkeys be associated with the `permies` topic.

```
{
  "kind": 1985,
  "tags": [
    ["L", "#t"],
    ["l", "permies", "#t"],
    ["p", <pubkey1>, <relay_ur1>],
    ["p", <pubkey2>, <relay_ur2>]
  ],
  // other fields...
}
```

A report flagging violence toward a human being as defined by ontology.example.com.

```
{
  "kind": 1985,
  "tags": [
    ["L", "com.example.ontology"],
    ["I", "VI-hum", "com.example.ontology"],
    ["p", <pubkey1>, <relay_url>],
    ["p", <pubkey2>, <relay_url>]
  ],
  // other fields...
}
```

A moderation suggestion for a chat event.

```
{
  "kind": 1985,
  "tags": [
    ["L", "nip28.moderation"],
    ["I", "approve", "nip28.moderation"],
    ["e", <kind40_event_id>, <relay_url>]
  ],
  // other fields...
}
```

Assignment of a license to an event.

```
{
  "kind": 1985,
  "tags": [
    ["L", "license"],
    ["I", "MIT", "license"],
    ["e", <event_id>, <relay_url>]
  ],
  // other fields...
}
```

Publishers can self-label by adding 1 tags to their own non-1985 events. In this case, the kind 1 event's author is labeling their note as being related to Milan, Italy using ISO 3166-2.

```
{
  "kind": 1,
  "tags": [
    ["L", "ISO-3166-2"],
    ["I", "IT-MI", "ISO-3166-2"]
  ],
  "content": "It's beautiful here in Milan!",
  // other fields...
}
```

Author is labeling their note language as English using ISO-639-1.

```
{
  "kind": 1,
  "tags": [
    ["L", "ISO-639-1"],
    ["I", "en", "ISO-639-1"]
  ],
  "content": "English text",
  // other fields...
}
```


Other Notes

When using this NIP to bulk-label many targets at once, events may be requested for deletion using NIP-09 and a replacement may be published. We have opted not to use addressable/replaceable events for this due to the complexity in coming up with a standard `d` tag. In order to avoid ambiguity when querying, publishers SHOULD limit labeling events to a single namespace.

Before creating a vocabulary, explore how your use case may have already been designed and imitate that design if possible. Reverse domain name notation is encouraged to avoid namespace clashes, but for the sake of interoperability all namespaces should be considered open for public use, and not proprietary. In other words, if there is a namespace that fits your use case, use it even if it points to someone else's domain name.

Vocabularies MAY choose to fully qualify all labels within a namespace (for example, `["l", "com.example.vocabulary:my-label"]`). This may be preferred when defining more formal vocabularies that should not be confused with another namespace when querying without an `L` tag. For these vocabularies, all labels SHOULD include the namespace (rather than mixing qualified and unqualified labels).

A good heuristic for whether a use case fits this NIP is whether labels would ever be unique. For example, many events might be labeled with a particular place, topic, or pubkey, but labels with specific values like "John Doe" or "3.18743" are not labels, they are values, and should be handled in some other way.

Appendix: Known Ontologies

Below is a non-exhaustive list of ontologies currently in widespread use.

- social ontology categories

NIP-51

Lists

draft optional

This NIP defines lists of things that users can create. Lists can contain references to anything, and these references can be **public** or **private**.

Public items in a list are specified in the event `tags` array, while private items are specified in a JSON array that mimics the structure of the event `tags` array, but stringified and encrypted using the same scheme from NIP-44 (the shared key is computed using the author's public and private key) and stored in the `.content`. An earlier version of this specification used NIP-04 for encryptions. Those are now deprecated. For backward compatibility, Clients can automatically discover if the encryption is NIP-04 or NIP-44 by searching for "iv" in the ciphertext and decrypting accordingly.

When new items are added to an existing list, clients SHOULD append them to the end of the list, so they are stored in chronological order.

Types of lists

Standard lists

Standard lists use normal replaceable events, meaning users may only have a single list of each kind. They have special meaning and clients may rely on them to augment a user's profile or browsing experience.

For example, *mute list* can contain the public keys of spammers and bad actors users don't want to see in their feeds or receive annoying notifications from.

name	kind	description	expected tag items
Follow list	3	microblogging basic follow list, see NIP-02	"p" (pubkeys – with optional relay hint and petname)
Mute list	10000	things the user doesn't want to see in their feeds	"p" (pubkeys), "t" (hashtags), "word" (lowercase string), "e" (threads)
Pinned notes	10001	events the user intends to showcase in their profile page	"e" (kind:1 notes)
Read/write relays	10002	where a user publishes to and where they expect mentions	see NIP-65
Bookmarks	10003	uncategorized, "global" list of things a user wants to save	"e" (kind:1 notes), "a" (kind:30023 articles)
Communities	10004	NIP-72 communities the user belongs to	"a" (kind:34550 community definitions)
Public chats	10005	NIP-28 chat channels the user is in	"e" (kind:40 channel definitions)
Blocked relays	10006	relays clients should never connect to	"relay" (relay URLs)
Search relays	10007	relays clients should use when performing search queries	"relay" (relay URLs)
Simple groups	10009	NIP-29 groups the user is in	"group" (NIP-29 group id + relay URL + optional group name), "r" for each relay in use

name	kind	description	expected tag items
Relay feeds	10012	user favorite browsable relays (and relay sets)	"relay" (relay URLs) and "a" (kind:30002 relay set)
Interests	10015	topics a user may be interested in and pointers	"t" (hashtags) and "a" (kind:30015 interest set)
Media follows	10020	multimedia (photos, short video) follow list	"p" (pubkeys – with optional relay hint and petname)
Emojis	10030	user preferred emojis and pointers to emoji sets	"emoji" (see NIP-30) and "a" (kind:30030 emoji set)
DM relays	10050	Where to receive NIP-17 direct messages	"relay" (see NIP-17)
Good wiki authors	10101	NIP-54 user recommended wiki authors	"p" (pubkeys)
Good wiki relays	10102	NIP-54 relays deemed to only host useful articles	"relay" (relay URLs)

Sets

Sets are lists with well-defined meaning that can enhance the functionality and the UI of clients that rely on them. Unlike standard lists, users are expected to have more than one set of each kind, therefore each of them must be assigned a different "d" identifier.

For example, *relay sets* can be displayed in a dropdown UI to give users the option to switch to which relays they will publish an event or from which relays they will read the replies to an event; *curation sets* can be used by apps to showcase curations made by others tagged to different topics.

Aside from their main identifier, the "d" tag, sets can optionally have a "title", an "image" and a "description" tags that can be used to enhance their UI.

name	kind	description	expected tag items
Follow sets	30000	categorized groups of users a client may choose to check out in different circumstances	"p" (pubkeys)
Relay sets	30002	user-defined relay groups the user can easily pick and choose from during various operations	"relay" (relay URLs)
Bookmark sets	30003	user-defined bookmarks categories, for when bookmarks must be in labeled separate groups	"e" (kind:1 notes), "a" (kind:30023 articles)
Curation sets	30004	groups of articles picked by users as interesting and/or belonging to the same category	"a" (kind:30023 articles), "e" (kind:1 notes)
Curation sets	30005	groups of videos picked by users as interesting and/or belonging to the same category	"e" (kind:21 videos)
Kind mute sets	30007	mute pubkeys by kinds "d" tag MUST be the kind string	"p" (pubkeys)

name	kind	description	expected tag items
Interest sets	30015	interest topics represented by a bunch of “hashtags”	"t" (hashtags)
Emoji sets	30030	categorized emoji groups	"emoji" (see NIP-30)
Release artifact sets	30063	group of artifacts of a software release	"e" (kind:1063 file metadata events), "a" (software application event)
App curation sets	30267	references to multiple software applications	"a" (software application event)
Calendar	31924	a set of events categorized in any way	"a" (calendar event event)
Starter packs	39089	a named set of profiles to be shared around with the goal of being followed together	"p" (pubkeys)
Media starter packs	39092	same as above, but specific to multimedia (photos, short video) clients	"p" (pubkeys)

Deprecated standard lists

Some clients have used these lists in the past, but they should work on transitioning to the standard formats above.

kind	"d" tag	use instead
30000	"mute"	kind 10000 <i>mute list</i>
30001	"pin"	kind 10001 <i>pin list</i>
30001	"bookmark"	kind 10003 <i>bookmarks list</i>
30001	"communities"	kind 10004 <i>communities list</i>

Examples

A *mute list* with some public items and some encrypted items

```
{
  "id": "a92a316b75e44cfdc19986c634049158d4206fcc0b7b9c7ccbcdabe28beebcd0",
  "pubkey": "854043ae8f1f97430ca8c1f1a090bdde6488bd5115c7a45307a2a212750ae4cb",
  "created_at": 1699597889,
  "kind": 10000,
  "tags": [
    ["p", "07caba282f76441955b695551c3c5c742e5b9202a3784780f8086fdcdc1da3a9"],
    ["p", "a55c15f5e41d5aebd236eca5e0142789c5385703f1a7485aa4b38d94fd18dcc4"]
  ],
  "content":
    "TJob1dQrf2ndsmdbGU+05HT5GMnBSx3fx8QdY/g3NvCa7klfzgaQCmRZuo1d3WQjHD0jzSY1+MgTK5WjewFFumCc0ZniWtQMSga9tJk1ky00tLoUzyLn1v",
  "sig":
    "1173822c53261f8cffe7efbf43ba4a97a9198b3e402c2a1df130f42a8985a2d0d3430f4de350db184141e45ca844ab4e5364ea80f11d720e36357e1853"
}
```

A *curation set* of articles and notes about yaks

```
{
  "id": "567b41fc9060c758c4216fe5f8d3df7c57daad7ae757fa4606f0c39d4dd220ef",
  "pubkey": "d6dc95542e18b8b7aec2f14610f55c335abebec76f3db9e58c254661d0593a0c",
}
```

```

"created_at": 1695327657,
"kind": 30004,
"tags": [
  ["d", "jvdy9i4"],
  ["title", "Yaks"],
  ["image", "https://cdn.britannica.com/40/188540-050-9AC748DE/Yak-Himalayas-Nepal.jpg"],
  ["description", "The domestic yak, also known as the Tartary ox, grunting ox, or hairy cattle, is a species of long-haired domesticated cattle found throughout the Himalayan region of the Indian subcontinent, the Tibetan Plateau, Gilgit-Baltistan, Tajikistan and as far north as Mongolia and Siberia."],
  ["a", "30023:26dc95542e18b8b7aec2f14610f55c335abebec76f3db9e58c254661d0593a0c:950DQzw3ajNoZ8SyMD0zQ"],
  ["a", "30023:54af95542e18b8b7aec2f14610f55c335abebec76f3db9e58c254661d0593a0c:1-MYP8dAhramH9J5gJWKx"],
  ["a", "30023:f8fe95542e18b8b7aec2f14610f55c335abebec76f3db9e58c254661d0593a0c:D2Tbd38bGrFvU0bIbvSmt"],
  ["e", "d78ba0d5dce22bfff9db0a9e996c9ef27e2c91051de0c4e1da340e0326b4941e"]
],
"content": "",
"sig":
  "a9a4e2192eede77e6c9d24ddfab95ba3ff7c03fbd07ad011fff245abea431fb4d3787c2d04aad001cb039cb8de91d83ce30e9a94f82ac3c5a2372aa129"
}

```

A release artifact set of an Example App

```

{
  "id": "567b41fc9060c758c4216fe5f8d3df7c57daad7ae757fa4606f0c39d4dd220ef",
  "pubkey": "d6dc95542e18b8b7aec2f14610f55c335abebec76f3db9e58c254661d0593a0c",
  "created_at": 1695327657,
  "kind": 30063,
  "content": "Release notes in markdown",
  "tags": [
    ["d", "com.example.app@0.0.1"],
    ["e", "d78ba0d5dce22bfff9db0a9e996c9ef27e2c91051de0c4e1da340e0326b4941e"], // Windows exe
    ["e", "f27e2c91051de0c4e1da0d5dce22bfff9db0a9340e0326b4941ed78bae996c9e"], // MacOS dmg
    ["e", "9d24ddfab95ba3ff7c03fbd07ad011fff245abea431fb4d3787c2d04aad02332"], // Linux AppImage
    ["e", "340e0326b340e0326b4941ed78ba340e0326b4941ed78ba340e0326b49ed78ba"], // PWA
    ["a", "32267:d6dc95542e18b8b7aec2f14610f55c335abebec76f3db9e58c254661d0593a0c:com.example.app"] // Reference to parent software application
  ],
  "content": "Example App is a decentralized marketplace for apps",
  "sig":
    "a9a4e2192eede77e6c9d24ddfab95ba3ff7c03fbd07ad011fff245abea431fb4d3787c2d04aad001cb039cb8de91d83ce30e9a94f82ac3c5a2372aa129"
}

```

An app curation set

```

{
  "id": "d8037fa866eb5acd2159960b3ada7284172f7d687b5289cc72a96ca2b431b611",
  "pubkey": "78ce6faa72264387284e647ba6938995735ec8c7d5c5a65737e55130f026307d",
  "sig":
    "c1ce0a04521c020ae7485307cd86285530c1f778766a3fd594d662a73e7c28f307d7cd9a9ab642ae749fce62abbabb3a32facfe8d19a21fba551b60f",
  "kind": 30267,
  "created_at": 1729302793,
  "content": "My nostr app selection",
  "tags": [
    ["d", "nostr"],
    ["a", "32267:7579076d9aff0a4cfdefa7e2045f2486c7e5d8bc63bfc6b45397233e1bbfcb19:com.example.app1"],
    ["a", "32267:045f2486c7e5d8bc63bfc6b45397233e1bbfcb197579076d9aff0a4cfdefa7e2:net.example.app2"],
    ["a", "32267:264387284e647ba6938995735ec8c7d5c5a6f026307d78ce6faa725737e55130:pl.code.app3"]
  ]
}

```

```
}
```

Encryption process pseudocode

```
val private_items = [  
  ["p", "07caba282f76441955b695551c3c5c742e5b9202a3784780f8086fdcdc1da3a9"],  
  ["a", "a55c15f5e41d5aebd236eca5e0142789c5385703f1a7485aa4b38d94fd18dcc4"],  
]  
val base64blob = nip44.encrypt(json.encode_to_string(private_items))  
event.content = base64blob
```

NIP-56

Reporting

optional

A report is a kind 1984 event that signals to users and relays that some referenced content is objectionable. The definition of objectionable is obviously subjective and all agents on the network (users, apps, relays, etc.) may consume and take action on them as they see fit.

The content MAY contain additional information submitted by the entity reporting the content.

Tags

The report event MUST include a p tag referencing the pubkey of the user you are reporting.

If reporting a note, an e tag MUST also be included referencing the note id.

A report type string MUST be included as the 3rd entry to the e, p or x tag being reported, which consists of the following report types:

- nudity - depictions of nudity, porn, etc.
- malware - virus, trojan horse, worm, robot, spyware, adware, back door, ransomware, rootkit, kidnapper, etc.
- profanity - profanity, hateful speech, etc.
- illegal - something which may be illegal in some jurisdiction
- spam - spam
- impersonation - someone pretending to be someone else
- other - for reports that don't fit in the above categories

Some report tags only make sense for profile reports, such as impersonation.

- x tags SHOULD be info hash of a blob which is intended to be report. when the x tag is represented client MUST include an e tag which is the id of the event that contains the mentioned blob. also, additionally these events can contain a server tag to point to media servers which may contain the mentioned media.

l and L tags MAY be also be used as defined in NIP-32 to support further qualification and querying.

Example events

```
{
  "kind": 1984,
  "tags": [
    ["p", "<pubkey>", "nudity"],
    ["L", "social.nos.ontology"],
    ["l", "NS-nud", "social.nos.ontology"]
  ],
  "content": "",
  // other fields...
}
```

```
{
  "kind": 1984,
  "tags": [
    ["e", "<eventId>", "illegal"],
    ["p", "<pubkey>"]
  ],
  "content": "He's insulting the king!",
  // other fields...
}
```

```
{
  "kind": 1984,
  "tags": [
    ["p", "<impersonator pubkey>", "impersonation"]
  ],
  "content": "Profile is impersonating nostr:<victim bech32 pubkey>",
  // other fields...
}
```

```
{
  "kind": 1984,
  "tags": [
    ["x", "<blob hash>", "malware"],
    ["e", "<event id which contains the blob on x tag>", "malware"],
    ["server", "https://you-may-find-the-blob-here.com/path-to-url.ext"]
  ],
  "content": "This file contains malware software in it.",
  // other fields...
}
```

Client behavior

Clients can use reports from friends to make moderation decisions if they choose to. For instance, if 3+ of your friends report a profile for nudity, clients can have an option to automatically blur photos from said account.

Relay behavior

It is not recommended that relays perform automatic moderation using reports, as they can be easily gamed. Admins could use reports from trusted moderators to takedown illegal or explicit content if the relay does not allow such things.

NIP-36

Sensitive Content / Content Warning

draft optional

The content-warning tag enables users to specify if the event's content needs to be approved by readers to be shown. Clients can hide the content until the user acts on it.

l and L tags MAY be also be used as defined in NIP-32 with the content-warning or other namespace to support further qualification and querying.

tag: content-warning

options:

- [reason]: optional

```
{
  "pubkey": "<pub-key>",
  "created_at": 1000000000,
  "kind": 1,
  "tags": [
    ["t", "hashtag"],
    ["L", "content-warning"],
    ["l", "reason", "content-warning"],
    ["L", "social.nos.ontology"],
    ["l", "NS-nud", "social.nos.ontology"],
    ["content-warning", "<optional reason>"]
  ],
  "content": "sensitive content with #hashtag\n",
  "id": "<event-id>"
}
```

NIP-72

Moderated Communities (Reddit Style)

draft optional

The goal of this NIP is to enable public communities. It defines the replaceable event kind:34550 to define the community and the current list of moderators/administrators. Users that want to post into the community, simply tag any Nostr event with the community's a tag. Moderators may issue an approval event kind:4550.

Community Definition

Kind:34550 SHOULD include any field that helps define the community and the set of moderators. relay tags MAY be used to describe the preferred relay to download requests and approvals. A community definition event's d tag MAY double as its name, but if a name tag is provided, it SHOULD be displayed instead of the d tag.

```
{
  "created_at": <Unix timestamp in seconds>,
  "kind": 34550,
  "tags": [
    ["d", "<community-d-identifier>"],
    ["name", "<Community name>"],
    ["description", "<Community description>"],
    ["image", "<Community image url>", "<Width>x<Height>"],

    //.. other tags relevant to defining the community

    // moderators
    ["p", "<32-bytes hex of a pubkey1>", "<optional recommended relay URL>", "moderator"],
    ["p", "<32-bytes hex of a pubkey2>", "<optional recommended relay URL>", "moderator"],
    ["p", "<32-bytes hex of a pubkey3>", "<optional recommended relay URL>", "moderator"],

    // relays used by the community (w/optional marker)
    ["relay", "<relay hosting author kind 0>", "author"],
    ["relay", "<relay where to send and receive requests>", "requests"],
    ["relay", "<relay where to send and receive approvals>", "approvals"],
    ["relay", "<relay where to post requests to and fetch approvals from>"]
  ],
  // other fields...
}
```

Posting to a community

NIP-22 kind 1111 events SHOULD be used for text notes posted to a community, with the A tag always scoped to the community definition.

Top-level posts

For top-level posts, the uppercase and lowercase NIP-22 tags should both refer to the community definition itself.

```

{
  "kind": 1111,
  "tags": [
    ["A", "34550:<community-author-pubkey>:<community-d-identifier>", "<optional-relay-url>"],
    ["a", "34550:<community-author-pubkey>:<community-d-identifier>", "<optional-relay-url>"],
    ["P", "<community-author-pubkey>", "<optional-relay-url>"],
    ["p", "<community-author-pubkey>", "<optional-relay-url>"],
    ["K", "34550"],
    ["k", "34550"],
  ],
  "content": "Hi everyone. It's great to be here!",
  // other fields...
}

```

Nested replies

For nested replies, the uppercase tags should still refer to the community definition, while the lowercase tags should refer to the parent post or reply.

```

{
  "kind": 1111,
  "tags": [
    // community definition itself
    ["A", "34550:<community-author-pubkey>:<community-d-identifier>", "<optional-relay-url>"],
    ["P", "<community-author-pubkey>", "<optional-relay-url>"],
    ["K", "34550"],

    // parent post or reply
    ["e", "<parent-event-id>", "<optional-relay-url>"],
    ["p", "<parent-event-author-pubkey>", "<optional-relay-url>"],
    ["k", "<parent-event-kind>"] // most likely "1111"
  ],
  "content": "Agreed! Welcome everyone!",
  // other fields...
}

```

Backwards compatibility note

Previously kind 1 events were used for posts in communities, with an “a” tag pointing to the community. For backwards compatibility, clients MAY still query for kind 1 events, but SHOULD NOT use them for new posts. Instead, clients SHOULD use kind 1111 events with the A and a tags as described above.

Moderation

Anyone may issue an approval event to express their opinion that a post is appropriate for a community. Clients MAY choose which approval events to honor, but SHOULD at least use ones published by the group’s defined moderators.

An approval event MUST include one or more community a tags, an e or a tag pointing to the post, and the p tag of the author of the post (for approval notifications). a tag prefixes can be used to disambiguate between community and replaceable event pointers (community a tags always begin with 34550).

The event SHOULD also include the JSON-stringified post request event inside the .content, and a k tag with the original post’s event kind to allow filtering of approved posts by kind.

Moderators MAY request deletion of their approval of a post at any time using NIP-09 event deletion requests.

```
{
  "pubkey": "<32-bytes lowercase hex-encoded public key of the event creator>",
  "kind": 4550,
  "tags": [
    ["a", "34550:<event-author-pubkey>:<community-d-identifier>", "<optional-relay-url>"],
    ["e", "<post-id>", "<optional-relay-url>"],
    ["p", "<port-author-pubkey>", "<optional-relay-url>"],
    ["k", "<post-request-kind>"]
  ],
  "content": "<the full approved event, JSON-encoded>",
  // other fields...
}
```

It's recommended that multiple moderators approve posts to avoid deleting them from the community when a moderator is removed from the owner's list. In case the full list of moderators must be rotated, the new moderator set must sign new approvals for posts in the past or the community will restart. The owner can also periodically copy and re-sign of each moderator's approval events to make sure posts don't disappear with moderators.

Approvals of replaceable events can be created in three ways:

1. By tagging the replaceable event as an e tag if moderators want to approve each individual change to the replaceable event
2. By tagging the replaceable event as an a tag if the moderator authorizes the replaceable event author to make changes without additional approvals and
3. By tagging the replaceable event with both its e and a tag which empowers clients to display the original and updated versions of the event, with appropriate remarks in the UI.

Since relays are instructed to delete old versions of a replaceable event, the content of an approval using an e tag MUST have the specific version of the event or clients might not be able to find that version of the content anywhere.

Clients SHOULD evaluate any non-34550:* a tag as posts to be approved for all 34550:* a tags.

Cross-posting

Clients MAY support cross-posting between communities by posting a NIP 18 kind 6 or kind 16 repost to one or more communities using a tags as described above. The content of the repost MUST be the original event, not the approval event.

NIP-13

Proof of Work

draft optional

This NIP defines a way to generate and interpret Proof of Work for nostr notes. Proof of Work (PoW) is a way to add a proof of computational work to a note. This is a bearer proof that all relays and clients can universally validate with a small amount of code. This proof can be used as a means of spam deterrence.

`difficulty` is defined to be the number of leading zero bits in the NIP-01 id. For example, an id of `000000000e9d97a1ab09fc38103` has a difficulty of 36 with 36 leading 0 bits.

`002f...` is `0000 0000 0010 1111...` in binary, which has 10 leading zeroes. Do not forget to count leading zeroes for hex digits `<= 7`.

Mining

To generate PoW for a NIP-01 note, a nonce tag is used:

```
{"content": "It's just me mining my own business", "tags": [{"nonce", "1", "21"}]}
```

When mining, the second entry to the nonce tag is updated, and then the id is recalculated (see NIP-01). If the id has the desired number of leading zero bits, the note has been mined. It is recommended to update the `created_at` as well during this process.

The third entry to the nonce tag **SHOULD** contain the target difficulty. This allows clients to protect against situations where bulk spammers targeting a lower difficulty get lucky and match a higher difficulty. For example, if you require 40 bits to reply to your thread and see a committed target of 30, you can safely reject it even if the note has 40 bits difficulty. Without a committed target difficulty you could not reject it. Committing to a target difficulty is something all honest miners should be ok with, and clients **MAY** reject a note matching a target difficulty if it is missing a difficulty commitment.

Example mined note

```
{
  "id": "000006d8c378af1779d2feebc7603a125d99eca0ccf1085959b307f64e5dd358",
  "pubkey": "a48380f4cfcc1ad5378294fcac36439770f9c878dd880ffa94bb74ea54a6f243",
  "created_at": 1651794653,
  "kind": 1,
  "tags": [
    ["nonce", "776797", "20"]
  ],
  "content": "It's just me mining my own business",
  "sig":
    "284622fc0a3f4f1303455d5175f7ba962a3300d136085b9566801bc2e0699de0c7e31e44c81fb40ad9049173742e904713c3594a1da0fc5d2382a25c11"
}
```

Validating

Here is some reference C code for calculating the difficulty (aka number of leading zero bits) in a nostr event id:

```
int zero_bits(unsigned char b)
{
    int n = 0;

    if (b == 0)
        return 8;

    while (b >= 1)
        n++;

    return 7-n;
}

/* find the number of leading zero bits in a hash */
int count_leading_zero_bits(unsigned char *hash)
{
    int bits, total, i;
    for (i = 0, total = 0; i < 32; i++) {
        bits = zero_bits(hash[i]);
        total += bits;
        if (bits != 8)
            break;
    }
    return total;
}
```

Here is some JavaScript code for doing the same thing:

```
// hex should be a hexadecimal string (with no 0x prefix)
function countLeadingZeroes(hex) {
    let count = 0;

    for (let i = 0; i < hex.length; i++) {
        const nibble = parseInt(hex[i], 16);
        if (nibble === 0) {
            count += 4;
        } else {
            count += Math.clz32(nibble) - 28;
            break;
        }
    }

    return count;
}
```

Delegated Proof of Work

Since the NIP-01 note id does not commit to any signature, PoW can be outsourced to PoW providers, perhaps for a fee. This provides a way for clients to get their messages out to PoW-restricted relays without having to do any work themselves, which is useful for energy-constrained devices like mobile phones.

Relays

NIP-11

Relay Information Document

draft optional

Relays may provide server metadata to clients to inform them of capabilities, administrative contacts, and various server attributes. This is made available as a JSON document over HTTP, on the same URI as the relay's websocket.

When a relay receives an HTTP(s) request with an Accept header of `application/nostr+json` to a URI supporting WebSocket upgrades, they SHOULD return a document with the following structure.

```
{
  "name": <string identifying relay>,
  "description": <string with detailed information>,
  "banner": <a link to an image (e.g. in .jpg, or .png format)>,
  "icon": <a link to an icon (e.g. in .jpg, or .png format)>,
  "pubkey": <administrative contact pubkey>,
  "self": <relay's own pubkey>,
  "contact": <administrative alternate contact>,
  "supported_nips": <a list of NIP numbers supported by the relay>,
  "software": <string identifying relay software URL>,
  "version": <string version identifier>,
  "privacy_policy": <a link to a text file describing the relay's privacy policy>,
  "terms_of_service": <a link to a text file describing the relay's term of service>,
}
```

Any field may be omitted, and clients MUST ignore any additional fields they do not understand. Relays MUST accept CORS requests by sending Access-Control-Allow-Origin, Access-Control-Allow-Headers, and Access-Control-Allow-Methods headers.

Field Descriptions

Name

A relay may select a name for use in client software. This is a string, and SHOULD be less than 30 characters to avoid client truncation.

Description

Detailed plain-text information about the relay may be contained in the description string. It is recommended that this contain no markup, formatting or line breaks for word wrapping, and simply use double newline characters to separate paragraphs. There are no limitations on length.

Banner

To make nostr relay management more user friendly, an effort should be made by relay owners to communicate with non-dev non-technical nostr end users. A banner is a visual representation of the relay. It should aim to visually communicate the brand of the relay, complementing the text Description. Here is an example banner mockup as visualized in Damus iOS relay view of the Damus relay.

Icon

Icon is a compact visual representation of the relay for use in UI with limited real estate such as a nostr user's relay list view. Below is an example URL pointing to an image to be used as an icon for the relay. Recommended to be squared in shape.

```
{
  "icon": "https://nostr.build/i/53866b44135a27d624e99c6165cabd76ac8f72797209700acb189fce75021f47.jpg",
  // other fields...
}
```

Pubkey

An administrative contact may be listed with a pubkey, in the same format as Nostr events (32-byte hex for a secp256k1 public key). If a contact is listed, this provides clients with a recommended address to send encrypted direct messages (See NIP-17) to a system administrator. Expected uses of this address are to report abuse or illegal content, file bug reports, or request other technical assistance.

Relay operators have no obligation to respond to direct messages.

Self

A relay MAY maintain an identity independent from its administrator using the `self` field, which MUST be a 32-byte hex public key. This allows relays to respond to requests with events published either in advance or on demand by their own key.

Contact

An alternative contact may be listed under the `contact` field as well, with the same purpose as `pubkey`. Use of a Nostr public key and direct message SHOULD be preferred over this. Contents of this field SHOULD be a URI, using schemes such as `mailto` or `https` to provide users with a means of contact.

Supported NIPs

As the Nostr protocol evolves, some functionality may only be available by relays that implement a specific NIP. This field is an array of the integer identifiers of NIPs that are implemented in the relay. Examples would include 1, for "NIP-01" and 9, for "NIP-09". Client-side NIPs SHOULD NOT be advertised, and can be ignored by clients.

Software

The relay server implementation MAY be provided in the `software` attribute. If present, this MUST be a URL to the project's homepage.

Version

The relay MAY choose to publish its software version as a string attribute. The string format is defined by the relay implementation. It is recommended this be a version number or commit identifier.

Privacy Policy

The relay owner/admin MAY choose to link to a privacy policy document, which describes how the relay utilizes user data. Data collection, data usage, data retention, monetization of data, and third party data sharing SHOULD be included.

Terms of Service

The relay owner/admin MAY choose to link to a terms of service document.

Extra Fields

Server Limitations

These are limitations imposed by the relay on clients. Your client should expect that requests exceed these *practical* limitations are rejected or fail immediately.

```
{
  "limitation": {
    "max_message_length": 16384,
    "max_subscriptions": 300,
    "max_limit": 5000,
    "max_subid_length": 100,
    "max_event_tags": 100,
    "max_content_length": 8196,
    "min_pow_difficulty": 30,
    "auth_required": true,
    "payment_required": true,
    "restricted_writes": true,
    "created_at_lower_limit": 31536000,
    "created_at_upper_limit": 3,
    "default_limit": 500
  },
  // other fields...
}
```

- `max_message_length`: the maximum number of bytes for incoming JSON that the relay will attempt to decode and act upon. When you send large subscriptions, you will be limited by this value. It also effectively limits the maximum size of any event. Value is calculated from [to] after UTF-8 serialization (so some unicode characters will cost 2-3 bytes). It is equal to the maximum size of the WebSocket message frame.
- `max_subscriptions`: total number of subscriptions that may be active on a single websocket connection to this relay. Authenticated clients with a (paid) relationship to the relay may have higher limits.
- `max_subid_length`: maximum length of subscription id as a string.
- `max_limit`: the relay server will clamp each filter's `limit` value to this number. This means the client won't be able to get more than this number of events from a single subscription filter. This clamping is typically done silently by the relay, but with this number, you can know that there are additional results if you narrow your filter's time range or other parameters.
- `max_event_tags`: in any event, this is the maximum number of elements in the `tags` list.
- `max_content_length`: maximum number of characters in the `content` field of any event. This is a count of unicode characters. After serializing into JSON it may be larger (in bytes), and is still subject to the `max_message_length`, if defined.

- `min_pow_difficulty`: new events will require at least this difficulty of PoW, based on NIP-13, or they will be rejected by this server.
- `auth_required`: this relay requires NIP-42 authentication to happen before a new connection may perform any other action. Even if set to `False`, authentication may be required for specific actions.
- `payment_required`: this relay requires payment before a new connection may perform any action.
- `restricted_writes`: this relay requires some kind of condition to be fulfilled to accept events (not necessarily, but including `payment_required` and `min_pow_difficulty`). This should only be set to `true` when users are expected to know the relay policy before trying to write to it – like belonging to a special pubkey-based whitelist or writing only events of a specific niche kind or content. Normal anti-spam heuristics, for example, do not qualify.
- `created_at_lower_limit`: ‘created_at’ lower limit
- `created_at_upper_limit`: ‘created_at’ upper limit
- `default_limit`: The maximum returned events if you send a filter without a limit.

Event Retention

There may be a cost associated with storing data forever, so relays may wish to state retention times. The values stated here are defaults for unauthenticated users and visitors. Paid users would likely have other policies.

Retention times are given in seconds, with `null` indicating infinity. If zero is provided, this means the event will not be stored at all, and preferably an error will be provided when those are received.

```
{
  "retention": [
    {"kinds": [0, 1, [5, 7], [40, 49]], "time": 3600},
    {"kinds": [[40000, 49999]], "time": 100},
    {"kinds": [[30000, 39999]], "count": 1000},
    {"time": 3600, "count": 10000}
  ],
  // other fields...
}
```

`retention` is a list of specifications: each will apply to either all kinds, or a subset of kinds. Ranges may be specified for the kind field as a tuple of inclusive start and end values. Events of indicated kind (or all) are then limited to a count and/or time period.

It is possible to effectively blacklist Nostr-based protocols that rely on a specific kind number, by giving a retention time of zero for those kind values. While that is unfortunate, it does allow clients to discover servers that will support their protocol quickly via a single HTTP fetch.

There is no need to specify retention times for *ephemeral events* since they are not retained.

Content Limitations

Some relays may be governed by the arbitrary laws of a nation state. This may limit what content can be stored in clear-text on those relays. All clients are encouraged to use encryption to work around this limitation.

It is not possible to describe the limitations of each country’s laws and policies which themselves are typically vague and constantly shifting.

Therefore, this field allows the relay operator to indicate which countries’ laws might end up being enforced on them, and then indirectly on their users’ content.

Users should be able to avoid relays in countries they don’t like, and/or select relays in more favorable zones. Exposing this flexibility is up to the client software.

```
{
  "relay_countries": [ "CA", "US" ],
  // other fields...
}
```

- `relay_countries`: a list of two-level ISO country codes (ISO 3166-1 alpha-2) whose laws and policies may affect this relay. EU may be used for European Union countries. A * can be used for global relays.

Remember that a relay may be hosted in a country which is not the country of the legal entities who own the relay, so it's very likely a number of countries are involved.

Community Preferences

For public text notes at least, a relay may try to foster a local community. This would encourage users to follow the global feed on that relay, in addition to their usual individual follows. To support this goal, relays MAY specify some of the following values.

```
{
  "language_tags": ["en", "en-419"],
  "tags": ["sfw-only", "bitcoin-only", "anime"],
  "posting_policy": "https://example.com/posting-policy.html",
  // other fields...
}
```

- `language_tags` is an ordered list of IETF language tags indicating the major languages spoken on the relay. A * can be used for global relays.
- `tags` is a list of limitations on the topics to be discussed. For example `sfw-only` indicates that only “Safe For Work” content is encouraged on this relay. This relies on assumptions of what the “work” “community” feels “safe” talking about. In time, a common set of tags may emerge that allow users to find relays that suit their needs, and client software will be able to parse these tags easily. The `bitcoin-only` tag indicates that any *altcoin*, *“crypto”* or *blockchain* comments will be ridiculed without mercy.
- `posting_policy` is a link to a human-readable page which specifies the community policies for the relay. In cases where `sfw-only` is `True`, it's important to link to a page which gets into the specifics of your posting policy.

The `description` field should be used to describe your community goals and values, in brief. The `posting_policy` is for additional detail and legal terms. Use the `tags` field to signify limitations on content, or topics to be discussed, which could be machine processed by appropriate client software.

Pay-to-Relay

Relays that require payments may want to expose their fee schedules.

```
{
  "payments_url": "https://my-relay/payments",
  "fees": {
    "admission": [{ "amount": 1000000, "unit": "msats" }],
    "subscription": [{ "amount": 5000000, "unit": "msats", "period": 2592000 }],
    "publication": [{ "kinds": [4], "amount": 100, "unit": "msats" }],
  },
  // other fields...
}
```

Examples

As of 25 March 2025 the following command provided these results:

```
curl -H "Accept: application/nostr+json" https://jellyfish.land | jq
```

```
{
  "name": "JellyFish",
  "description": "Stay Immortal!",
  "banner": "https://image.nostr.build/7fdefea2dec1f1ec25b8ce69362566c13b2b7f13f1726c2e4584f05f64f62496.jpg",
  "pubkey": "bf2bee5281149c7c350f5d12ae32f514c7864ff10805182f4178538c2c421007",
  "contact": "hi@dezh.tech",
  "software": "https://github.com/dezh-tech/immortal",
  "supported_nips": [
    1,
    9,
    11,
    13,
    17,
    40,
    42,
    59,
    62,
    70
  ],
  "version": "immortal - 0.0.9",
  "relay_countries": [
    "*"
  ],
  "language_tags": [
    "*"
  ],
  "tags": [],
  "posting_policy": "https://jellyfish.land/tos.txt",
  "payments_url": "https://jellyfish.land/relay",
  "icon": "https://image.nostr.build/2547e9ec4b23589e09bc7071e0806c3d4293f76284c58ff331a64bce978aaee8.jpg",
  "retention": [],
  "fees": {
    "subscription": [
      {
        "amount": 3000,
        "period": 2628003,
        "unit": "sats"
      },
      {
        "amount": 8000,
        "period": 7884009,
        "unit": "sats"
      },
      {
        "amount": 15000,
        "period": 15768018,
        "unit": "sats"
      },
      {
        "amount": 28000,
```

```

        "period": 31536036,
        "unit": "sats"
    }
]
},
"limitation": {
    "auth_required": false,
    "max_message_length": 70000,
    "max_subid_length": 256,
    "max_subscriptions": 350,
    "min_pow_difficulty": 0,
    "payment_required": true,
    "restricted_writes": true,
    "max_event_tags": 2000,
    "max_content_length": 70000,
    "created_at_lower_limit": 0,
    "created_at_upper_limit": 2147483647,
    "default_limit": 500,
    "max_limit": 5000
}
}

```

NIP-42

Authentication of clients to relays

draft optional

This NIP defines a way for clients to authenticate to relays by signing an ephemeral event.

Motivation

A relay may want to require clients to authenticate to access restricted resources. For example,

- A relay may request payment or other forms of whitelisting to publish events – this can naïvely be achieved by limiting publication to events signed by the whitelisted key, but with this NIP they may choose to accept any events as long as they are published from an authenticated user;
- A relay may limit access to kind: 4 DMs to only the parties involved in the chat exchange, and for that it may require authentication before clients can query for that kind.
- A relay may limit subscriptions of any kind to paying users or users whitelisted through any other means, and require authentication.

Definitions

New client-relay protocol messages

This NIP defines a new message, AUTH, which relays CAN send when they support authentication and clients can send to relays when they want to authenticate. When sent by relays the message has the following form:

```
["AUTH", <challenge-string>]
```

And, when sent by clients, the following form:

```
["AUTH", <signed-event-json>]
```

Clients MAY provide signed events from multiple pubkeys in a sequence of AUTH messages. Relays MUST treat all pubkeys as authenticated accordingly.

AUTH messages sent by clients MUST be answered with an OK message, like any EVENT message.

Canonical authentication event

The signed event is an ephemeral event not meant to be published or queried, it must be of kind: 22242 and it should have at least two tags, one for the relay URL and one for the challenge string as received from the relay. Relays MUST exclude kind: 22242 events from being broadcasted to any client. `created_at` should be the current time. Example:

```
{
  "kind": 22242,
  "tags": [
    ["relay", "wss://relay.example.com/"],
    ["challenge", "challengestringhere"]
  ]
}
```

```

],
// other fields...
}

```

OK and CLOSED machine-readable prefixes

This NIP defines two new prefixes that can be used in OK (in response to event writes by clients) and CLOSED (in response to rejected subscriptions by clients):

- "auth-required: " - for when a client has not performed AUTH and the relay requires that to fulfill the query or write the event.
- "restricted: " - for when a client has already performed AUTH but the key used to perform it is still not allowed by the relay or is exceeding its authorization.

Protocol flow

At any moment the relay may send an AUTH message to the client containing a challenge. The challenge is valid for the duration of the connection or until another challenge is sent by the relay. The client MAY decide to send its AUTH event at any point and the authenticated session is valid afterwards for the duration of the connection.

auth-required in response to a REQ message

Given that a relay is likely to require clients to perform authentication only for certain jobs, like answering a REQ or accepting an EVENT write, these are some expected common flows:

```

relay: ["AUTH", "<challenge>"]
client: ["REQ", "sub_1", {"kinds": [4]}]
relay: ["CLOSED", "sub_1", "auth-required: we can't serve DMs to unauthenticated users"]
client: ["AUTH", {"id": "abcdef...", ...}]
client: ["AUTH", {"id": "abcde2...", ...}]
relay: ["OK", "abcdef...", true, ""]
relay: ["OK", "abcde2...", true, ""]
client: ["REQ", "sub_1", {"kinds": [4]}]
relay: ["EVENT", "sub_1", {...}]
relay: ["EVENT", "sub_1", {...}]
relay: ["EVENT", "sub_1", {...}]
relay: ["EVENT", "sub_1", {...}]
...

```

In this case, the AUTH message from the relay could be sent right as the client connects or it can be sent immediately before the CLOSED is sent. The only requirement is that *the client must have a stored challenge associated with that relay* so it can act upon that in response to the auth-required CLOSED message.

auth-required in response to an EVENT message

The same flow is valid for when a client wants to write an EVENT to the relay, except now the relay sends back an OK message instead of a CLOSED message:

```

relay: ["AUTH", "<challenge>"]
client: ["EVENT", {"id": "012345...", ...}]
relay: ["OK", "012345...", false, "auth-required: we only accept events from registered users"]

```



```
client: ["AUTH", {"id": "abcdef...", ...}]
relay: ["OK", "abcdef...", true, ""]
client: ["EVENT", {"id": "012345...", ...}]
relay: ["OK", "012345...", true, ""]
```

Signed Event Verification

To verify AUTH messages, relays must ensure:

- that the kind is 22242;
- that the event created_at is close (e.g. within ~10 minutes) of the current time;
- that the "challenge" tag matches the challenge sent before;
- that the "relay" tag matches the relay URL:
 - URL normalization techniques can be applied. For most cases just checking if the domain name is correct should be enough.

NIP-50

Search Capability

draft optional

Abstract

Many Nostr use cases require some form of general search feature, in addition to structured queries by tags or ids. Specifics of the search algorithms will differ between event kinds, this NIP only describes a general extensible framework for performing such queries.

search filter field

A new search field is introduced for REQ messages from clients:

```
{
  // other fields on filter object...
  "search": <string>
}
```

search field is a string describing a query in a human-readable form, i.e. "best nostr apps". Relays SHOULD interpret the query to the best of their ability and return events that match it. Relays SHOULD perform matching against content event field, and MAY perform matching against other fields if that makes sense in the context of a specific kind.

Results SHOULD be returned in descending order by quality of search result (as defined by the implementation), not by the usual .created_at. The limit filter SHOULD be applied after sorting by matching score. A query string may contain key:value pairs (two words separated by colon), these are extensions, relays SHOULD ignore extensions they don't support.

Clients may specify several search filters, i.e. ["REQ", "", { "search": "orange" }, { "kinds": [1, 2], "search": "purple" }]. Clients may include kinds, ids and other filter field to restrict the search results to particular event kinds.

Clients SHOULD use the supported_nips field to learn if a relay supports search filter. Clients MAY send search filter queries to any relay, if they are prepared to filter out extraneous responses from relays that do not support this NIP.

Clients SHOULD query several relays supporting this NIP to compensate for potentially different implementation details between relays.

Clients MAY verify that events returned by a relay match the specified query in a way that suits the client's use case, and MAY stop querying relays that have low precision.

Relays SHOULD exclude spam from search results by default if they support some form of spam filtering.

Extensions

Relay MAY support these extensions: - include:spam - turn off spam filtering, if it was enabled by default - domain:<domain> - include only events from users whose valid nip05 domain matches the domain - language:<two letter ISO 639-1 language code> - include only events of a specified language - sentiment:<negative/neutral/positive> - include only events of a specific sentiment - nsfw:<true/false> - include or exclude nsfw events (default: true)

NIP-45

Event Counts

draft optional

Relays may support the verb COUNT, which provides a mechanism for obtaining event counts.

Motivation

Some queries a client may want to execute against connected relays are prohibitively expensive, for example, in order to retrieve follower counts for a given pubkey, a client must query all kind-3 events referring to a given pubkey only to count them. The result may be cached, either by a client or by a separate indexing server as an alternative, but both options erode the decentralization of the network by creating a second-layer protocol on top of Nostr.

Filters and return values

This NIP defines the verb COUNT, which accepts a query id and filters as specified in NIP 01 for the verb REQ. Multiple filters are OR'd together and aggregated into a single count result.

```
["COUNT", <query_id>, <filters JSON>...]
```

Counts are returned using a COUNT response in the form {"count": <integer>}. Relays may use probabilistic counts to reduce compute requirements. In case a relay uses probabilistic counts, it MAY indicate it in the response with approximate key i.e. {"count": <integer>, "approximate": <true|false>}.

```
["COUNT", <query_id>, {"count": <integer>}]
```

Whenever the relay decides to refuse to fulfill the COUNT request, it MUST return a CLOSED message.

Examples

Followers count

```
["COUNT", <query_id>, {"kinds": [3], "#p": [<pubkey>]}]
["COUNT", <query_id>, {"count": 238}]
```

Count posts and reactions

```
["COUNT", <query_id>, {"kinds": [1, 7], "authors": [<pubkey>]}]
["COUNT", <query_id>, {"count": 5}]
```

Count posts approximately

```
["COUNT", <query_id>, {"kinds": [1]}]  
["COUNT", <query_id>, {"count": 93412452, "approximate": true}]
```

Relay refuses to count

```
["COUNT", <query_id>, {"kinds": [1059], "#p": [<pubkey>]}]  
["CLOSED", <query_id>, "auth-required: cannot count other people's DMs"]
```

NIP-65

Relay List Metadata

draft optional

Defines a replaceable event using `kind:10002` to advertise relays where the user generally **writes** to and relays where the user generally **reads** mentions.

The event MUST include a list of `r` tags with relay URLs as value and an optional `read` or `write` marker. If the marker is omitted, the relay is both **read** and **write**.

```
{
  "kind": 10002,
  "tags": [
    ["r", "wss://alicerelay.example.com"],
    ["r", "wss://brando-relay.com"],
    ["r", "wss://expensive-relay.example2.com", "write"],
    ["r", "wss://nostr-relay.example.com", "read"]
  ],
  "content": "",
  // other fields...
}
```

When downloading events **from** a user, clients SHOULD use the **write** relays of that user.

When downloading events **about** a user, where the user was tagged (mentioned), clients SHOULD use the user's **read** relays.

When publishing an event, clients SHOULD:

- Send the event to the **write** relays of the author
- Send the event to all **read** relays of each tagged user
- Send the author's `kind:10002` event to all relays the event was published to

Size

Clients SHOULD guide users to keep `kind:10002` lists small (2-4 relays of each category).

Discoverability

Clients SHOULD spread an author's `kind:10002` event to as many relays as viable, paying attention to relays that, at any moment, serve naturally as well-known public indexers for these relay lists (where most other clients and users are connecting to in order to publish and fetch those).

NIP-48

Proxy Tags

draft optional

Nostr events bridged from other protocols such as ActivityPub can link back to the source object by including a "proxy" tag, in the form:

```
["proxy", <id>, <protocol>]
```

Where:

- <id> is the ID of the source object. The ID format varies depending on the protocol. The ID must be universally unique, regardless of the protocol.
- <protocol> is the name of the protocol, e.g. "activitypub".

Clients may use this information to reconcile duplicated content bridged from other protocols, or to display a link to the source object.

Proxy tags may be added to any event kind, and doing so indicates that the event did not originate on the Nostr protocol, and instead originated elsewhere on the web.

Supported protocols

This list may be extended in the future.

Protocol	ID format	Example
activitypub	URL	https://gleasonator.com/objects/9f5248
atproto	AT URI	at://did:plc:zhbjlbmir5dganqhueg7y4i3/
rss	URL with guid fragment	https://soapbox.pub/rss/feed.xml#https
web	URL	https://twitter.com/jack/status/20

Examples

ActivityPub object:

```
{
  "kind": 1,
  "content": "I'm vegan btw",
  "tags": [
    [
      "proxy",
      "https://gleasonator.com/objects/8f6fac53-4f66-4c6e-ac7d-92e5e78c3e79",
      "activitypub"
    ]
  ],
}
```

```
"pubkey": "79c2cae114ea28a981e7559b4fe7854a473521a8d22a66bbab9fa248eb820ff6",
"created_at": 1691091365,
"id": "55920b758b9c7b17854b6e3d44e6a02a83d1cb49e1227e75a30426dea94d4cb2",
"sig":
  "a72f12c08f18e85d98fb92ae89e2fe63e48b8864c5e10fbdd5335f3c9f936397a6b0a7350efe251f8168b1601d7012d4a6d0ee6eec958067cf22a14f5a
}
```

See also

- FEP-ffff: Proxy Objects
- Mostr bridge

Clients

NIP-21

nostr: URI scheme

draft optional

This NIP standardizes the usage of a common URI scheme for maximum interoperability and openness in the network.

The scheme is `nostr`:

The identifiers that come after are expected to be the same as those defined in NIP-19 (except `nsec`).

Examples

- `nostr:npub1sn0wdenkukak0d9dfczzeacvkhrgz92ak56egt7vdgzn8pv2wfqqhrjdv9`
- `nostr:nprofile1qqsrhuxx8l9ex335q7he0f09aej04zpazpl0ne2cgukyawd24mayt8gpp4mhxue69uhhytnc9e3k7mgpz4mhxue69`
- `nostr:note1fntxtkcy9pjwucqwa9mddn7v03wwwsu9j330jj350nvhpky2tuaspk6nqc`
- `nostr:nevent1qqstna2yrezu5wghjvswqqculvwxsrcvu7uc0f78gan4xqhvz49d9spr3mhxue69uhkummnw3ez6un9d3shjtn4de`

Linking HTML pages to Nostr entities

`<link>` tags with `rel="alternate"` can be used to associate webpages to Nostr events, in cases where the same content is served via the two mediums (for example, a web server that exposes Markdown articles both as HTML pages and as `kind:30023` events served under itself as a relay or through some other relay). For example:

```
<head>
  <link rel="alternate"
        href="nostr:naddr1qqyrzwrxc6ngvfkqyghwum8ghj7enf96x5ctx9e3k7mgzyqalp33lwf5vdq847t6te0wnags0gs0mu72kz8938tn24wl fze6qcyq"
        />
</head>
```

Likewise, `<link>` tags with `rel="me"` or `rel="author"` can be used to assign authorship of webpages to Nostr profiles. For example:

```
<head>
  <link rel="me"
        href="nostr:nprofile1qyxhwum8ghj7mn0vwxcmvqyd8wum8ghj7un9d3shjtnhv4ehgetjde38gcewdhk6qpq80cvw07tjdrngpa0j7j7tmyl2yr6y"
        />
</head>
```

NIP-19

bech32-encoded entities

draft optional

This NIP standardizes bech32-formatted strings that can be used to display keys, ids and other information in clients. These formats are not meant to be used anywhere in the core protocol, they are only meant for displaying to users, copy-pasting, sharing, rendering QR codes and inputting data.

It is recommended that ids and keys are stored in either hex or binary format, since these formats are closer to what must actually be used the core protocol.

Bare keys and ids

To prevent confusion and mixing between private keys, public keys and event ids, which are all 32 byte strings. bech32-(not-m) encoding with different prefixes can be used for each of these entities.

These are the possible bech32 prefixes:

- npub: public keys
- nsec: private keys
- note: note ids

Example: the hex public key 3bf0c63fcb93463407af97a5e5ee64fa883d107ef9e558472c4eb9aaefa459d translates to npub180cvv07tjdrnga0j7j7tmnyl2yr6yr7l8j4s3evf6u64th6gkwsyjh6w6.

The bech32 encodings of keys and ids are not meant to be used inside the standard NIP-01 event formats or inside the filters, they're meant for human-friendlier display and input only. Clients should still accept keys in both hex and npub format for now, and convert internally.

Shareable identifiers with extra metadata

When sharing a profile or an event, an app may decide to include relay information and other metadata such that other apps can locate and display these entities more easily.

For these events, the contents are a binary-encoded list of TLV (type-length-value), with T and L being 1 byte each (uint8, i.e. a number in the range of 0-255), and V being a sequence of bytes of the size indicated by L.

These are the possible bech32 prefixes with TLV:

- nprofile: a nostr profile
- nevent: a nostr event
- naddr: a nostr *addressable event* coordinate
- nrelay: a nostr relay (deprecated)

These possible standardized TLV types are indicated here:

- 0: special
 - depends on the bech32 prefix:
 - * for nprofile it will be the 32 bytes of the profile public key
 - * for nevent it will be the 32 bytes of the event id
 - * for naddr, it is the identifier (the "d" tag) of the event being referenced. For normal replaceable events use an empty string.
- 1: relay
 - for nprofile, nevent and naddr, *optionally*, a relay in which the entity (profile or event) is more likely to be found, encoded as ascii
 - this may be included multiple times
- 2: author
 - for naddr, the 32 bytes of the pubkey of the event
 - for nevent, *optionally*, the 32 bytes of the pubkey of the event
- 3: kind
 - for naddr, the 32-bit unsigned integer of the kind, big-endian

- for nevent, *optionally*, the 32-bit unsigned integer of the kind, big-endian

Examples

- npub10elfcs4fr0l0r8af98jlmgdh9c8tcxjvz9qkw038js35mp4dma8qzvjpg should decode into the public key hex 7e7e9c42a91bfef19fa929e5fda1b72e0ebc1a4c1141673e2794234d86addf4e and vice-versa
- nsec1vl029mgpspedva04g90vltk6fvh240zqtv9k0t9af8935ke9laqsnlfe5 should decode into the private key hex 67dea2ed018072d675f5415ecfaed7d2597555e202d85b3d65ea4e58d2d92ffa and vice-versa
- nprofile1qqsrhuxx8l9ex335q7he0f09aej04zpazpl0ne2cgukyawd24mayt8gpp4mxxue69uhhytnc9e3k7mgpz4mxxue69uhkg6 should decode into a profile with the following TLV items:
 - pubkey: 3bf0c63fcb93463407af97a5e5ee64fa883d107ef9e558472c4eb9aaefa459d
 - relay: wss://r.x.com
 - relay: wss://djbas.sadkb.com

Notes

- npub keys MUST NOT be used in NIP-01 events or in NIP-05 JSON responses, only the hex format is supported there.
- When decoding a bech32-formatted string, TLVs that are not recognized or supported should be ignored, rather than causing an error.

NIP-03

OpenTimestamps Attestations for Events

draft optional

This NIP defines an event with kind:1040 that can contain an OpenTimestamps proof for any other event:

```
{
  "kind": 1040
  "tags": [
    ["e", <target-event-id>, <relay-url>],
    ["k", "<target-event-kind>"]
  ],
  "content": <base64-encoded OTS file data>
}
```

- The OpenTimestamps proof MUST prove the referenced e event id as its digest.
- The content MUST be the full content of an .ots file containing at least one Bitcoin attestation. This file SHOULD contain a **single** Bitcoin attestation (as not more than one valid attestation is necessary and less bytes is better than more) and no reference to “pending” attestations since they are useless in this context.

Example OpenTimestamps proof verification flow

Using nak, jq and ots:

```
~> nak req -i e71c6ea722987debd60f81f9ea4f604b5ac0664120dd64fb9d23abc4ec7c323 wss://nostr-pub.wellorder.net | jq -r
  .content | ots verify
> using an esplora server at https://blockstream.info/api
- sequence ending on block 810391 is valid
timestamp validated at block [810391]
```

Payments

NIP-57

Lightning Zaps

draft optional

This NIP defines two new event types for recording lightning payments between users. 9734 is a `zap request`, representing a payer's request to a recipient's lightning wallet for an invoice. 9735 is a `zap receipt`, representing the confirmation by the recipient's lightning wallet that the invoice issued in response to a `zap request` has been paid.

Having lightning receipts on nostr allows clients to display lightning payments from entities on the network. These can be used for fun or for spam deterrence.

Protocol flow

1. Client calculates a recipient's `lnurl pay` request url from the `zap` tag on the event being zapped (see Appendix G), or by decoding their `lud16` field on their profile according to the `lnurl` specifications. The client **MUST** send a GET request to this url and parse the response. If `allowsNostr` exists and it is `true`, and if `nostrPubkey` exists and is a valid BIP 340 public key in hex, the client should associate this information with the user, along with the response's `callback`, `minSendable`, and `maxSendable` values.
2. Clients may choose to display a lightning zap button on each post or on a user's profile. If the user's `lnurl pay` request endpoint supports `nostr`, the client **SHOULD** use this NIP to request a `zap receipt` rather than a normal `lnurl invoice`.
3. When a user (the "sender") indicates they want to send a zap to another user (the "recipient"), the client should create a `zap request` event as described in Appendix A of this NIP and sign it.
4. Instead of publishing the `zap request`, the 9734 event should instead be sent to the `callback` url received from the `lnurl pay` endpoint for the recipient using a GET request. See Appendix B for details and an example.
5. The recipient's `lnurl` server will receive this `zap request` and validate it. See Appendix C for details on how to properly configure an `lnurl` server to support zaps, and Appendix D for details on how to validate the `nostr` query parameter.
6. If the `zap request` is valid, the server should fetch a description hash invoice where the description is this `zap request` note and this note only. No additional `lnurl` metadata is included in the description. This will be returned in the response according to LUD06.
7. On receiving the invoice, the client **MAY** pay it or pass it to an app that can pay the invoice.
8. Once the invoice is paid, the recipient's `lnurl` server **MUST** generate a `zap receipt` as described in Appendix E, and publish it to the relays specified in the `zap request`.
9. Clients **MAY** fetch `zap receipts` on posts and profiles, but **MUST** authorize their validity as described in Appendix F. If the `zap request` note contains a non-empty content, it may display a zap comment. Generally clients should show users the `zap request` note, and use the `zap receipt` to show "zap authorized by ..." but this is optional.

Reference and examples

Appendix A: Zap Request Event

A `zap request` is an event of kind 9734 that is *not* published to relays, but is instead sent to a recipient's `lnurl pay` `callback` url. This event's content **MAY** be an optional message to send along with the payment. The event **MUST** include the following tags:

- `relays` is a list of relays the recipient's wallet should publish its `zap receipt` to. Note that relays should not be nested in an additional list, but should be included as shown in the example below.
- `amount` is the amount in *millisats* the sender intends to pay, formatted as a string. This is recommended, but optional.
- `lnurl` is the `lnurl pay` url of the recipient, encoded using bech32 with the prefix `lnurl`. This is recommended, but optional.
- `p` is the hex-encoded pubkey of the recipient.

In addition, the event **MAY** include the following tags:

- `e` is an optional hex-encoded event id. Clients MUST include this if zapping an event rather than a person.
- `a` is an optional event coordinate that allows tipping addressable events such as NIP-23 long-form notes.
- `k` is the stringified kind of the target event.

Example:

```
{
  "kind": 9734,
  "content": "Zap!",
  "tags": [
    ["relays", "wss://nostr-pub.wellorder.com", "wss://anotherrelay.example.com"],
    ["amount", "21000"],
    ["lnurl", "lnurl1dp68gurn8ghj7um5v93kktj9ehx2amr9uh8wetvdskkkmn0wahz7mrww4excup0dajx2mr92x9xp"],
    ["p", "04c915daefee38317fa734444acee390a8269fe5810b2241e5e6dd343dfbecc9"],
    ["e", "9ae37aa68f48645127299e9453eb5d908a0cbb6058ff340d528ed4d37c8994fb"],
    ["k", "1"]
  ],
  "pubkey": "97c70a44366a6535c145b333f973ea86dfdc2d7a99da618c40c64705ad98e322",
  "created_at": 1679673265,
  "id": "30efed56a035b2549fcaec0bf2c1595f9a9b3bb4b1a38abaf8ee9041c4b7d93",
  "sig":
    "f2cb581a84ed10e4dc84937bd98e27acac71ab057255f6aa8dfa561808c981fe8870f4a03c1e3666784d82a9c802d3704e174371aa13d63e2aeaf24ff5"
}
```

Appendix B: Zap Request HTTP Request

A signed zap request event is not published, but is instead sent using a HTTP GET request to the recipient's callback url, which was provided by the recipient's `lnurl` pay endpoint. This request should have the following query parameters defined:

- `amount` is the amount in *millisats* the sender intends to pay
- `nostr` is the 9734 zap request event, JSON encoded then URI encoded
- `lnurl` is the `lnurl` pay url of the recipient, encoded using bech32 with the prefix `lnurl`

This request should return a JSON response with a `pr` key, which is the invoice the sender must pay to finalize their zap. Here is an example flow in javascript:

```
const senderPubkey // The sender's pubkey
const recipientPubkey = // The recipient's pubkey
const callback = // The callback received from the recipients lnurl pay endpoint
const lnurl = // The recipient's lightning address, encoded as a lnurl
const sats = 21

const amount = sats * 1000
const relays = ['wss://nostr-pub.wellorder.net']
const event = encodeURI(JSON.stringify(await signEvent({
  kind: 9734,
  content: "",
  pubkey: senderPubkey,
  created_at: Math.round(Date.now() / 1000),
  tags: [
    ["relays", ...relays],
    ["amount", amount.toString()],
    ["lnurl", lnurl],
    ["p", recipientPubkey],
  ],
})))

const {pr: invoice} = await fetchJson(`${callback}?amount=${amount}&nostr=${event}&lnurl=${lnurl}`)
```

Appendix C: LNURL Server Configuration

The lnurl server will need some additional pieces of information so that clients can know that zap invoices are supported:

1. Add a nostrPubkey to the lnurl-pay static endpoint /.well-known/lnurlp/<user>, where nostrPubkey is the nostr pubkey your server will use to sign zap receipt events. Clients will use this to validate zap receipts.
2. Add an allowsNostr field and set it to true.

Appendix D: LNURL Server Zap Request Validation

When a client sends a zap request event to a server's lnurl-pay callback URL, there will be a nostr query parameter whose value is that event which is URI- and JSON-encoded. If present, the zap request event must be validated in the following ways:

1. It MUST have a valid nostr signature
2. It MUST have tags
3. It MUST have only one p tag
4. It MUST have 0 or 1 e tags
5. There should be a relays tag with the relays to send the zap receipt to.
6. If there is an amount tag, it MUST be equal to the amount query parameter.
7. If there is an a tag, it MUST be a valid event coordinate
8. There MUST be 0 or 1 P tags. If there is one, it MUST be equal to the zap receipt's pubkey.

The event MUST then be stored for use later, when the invoice is paid.

Appendix E: Zap Receipt Event

A zap receipt is created by a lightning node when an invoice generated by a zap request is paid. Zap receipts are only created when the invoice description (committed to the description hash) contains a zap request note.

When receiving a payment, the following steps are executed:

1. Get the description for the invoice. This needs to be saved somewhere during the generation of the description hash invoice. It is saved automatically for you with CLN, which is the reference implementation used here.
2. Parse the bolt11 description as a JSON nostr event. This SHOULD be validated based on the requirements in Appendix D, either when it is received, or before the invoice is paid.
3. Create a nostr event of kind 9735 as described below, and publish it to the relays declared in the zap request.

The following should be true of the zap receipt event:

- The content SHOULD be empty.
- The created_at date SHOULD be set to the invoice paid_at date for idempotency.
- tags MUST include the p tag (zap recipient) AND optional e tag from the zap request AND optional a tag from the zap request AND optional P tag from the pubkey of the zap request (zap sender).
- The zap receipt MUST have a bolt11 tag containing the description hash bolt11 invoice.
- The zap receipt MUST contain a description tag which is the JSON-encoded zap request.
- SHA256(description) SHOULD match the description hash in the bolt11 invoice.
- The zap receipt MAY contain a preimage tag to match against the payment hash of the bolt11 invoice. This isn't really a payment proof, there is no real way to prove that the invoice is real or has been paid. You are trusting the author of the zap receipt for the legitimacy of the payment.

The zap receipt is not a proof of payment, all it proves is that some nostr user fetched an invoice. The existence of the zap receipt implies the invoice as paid, but it could be a lie given a rogue implementation.

A reference implementation for a zap-enabled lnurl server can be found [here](#).

Example zap receipt:

```
{
  "id": "67b48a14fb66c60c8f9070bdeb37afdfcc3d08ad01989460448e4081eddda446",
  "pubkey": "9630f464cca6a5147aa8a35f0bccdd3ce485324e732fd39e09233b1d848238f31",
  "created_at": 1674164545,
```



```

"kind": 9735,
"tags": [
  ["p", "32e1827635450ebb3c5a7d12c1f8e7b2b514439ac10a67eef3d9fd9c5c68e245"],
  ["P", "97c70a44366a6535c145b333f973ea86dfdc2d7a99da618c40c64705ad98e322"],
  ["e", "3624762a1274dd9636e0c552b53086d70bc88c165bc4dc0f9e836a1eaf86c3b8"],
  ["k", "1"],
  ["bolt11",
    "lnbc10u1p3unwfp5t9r3yymhpfqcu1x78u027lxspgxc2n2987mx2j55nnfs95nxnzapp5jmrh92pfl78spqs78v9euf2385t83uwpk9l1dr1vf6ch"],
  ["description",
    "{\n\"pubkey\": \"97c70a44366a6535c145b333f973ea86dfdc2d7a99da618c40c64705ad98e322\", \"content\": \"\", \"id\": \"d9cc14d50fcd\"}"],
  ["preimage", "5d006d2cf1e73c7148e7519a4c68adc81642ce0e25a432b2434c99f97344c15f"]
],
"content": "",
}

```

Appendix F: Validating Zap Receipts

A client can retrieve zap receipts on events and pubkeys using a NIP-01 filter, for example `{"kinds": [9735], "#e": [...]}`. Zaps MUST be validated using the following steps:

- The zap receipt event's pubkey MUST be the same as the recipient's lnurl provider's nostrPubkey (retrieved in step 1 of the protocol flow).
- The invoiceAmount contained in the bolt11 tag of the zap receipt MUST equal the amount tag of the zap request (if present).
- The lnurl tag of the zap request (if present) SHOULD equal the recipient's lnurl.

Appendix G: zap tag on other events

When an event includes one or more zap tags, clients wishing to zap it SHOULD calculate the lnurl pay request based on the tags value instead of the event author's profile field. The tag's second argument is the hex string of the receiver's pub key and the third argument is the relay to download the receiver's metadata (Kind-0). An optional fourth parameter specifies the weight (a generalization of a percentage) assigned to the respective receiver. Clients should parse all weights, calculate a sum, and then a percentage to each receiver. If weights are not present, CLIENTS should equally divide the zap amount to all receivers. If weights are only partially present, receivers without a weight should not be zapped (weight = 0).

```

{
  "tags": [
    ["zap", "82341f882b6eabcd2ba7f1ef90aad961cf074af15b9ef44a09f9d2a8fbfb6a2", "wss://nostr.oxtr.dev", "1" ],
    // 25%
    ["zap", "fa984bd7dbb282f07e16e7ae87b26a2a7b9b90b7246a44771f0cf5ae58018f52", "wss://nostr.wine/", "1" ],
    // 25%
    ["zap", "460c25e682fda7832b52d1f22d3d22b3176d972f60dc3212ed8c92ef85065c", "wss://nos.lol/", "2" ]
    // 50%
  ]
}

```

Clients MAY display the zap split configuration in the note.

Future Work

Zaps can be extended to be more private by encrypting zap request notes to the target user, but for simplicity it has been left out of this initial draft.

NIP-47

Nostr Wallet Connect (NWC)

draft optional

Rationale

This NIP describes a way for clients to access a remote lightning wallet through a standardized protocol. Custodians may implement this, or the user may run a bridge that bridges their wallet/node and the Nostr Wallet Connect protocol.

Terms

- **client**: Nostr app on any platform that wants to interact with a lightning wallet.
- **user**: The person using the **client**, and wants to connect their wallet to their **client**.
- **wallet service**: Nostr app that typically runs on an always-on computer (eg. in the cloud or on a Raspberry Pi). This app has access to the APIs of the wallets it serves.

Theory of Operation

Fundamentally NWC is communication between a **client** and **wallet service** by the means of E2E-encrypted direct messages over a nostr relay. The relay knows the kinds and tags of notes, but not the content of the encrypted payloads. The **user's** identity key is not used to avoid linking payment activity to the user. Ideally unique keys are used for each individual connection.

1. **Users** who wish to use this NIP to allow **client(s)** to interact with their wallet must first acquire a special "connection" URI from their NIP-47 compliant wallet application. The wallet application may provide this URI using a QR screen, or a pasteable string, or some other means.
2. The **user** should then copy this URI into their **client(s)** by pasting, or scanning the QR, etc. The **client(s)** should save this URI and use it later whenever the **user** (or the **client** on the user's behalf) wants to interact with the wallet. The **client** should then request an info (13194) event from the relay(s) specified in the URI. The **wallet service** will have sent that event to those relays earlier, and the relays will hold it as a replaceable event.
3. When the **user** initiates a payment their nostr **client** create a `pay_invoice` request, encrypts it using a token from the URI, and sends it (kind 23194) to the relay(s) specified in the connection URI. The **wallet service** will be listening on those relays and will decrypt the request and then contact the **user's** wallet application to send the payment. The **wallet service** will know how to talk to the wallet application because the connection URI specified relay(s) that have access to the wallet app API.
4. Once the payment is complete the **wallet service** will send an encrypted response (kind 23195) to the **user** over the relay(s) in the URI.
5. The **wallet service** may send encrypted notifications (kind 23197 or 23196) of wallet events (such as a received payment) to the **client**.

Events

There are four event kinds:

- NIP-47 info event: 13194
- NIP-47 request: 23194
- NIP-47 response: 23195
- NIP-47 notification event: 23197 (23196 for backwards compatibility with NIP-04)

Info Event

The info event should be a replaceable event that is published by the **wallet service** on the relay to indicate which capabilities it supports.

The content should be a plaintext string with the supported capabilities space-separated, eg. `pay_invoice get_balance notifications`.

If the **wallet service** supports notifications, the info event SHOULD contain a `notifications` tag with the supported notification types space-separated, eg. `payment_received payment_sent`.

It should also contain supported encryption modes as described in the Encryption section. For example:

```
{
  "kind": 13194,
  "tags": [
    ["encryption", "nip44_v2 nip04"], // List of supported encryption schemes as described in the Encryption
    ["notifications", "payment_received payment_sent"]
    // ...
  ],
  "content": "pay_invoice get_balance make_invoice lookup_invoice list_transactions get_info notifications",
  // ...
}
```

Request and Response Events

Both the request and response events SHOULD contain one `p` tag, containing the public key of the **wallet service** if this is a request, and the public key of the **client** if this is a response. The response event SHOULD contain an `e` tag with the id of the request event it is responding to. Optionally, a request can have an `expiration` tag that has a unix timestamp in seconds. If the request is received after this timestamp, it should be ignored.

The content of requests and responses is encrypted with NIP44, and is a JSON-RPCish object with a semi-fixed structure.

Important note for backwards-compatibility: The initial version of the protocol used NIP04. If a **wallet service** or client app does not include the `encryption` tag in the `info` or request events, it should be assumed that the connection is using NIP04 for encryption. See the Encryption section for more information.

Example request:

```
{
  "kind": 23194,
  "tags": [
    ["encryption", "nip44_v2"],
    ["p", "03..." ] // public key of the wallet service.
    // ...
  ],
  "content": nip44_encrypt({ // Encryption type corresponds to the `encryption` tag.
    "method": "pay_invoice", // method, string
    "params": { // params, object
      "invoice": "lnbc50n1..." // command-related data
    }
  }),
}
```

Example response:

```
{
  "kind": 23195,
  "tags": [
    ["p", "03..." ] // public key of the requesting client app
    ["e", "1234"] // id of the request event this is responding to
    // ...
  ],
  "content": nip44_encrypt({ // Encrypted using the scheme requested by the client.
```

```

    "result_type": "pay_invoice", //indicates the structure of the result field
    "error": { //object, non-null in case of error
        "code": "UNAUTHORIZED", //string error code, see below
        "message": "human readable error message"
    },
    "result": { // result, object. null in case of error.
        "preimage": "0123456789abcdef..." // command-related data
    }
}
// ...
}

```

The `result_type` field **MUST** contain the name of the method that this event is responding to. The `error` field **MUST** contain a `message` field with a human readable error message and a `code` field with the error code if the command was not successful. If the command was successful, the `error` field must be null.

Notification Events

The notification event is a kind 23197 event **SHOULD** contain one `p` tag, the public key of the **client**.

The content of notifications is encrypted with NIP44 (or NIP-04 for legacy client apps), and is a JSON-RPCish object with a semi-fixed structure:

```

{
  "notification_type": "payment_received", //indicates the structure of the notification field
  "notification": {
    "payment_hash": "0123456789abcdef..." // notification-related data
  }
}

```

Note on backwards-compatibility: If a **wallet service** supports both nip44 and nip04 for legacy client apps, it should publish both notification events for each notification - kind 23196 encrypted with NIP-04, and kind 23197 encrypted with NIP-44. It is up to the **client** to decide which event to listen to based on its supported encryption and declared supported encryption schemes of the **wallet service** in the `info` event.

Error codes

- **RATE_LIMITED:** The client is sending commands too fast. It should retry in a few seconds.
- **NOT_IMPLEMENTED:** The command is not known or is intentionally not implemented.
- **INSUFFICIENT_BALANCE:** The wallet does not have enough funds to cover a fee reserve or the payment amount.
- **QUOTA_EXCEEDED:** The wallet has exceeded its spending quota.
- **RESTRICTED:** This public key is not allowed to do this operation.
- **UNAUTHORIZED:** This public key has no wallet connected.
- **INTERNAL:** An internal error.
- **UNSUPPORTED_ENCRYPTION:** The encryption type of the request is not supported by the wallet service.
- **OTHER:** Other error.

Nostr Wallet Connect URI

Communication between the **client** and **wallet service** requires two keys in order to encrypt and decrypt messages. The connection URI includes the secret key of the **client** and only the public key of the **wallet service**.

The **client** discovers **wallet service** by scanning a QR code, handling a deeplink or pasting in a URI.

The **wallet service** generates this connection URI with protocol `nostr+walletconnect://` and base path its 32-byte hex-encoded pubkey, which **SHOULD** be unique per client connection.

The connection URI contains the following query string parameters:

- **relay Required.** URL of the relay where the **wallet service** is connected and will be listening for events. May be more than one.

- **secret** Required. 32-byte randomly generated hex encoded string. The **client** MUST use this to sign events and encrypt payloads when communicating with the **wallet service**. The **wallet service** MUST use the corresponding public key of this secret to communicate with the **client**.
 - Authorization does not require passing keys back and forth.
 - The user can have different keys for different applications. Keys can be revoked and created at will and have arbitrary constraints (eg. budgets).
 - The key is harder to leak since it is not shown to the user and backed up.
 - It improves privacy because the user's main key would not be linked to their payments.
- **lud16** Recommended. A lightning address that clients can use to automatically setup the lud16 field on the user's profile if they have none configured.

The **client** should then store this connection and use it when the user wants to perform actions like paying an invoice. Due to this NIP using ephemeral events, it is recommended to pick relays that do not close connections on inactivity to not drop events, and ideally retain the events until they are either consumed or become stale.

- When the **client** sends or receives a message it will use the **secret** from the connection URI and **wallet service's** pubkey to encrypt or decrypt.
- When the **wallet service** sends or receives a message it will use its own secret and the corresponding pubkey of the **client's** secret to encrypt or decrypt. The **wallet service** SHOULD NOT store the secret it generates for the client and MUST NOT rely on the knowing the **client** secret for general operation.

Example connection string

nostr+walletconnect://b889ff5b1513b641e2a139f661a661364979c5beee91842f8f0ef42ab558e9d4?relay=wss%3A%2F%2Frelay.damus.io&secret=71a

Commands

pay_invoice

Description: Requests payment of an invoice.

Request:

```
{
  "method": "pay_invoice",
  "params": {
    "invoice": "lnbc50n1...", // bolt11 invoice
    "amount": 123, // invoice amount in msats, optional
    "metadata": {} // generic metadata that can be used to add things like zap/boostagram details for a payer
                    name/comment/etc, optional
  }
}
```

Response:

```
{
  "result_type": "pay_invoice",
  "result": {
    "preimage": "0123456789abcdef...", // preimage of the payment
    "fees_paid": 123, // value in msats, optional
  }
}
```

Errors: - **PAYMENT_FAILED**: The payment failed. This may be due to a timeout, exhausting all routes, insufficient capacity or similar.

multi_pay_invoice

Description: Requests payment of multiple invoices.

Request:

```
{
  "method": "multi_pay_invoice",
  "params": {
    "invoices": [
      {"id": "4da52c32a1", "invoice": "lnbc1...", "amount": 123}, // bolt11 invoice and amount in msats, amount
        is optional
      {"id": "3da52c32a1", "invoice": "lnbc50n1...", "metadata": {} }, // generic metadata that can be used to
        add things like zap/boostagram details for a payer name/comment/etc, optional
    ],
  }
}
```

Response:

For every invoice in the request, a separate response event is sent. To differentiate between the responses, each response event contains a `d` tag with the id of the invoice it is responding to; if no id was given, then the payment hash of the invoice should be used.

```
{
  "result_type": "multi_pay_invoice",
  "result": {
    "preimage": "0123456789abcdef...", // preimage of the payment
    "fees_paid": 123, // value in msats, optional
  }
}
```

Errors: - `PAYMENT_FAILED`: The payment failed. This may be due to a timeout, exhausting all routes, insufficient capacity or similar.

`pay_keysend`

Request:

```
{
  "method": "pay_keysend",
  "params": {
    "amount": 123, // invoice amount in msats, required
    "pubkey": "03...", // payee pubkey, required
    "preimage": "0123456789abcdef...", // preimage of the payment, optional
    "tlv_records": [ // tlv records, optional
      {
        "type": 5482373484, // tlv type
        "value": "0123456789abcdef" // hex encoded tlv value
      }
    ]
  }
}
```

Response:

```
{
  "result_type": "pay_keysend",
  "result": {
    "preimage": "0123456789abcdef...", // preimage of the payment
    "fees_paid": 123, // value in msats, optional
  }
}
```

Errors: - `PAYMENT_FAILED`: The payment failed. This may be due to a timeout, exhausting all routes, insufficient capacity or similar.

multi_pay_keysend

Description: Requests multiple keysend payments.

Has an array of keysends, these follow the same semantics as pay_keysend, just done in a batch

Request:

```
{
  "method": "multi_pay_keysend",
  "params": {
    "keysends": [
      {"id": "4c5b24a351", "pubkey": "03...", "amount": 123},
      {"id": "3da52c32a1", "pubkey": "02...", "amount": 567, "preimage": "abc123..", "tlv_records": [{"type": 696969, "value": "77616c5f6872444873305242454d353736"}]},
    ],
  }
}
```

Response:

For every keysend in the request, a separate response event is sent. To differentiate between the responses, each response event contains a d tag with the id of the keysend it is responding to; if no id was given, then the pubkey should be used.

```
{
  "result_type": "multi_pay_keysend",
  "result": {
    "preimage": "0123456789abcdef...", // preimage of the payment
    "fees_paid": 123, // value in msats, optional
  }
}
```

Errors: - PAYMENT_FAILED: The payment failed. This may be due to a timeout, exhausting all routes, insufficient capacity or similar.

make_invoice

Request:

```
{
  "method": "make_invoice",
  "params": {
    "amount": 123, // value in msats
    "description": "string", // invoice's description, optional
    "description_hash": "string", // invoice's description hash, optional
    "expiry": 213, // expiry in seconds from time invoice is created, optional
    "metadata": {} // generic metadata that can be used to add things like zap/boostagram details for a payer
                    name/comment/etc, optional
  }
}
```

Response:

```
{
  "result_type": "make_invoice",
  "result": {
    "type": "incoming", // "incoming" for invoices, "outgoing" for payments
    "state": "pending", // optional
    "invoice": "string", // encoded invoice, optional
    "description": "string", // invoice's description, optional
  }
}
```

```

    "description_hash": "string", // invoice's description hash, optional
    "preimage": "string", // payment's preimage, optional if unpaid
    "payment_hash": "string", // Payment hash for the payment
    "amount": 123, // value in msats
    "fees_paid": 123, // value in msats
    "created_at": unixtimestamp, // invoice/payment creation time
    "expires_at": unixtimestamp, // invoice expiration time, optional if not applicable
    "metadata": {} // generic metadata that can be used to add things like zap/boostagram details for a payer
                    name/comment/etc.
  }
}

```

lookup_invoice

Request:

```

{
  "method": "lookup_invoice",
  "params": {
    "payment_hash": "31afdf1..", // payment hash of the invoice, one of payment_hash or invoice is required
    "invoice": "lnbc50n1.." // invoice to lookup
  }
}

```

Response:

```

{
  "result_type": "lookup_invoice",
  "result": {
    "type": "incoming", // "incoming" for invoices, "outgoing" for payments
    "state": "pending", // can be "pending", "settled", "expired" (for invoices) or "failed" (for payments),
                      optional
    "invoice": "string", // encoded invoice, optional
    "description": "string", // invoice's description, optional
    "description_hash": "string", // invoice's description hash, optional
    "preimage": "string", // payment's preimage, optional if unpaid
    "payment_hash": "string", // Payment hash for the payment
    "amount": 123, // value in msats
    "fees_paid": 123, // value in msats
    "created_at": unixtimestamp, // invoice/payment creation time
    "expires_at": unixtimestamp, // invoice expiration time, optional if not applicable
    "settled_at": unixtimestamp, // invoice/payment settlement time, optional if unpaid
    "metadata": {} // generic metadata that can be used to add things like zap/boostagram details for a payer
                    name/comment/etc.
  }
}

```

Errors: - NOT_FOUND: The invoice could not be found by the given parameters.

list_transactions

Lists invoices and payments. If type is not specified, both invoices and payments are returned. The from and until parameters are timestamps in seconds since epoch. If from is not specified, it defaults to 0. If until is not specified, it defaults to the current time. Transactions are returned in descending order of creation time.

Request:

```

{
  "method": "list_transactions",

```



```

    "params": {
      "from": 1693876973, // starting timestamp in seconds since epoch (inclusive), optional
      "until": 1703225078, // ending timestamp in seconds since epoch (inclusive), optional
      "limit": 10, // maximum number of invoices to return, optional
      "offset": 0, // offset of the first invoice to return, optional
      "unpaid": true, // include unpaid invoices, optional, default false
      "type": "incoming", // "incoming" for invoices, "outgoing" for payments, undefined for both
    }
  }
}

```

Response:

```

{
  "result_type": "list_transactions",
  "result": {
    "transactions": [
      {
        "type": "incoming", // "incoming" for invoices, "outgoing" for payments
        "state": "pending", // can be "pending", "settled", "expired" (for invoices) or "failed" (for
          payments), optional
        "invoice": "string", // encoded invoice, optional
        "description": "string", // invoice's description, optional
        "description_hash": "string", // invoice's description hash, optional
        "preimage": "string", // payment's preimage, optional if unpaid
        "payment_hash": "string", // Payment hash for the payment
        "amount": 123, // value in msats
        "fees_paid": 123, // value in msats
        "created_at": unixtimestamp, // invoice/payment creation time
        "expires_at": unixtimestamp, // invoice expiration time, optional if not applicable
        "settled_at": unixtimestamp, // invoice/payment settlement time, optional if unpaid
        "metadata": {} // generic metadata that can be used to add things like zap/boostagram details for a
          payer name/comment/etc.
      }
    ],
  },
}

```

get_balance

Request:

```

{
  "method": "get_balance",
  "params": {}
}

```

Response:

```

{
  "result_type": "get_balance",
  "result": {
    "balance": 10000, // user's balance in msats
  }
}

```

get_info

Request:

```
{
  "method": "get_info",
  "params": {}
}
```

Response:

```
{
  "result_type": "get_info",
  "result": {
    "alias": "string",
    "color": "hex string",
    "pubkey": "hex string",
    "network": "string", // mainnet, testnet, signet, or regtest
    "block_height": 1,
    "block_hash": "hex string",
    "methods": ["pay_invoice", "get_balance", "make_invoice", "lookup_invoice", "list_transactions",
      "get_info"], // list of supported methods for this connection
    "notifications": ["payment_received", "payment_sent"], // list of supported notifications for this
      connection, optional.
  }
}
```

Notifications

payment_received

Description: A payment was successfully received by the wallet.

Notification:

```
{
  "notification_type": "payment_received",
  "notification": {
    "type": "incoming",
    "state": "settled", // optional
    "invoice": "string", // encoded invoice
    "description": "string", // invoice's description, optional
    "description_hash": "string", // invoice's description hash, optional
    "preimage": "string", // payment's preimage
    "payment_hash": "string", // Payment hash for the payment
    "amount": 123, // value in msats
    "fees_paid": 123, // value in msats
    "created_at": unixtimestamp, // invoice/payment creation time
    "expires_at": unixtimestamp, // invoice expiration time, optional if not applicable
    "settled_at": unixtimestamp, // invoice/payment settlement time
    "metadata": {} // generic metadata that can be used to add things like zap/boostagram details for a payer
      name/comment/etc.
  }
}
```

payment_sent

Description: A payment was successfully sent by the wallet.

Notification:

```
{
  "notification_type": "payment_sent",
```

```

"notification": {
  "type": "outgoing",
  "state": "settled", // optional
  "invoice": "string", // encoded invoice
  "description": "string", // invoice's description, optional
  "description_hash": "string", // invoice's description hash, optional
  "preimage": "string", // payment's preimage
  "payment_hash": "string", // Payment hash for the payment
  "amount": 123, // value in msats
  "fees_paid": 123, // value in msats
  "created_at": unixtimestamp, // invoice/payment creation time
  "expires_at": unixtimestamp, // invoice expiration time, optional if not applicable
  "settled_at": unixtimestamp, // invoice/payment settlement time
  "metadata": {} // generic metadata that can be used to add things like zap/boostagram details for a payer
                    name/comment/etc.
}
}

```

Example pay invoice flow

0. The user scans the QR code generated by the **wallet service** with their **client** application, they follow a `nostr+walletconnect://` deeplink or configure the connection details manually.
1. **client** sends an event to the **wallet service** with kind 23194. The content is a `pay_invoice` request. The private key is the secret from the connection string above.
2. **wallet service** verifies that the author's key is authorized to perform the payment, decrypts the payload and sends the payment.
3. **wallet service** responds to the event by sending an event with kind 23195 and content being a response either containing an error message or a preimage.

Encryption

The initial version of NWC used NIP-04 for encryption which has been deprecated and replaced by NIP-44. NIP-44 should always be preferred for encryption, but there may be legacy cases where the **wallet service** or **client** has not yet migrated to NIP-44. The **wallet service** and **client** should negotiate the encryption method to use based on the encryption tag in the info event.

The encryption tag can contain either `nip44_v2` or `nip04`. The absence of this tag implies that the wallet only supports `nip04`.

Encryption code	Use	Notes
<code>nip44_v2</code>	NIP-44	Required
<code>nip04</code>	NIP-04	Deprecated and only here for backward compatibility
<code><not present></code>	NIP-04	Deprecated and only here for backward compatibility

The negotiation works as follows.

1. The **wallet service** includes an encryption tag in the info event. This tag contains a space-separated list of encryption schemes that the **wallet service** supports (eg. `nip44_v2 nip04`)
2. The **client application** includes an encryption tag in each request event. This tag contains the encryption scheme which should be used for the request. The **client application** should always prefer `nip44` if supported by the **wallet service**.

Info event

First, the **wallet service** adds an encryption tag to its info event containing a space-separated list of encryption schemes it supports. For example, if a wallet service supports `nip44`, but also allows backwards-compatibility to `nip04` client applications, its encryption tag in the info event might look something like:

```
{
  "kind": 13194,
  "tags": [
    ["encryption", "nip44_v2 nip04"],
    // ...
  ],
  "content": "pay_invoice get_balance make_invoice lookup_invoice list_transactions get_info",
  // ...
}
```

When a **client application** establishes a connection, it should read the info event and look for the encryption tag.

Absence of this tag implies that the wallet only supports nip04.

If the encryption tag is present, the **client application** will choose optimal encryption supported by both itself, and the **wallet service**, which should always be nip44 if possible.

Request events

When a **client application** sends a request event, it should include a encryption tag with the encryption scheme it is using. The scheme **MUST** be supported by the **wallet service** as indicated by the info event. For example, if the client application supports nip44, the request event might look like:

```
{
  "kind": 23194,
  "tags": [
    ["encryption", "nip44_v2"],
    // ...
  ],
  // ...
}
```

If the **wallet service** does not support the specified encryption scheme, it will return an `UNSUPPORTED_ENCRYPTION` error. Absence of the encryption tag indicates use of nip04 for encryption.

Notification events

As described above in the Notifications section, if a **wallet service** supports both nip04 and nip44, it should publish two notification events for each notification - kind 23196 encrypted with NIP-04, and kind 23197 encrypted with NIP-44. If the **wallet service** only supports nip44, it should only publish kind 23197 events.

The **client** should check the encryption tag in the info event to determine which encryption schemes the **wallet service** supports, and listen to the appropriate notification event.

Using a dedicated relay

This NIP does not specify any requirements on the type of relays used. However, if the user is using a custodial service it might make sense to use a relay that is hosted by the custodial service. The relay may then enforce authentication to prevent metadata leaks. Not depending on a 3rd party relay would also improve reliability in this case.

Metadata

Metadata **MAY** be stored by the **wallet service** alongside invoices and payments. The metadata **MUST** be no more than 4096 characters, otherwise **MUST** be dropped. This is to ensure transactions do not get too large to be relayed.

NWC relays **SHOULD** allow at least a payload size of 64KB and **clients** **SHOULD** fetch small page sizes (maximum of 20 transactions per page) otherwise there is risk of `list_transactions` responses being rejected.

Here are some properties that are recognized by some NWC clients:

```

{
  "comment": "string", // LUD-12 comment
  "payer_data": {
    "email": "string",
    "name": "string",
    "pubkey": "string",
  }, // LUD-18 payer data
  "recipient_data": {
    "identifier": "string"
  }, // similar to LUD-18 payer data, but to record recipient data e.g. the lightning address of the recipient
  "nostr": {
    "pubkey": "string",
    "tags": [],
    // ... rest of zap request event
  }, // NIP-57 Zap Request event (kind 9734)
  "tlv_records": [
    {
      "type": 5482373484, // tlv type
      "value": "0123456789abcdef" // hex encoded tlv value
    }
  ] // keysend TLV records (e.g. for podcasting 2.0 boostagrams)
} & Record<string, unknown>;

```

Appendix

Example NIP-47 info event

```

{
  "id": "df467db0a9f9ec77ffe6f561811714ccaa2e26051c20f58f33c3d66d6c2b4d1c",
  "pubkey": "c04ccd5c82fc1ea3499b9c6a5c0a7ab627fbc00a0116110d4c750faeaecha1e2",
  "created_at": 1713883677,
  "kind": 13194,
  "tags": [
    [ "encryption", "nip44_v2 nip04" ],
    [
      "notifications",
      "payment_received payment_sent"
    ]
  ],
  "content": "pay_invoice pay_keysend get_balance get_info make_invoice lookup_invoice list_transactions\nmulti_pay_invoice multi_pay_keysend sign_message notifications",
  "sig":
    "31f57b369459b5306a5353aa9e03be7fbde169bc881c3233625605dd12f53548179def16b9fe1137e6465d7e4d5bb27ce81fd6e75908c46b06269f4233"
}

```

Deep-links

Wallet applications can register deeplinks in mobile systems to make it possible to create a linking UX that doesn't require the user scanning a QR code or pasting some code.

`nostrnwc://connect` and `nostrnwc+{app_name}://connect` can be registered by wallet apps and queried by apps that want to receive an NWC pairing code.

All URI parameters, MUST be URI-encoded.

URI parameters: * `appicon` – URL to an icon of the client that wants to create a connection. * `appname` – Name of the client that wants to create a connection. * `callback` – URI schema the wallet should open with the connection string

Once a connection has been created by the wallet, it should be returned to the client by opening the callback with the following parameters * value – NWC pairing code (e.g. `nostr+walletconnect://...`)

NIP-75

Zap Goals

draft optional

This NIP defines an event for creating fundraising goals. Users can contribute funds towards the goal by zapping the goal event.

Nostr Event

A kind:9041 event is used.

The .content contains a human-readable description of the goal.

The following tags are defined as REQUIRED.

- amount - target amount in milisats.
- relays - a list of relays the zaps to this goal will be sent to and tallied from.

Example event:

```
{
  "kind": 9041,
  "tags": [
    ["relays", "wss://alicerelay.example.com", "wss://bobrelay.example.com", /*...*/],
    ["amount", "210000"],
  ],
  "content": "Nostrasia travel expenses",
  // other fields...
}
```

The following tags are OPTIONAL.

- closed_at - timestamp for determining which zaps are included in the tally. Zap receipts published after the closed_at timestamp SHOULD NOT count towards the goal progress.
- image - an image for the goal
- summary - a brief description

```
{
  "kind": 9041,
  "tags": [
    ["relays", "wss://alicerelay.example.com", "wss://bobrelay.example.com", /*...*/],
    ["amount", "210000"],
    ["closed_at", "<unix timestamp in seconds>"],
    ["image", "<image URL>"],
    ["summary", "<description of the goal>"],
  ],
  "content": "Nostrasia travel expenses",
  // other fields...
}
```

The goal MAY include an r or a tag linking to a URL or addressable event.

The goal MAY include multiple beneficiary pubkeys by specifying zap tags.

Addressable events can link to a goal by using a goal tag specifying the event id and an optional relay hint.

```
{
  "kind": 3xxxx,
  "tags": [
    ["goal", "<event id>", "<Relay URL (optional)>"],
    // rest of tags...
  ]
}
```

```
],  
  // other fields...  
}
```

Client behavior

Clients MAY display funding goals on user profiles.

When zapping a goal event, clients MUST include the relays in the relays tag of the goal event in the zap request relays tag.

When zapping an addressable event with a goal tag, clients SHOULD tag the goal event id in the e tag of the zap request.

Use cases

- Fundraising clients
- Adding funding goals to events such as long form posts, badges or live streams

Third Parties

Warning unrecommended: adds unnecessary burden for little gain

NIP-26

Delegated Event Signing

draft optional

This NIP defines how events can be delegated so that they can be signed by other keypairs.

Another application of this proposal is to abstract away the use of the ‘root’ keypairs when interacting with clients. For example, a user could generate new keypairs for each client they wish to use and authorize those keypairs to generate events on behalf of their root pubkey, where the root keypair is stored in cold storage.

Introducing the ‘delegation’ tag This NIP introduces a new tag: `delegation` which is formatted as follows:

```
[
  "delegation",
  <pubkey of the delegator>,
  <conditions query string>,
  <delegation token: 64-byte Schnorr signature of the sha256 hash of the delegation string>
]
```

Delegation Token The **delegation token** should be a 64-byte Schnorr signature of the sha256 hash of the following string:

```
nostr:delegation:<pubkey of publisher (delegatee)>:<conditions query string>
```

Conditions Query String The following fields and operators are supported in the above query string:

Fields: 1. `kind` - *Operators:* `=` `{KIND_NUMBER}` - delegatee may only sign events of this kind 2. `created_at` - *Operators:* `<` `{TIMESTAMP}` - delegatee may only sign events created *before* the specified timestamp `>` `{TIMESTAMP}` - delegatee may only sign events created *after* the specified timestamp

In order to create a single condition, you must use a supported field and operator. Multiple conditions can be used in a single query string, including on the same field. Conditions must be combined with `&`.

For example, the following condition strings are valid:

- `kind=1&created_at<1675721813`
- `kind=0&kind=1&created_at>1675721813`
- `kind=1&created_at>1674777689&created_at<1675721813`

For the vast majority of use-cases, it is advisable that: 1. Query strings should include a `created_at` *after* condition reflecting the current time, to prevent the delegatee from publishing historic notes on the delegator’s behalf. 2. Query strings should include a `created_at` *before* condition that is not empty and is not some extremely distant time in the future. If delegations are not limited in time scope, they expose similar security risks to simply using the root key for authentication.

Delegator:

```
privkey: ee35e8bb71131c02c1d7e73231daa48e9953d329a4b701f7133c8f46dd21139c
pubkey: 8e0d3d3eb2881ec137a11debe736a9086715a8c8beeda615780064d68bc25dd
```

Delegatee:

```
privkey: 777e4f60b4aa87937e13acc84f7abcc3c93cc035cb4c1e9f7a9086dd78fffc1
pubkey: 477318cfb5427b9cfc66a9fa376150c1ddbc62115ae27cef72417eb959691396
```

Delegation string to grant note publishing authorization to the delegatee (477318cf) from now, for the next 30 days, given the current timestamp is 1674834236.

nostr:delegation:477318cfb5427b9cfc66a9fa376150c1ddbc62115ae27cef72417eb959691396:kind=1&created_at>1674834236&created_at<1677426236

The delegator (8e0d3d3e) then signs a SHA256 hash of the above delegation string, the result of which is the delegation token:

6f44d7fe4f1c09f3954640fb58bd12bae8bb8ff4120853c4693106c82e920e2b898f1f9ba9bd65449a987c39c0423426ab7b53910c0c6abfb41b30bc16e5f524

The delegatee (477318cf) can now construct an event on behalf of the delegator (8e0d3d3e). The delegatee then signs the event with its own private key and publishes.

```
{
  "id": "e93c6095c3db1c31d15ac771f8fc5fb672f6e52cd25505099f62cd055523224f",
  "pubkey": "477318cfb5427b9cfc66a9fa376150c1ddbc62115ae27cef72417eb959691396",
  "created_at": 1677426298,
  "kind": 1,
  "tags": [
    [
      "delegation",
      "8e0d3d3eb2881ec137a11debe736a9086715a8c8beeda615780064d68bc25dd",
      "kind=1&created_at>1674834236&created_at<1677426236",
      "6f44d7fe4f1c09f3954640fb58bd12bae8bb8ff4120853c4693106c82e920e2b898f1f9ba9bd65449a987c39c0423426ab7b53910c0c6abfb41b30bc16e5f524"
    ]
  ],
  "content": "Hello, world!",
  "sig":
    "633db60e2e7082c13a47a6b19d663d45b2a2ebdeaf0b4c35ef83be2738030c54fc7fd56d139652937cdca875ee61b51904a1d0d0588a6acd6168d7be29"
}
```

The event should be considered a valid delegation if the conditions are satisfied (kind=1, created_at>1674834236 and created_at<1677426236 in this example) and, upon validation of the delegation token, are found to be unchanged from the conditions in the original delegation string.

Clients should display the delegated note as if it was published directly by the delegator (8e0d3d3e).

Relay & Client Support Relays should answer requests such as ["REQ", "", {"authors": ["A"]}]] by querying both the pubkey and delegation tags [1] value.

Relays SHOULD allow the delegator (8e0d3d3e) to delete the events published by the delegatee (477318cf).

NIP-59

Gift Wrap

optional

This NIP defines a protocol for encapsulating any nostr event. This makes it possible to obscure most metadata for a given event, perform collaborative signing, and more.

This NIP *does not* define any messaging protocol. Applications of this NIP should be defined separately.

This NIP relies on NIP-44's versioned encryption algorithms.

Overview

This protocol uses three main concepts to protect the transmission of a target event: rumors, seals, and gift wraps.

- A rumor is a regular nostr event, but is **not signed**. This means that if it is leaked, it cannot be verified.
- A rumor is serialized to JSON, encrypted, and placed in the content field of a seal. The seal is then signed by the author of the note. The only information publicly available on a seal is who signed it, but not what was said.
- A seal is serialized to JSON, encrypted, and placed in the content field of a gift wrap.

This allows the isolation of concerns across layers:

- A rumor carries the content but is unsigned, which means if leaked it will be rejected by relays and clients, and can't be authenticated. This provides a measure of deniability.
- A seal identifies the author without revealing the content or the recipient.
- A gift wrap can add metadata (recipient, tags, a different author) without revealing the true author.

Protocol Description

1. The Rumor Event Kind

A rumor is the same thing as an unsigned event. Any event kind can be made a rumor by removing the signature.

2. The Seal Event Kind

A seal is a kind:13 event that wraps a rumor with the sender's regular key. The seal is **always** encrypted to a receiver's pubkey but there is no p tag pointing to the receiver. There is no way to know who the rumor is for without the receiver's or the sender's private key. The only public information in this event is who is signing it.

```
{
  "id": "<id>",
  "pubkey": "<real author's pubkey>",
  "content": "<encrypted rumor>",
  "kind": 13,
  "created_at": 1686840217,
  "tags": [],
  "sig": "<real author's pubkey signature>"
}
```

Tags MUST always be empty in a kind:13. The inner event MUST always be unsigned.

3. Gift Wrap Event Kind

A gift wrap event is a kind:1059 event that wraps any other event. tags SHOULD include any information needed to route the event to its intended recipient, including the recipient's p tag or NIP-13 proof of work.

```
{
  "id": "<id>",
  "pubkey": "<random, one-time-use pubkey>",
```

```

"content": "<encrypted kind 13>",
"kind": 1059,
"created_at": 1686840217,
"tags": [["p", "<recipient pubkey>"]],
"sig": "<random, one-time-use pubkey signature>"
}

```

Encrypting Payloads

Encryption is done following NIP-44 on the JSON-encoded event. Place the encryption payload in the `.content` of the wrapper event (either a `seal` or a `gift wrap`).

Other Considerations

If a rumor is intended for more than one party, or if the author wants to retain an encrypted copy, a single rumor may be wrapped and addressed for each recipient individually.

The canonical `created_at` time belongs to the rumor. All other timestamps **SHOULD** be tweaked to thwart time-analysis attacks. Note that some relays don't serve events dated in the future, so all timestamps **SHOULD** be in the past.

Relays may choose not to store gift wrapped events due to them not being publicly useful. Clients **MAY** choose to attach a certain amount of proof-of-work to the wrapper event per NIP-13 in a bid to demonstrate that the event is not spam or a denial-of-service attack.

To protect recipient metadata, relays **SHOULD** guard access to kind 1059 events based on user AUTH. When possible, clients should only send wrapped events to relays that offer this protection.

To protect recipient metadata, relays **SHOULD** only serve kind 1059 events intended for the marked recipient. When possible, clients should only send wrapped events to read relays for the recipient that implement AUTH, and refuse to serve wrapped events to non-recipients.

When adding expiration tags to both `seal` and `gift wrap` layers, implementations **SHOULD** use independent random timestamps for each layer. Using different `created_at` values increases timing variance and helps protect against metadata correlation attacks.

Since signing keys are random, relays **SHOULD** delete kind 1059 events whose p-tag matches the signer of NIP-09 deletions or NIP-62 vanish requests.

An Example

Let's send a wrapped kind 1 message between two parties asking "Are you going to the party tonight?"

- Author private key: 0beebd062ec8735f4243466049d7747ef5d6594ee838de147f8aab842b15e273
- Recipient private key: e108399bd8424357a710b606ae0c13166d853d327e47a6e5e038197346bdbf45
- Ephemeral wrapper key: 4f02eac59266002db5801adc5270700ca69d5b8f761d8732fab2fbf233c90cbd

Note that this messaging protocol should not be used in practice, this is just an example. Refer to other NIPs for concrete messaging protocols that depend on gift wraps.

1. Create an event

Create a kind 1 event with the message, the receivers, and any other tags you want, signed by the author. Do not sign the event.

```

{
  "created_at": 1691518405,
  "content": "Are you going to the party tonight?",
  "tags": [],
  "kind": 1,
  "pubkey": "611df01bfcf85c26ae65453b772d8f1dfd25c264621c0277e1fc1518686faef9",

```

```

    "id": "9dd003c6d3b73b74a85a9ab099469ce251653a7af76f523671ab828acd2a0ef9"
  }

```

2. Seal the rumor

Encrypt the JSON-encoded rumor with a conversation key derived using the author's private key and the recipient's public key. Place the result in the content field of a kind 13 seal event. Sign it with the author's key.

```

{
  "content":
    "AqBCdwoS7/tPK+QGkPCadJTn8FxGkd24iApo3BR9/M0uw6n4RFAFSPAKIMgkzVMbRyR3ZS/aqATDFvoZJ0kE9cPG/AzmyZvr/WUIS8kLmuI1dCA+i tFF6+ULZ",
  "kind": 13,
  "created_at": 1703015180,
  "pubkey": "611df01bfcf85c26ae65453b772d8f1dfd25c264621c0277e1fc1518686faef9",
  "tags": [],
  "id": "28a87d7c074d94a58e9e89bb3e9e4e813e2189f285d797b1c56069d36f59eaa7",
  "sig":
    "02fc3facf6621196c32912b1ef53bac8f8bfe9db51c0e7102c073103586b0d29c3f39bdaa1e62856c20e90b6c7cc5dc34ca8bb6a528872cf6e65e62845"
}

```

3. Wrap the seal

Encrypt the JSON-encoded kind 13 event with your ephemeral, single-use random key. Place the result in the content field of a kind 1059. Add a single p tag containing the recipient's public key. Sign the gift wrap using the random key generated in the previous step.

```

{
  "content":
    "AhC3Qj/QsKJFWuf6xroiYip+ZyK95qPwJjVvFujhzSguJWb/6TLpPBW0CGFwfufCs2Zyb0JeuLmZhNlnqecAAa1C4ZCugB+I9ViA5pxLyFfQjs1lcE6KdX3euC",
  "kind": 1059,
  "created_at": 1703021488,
  "pubkey": "18b1a75918f1f2c90c23da616bce317d36e348bcf5f7ba55e75949319210c87c",
  "id": "5c005f3ccf01950aa8d131203248544fb1e41a0d698e846bd419cec3890903ac",
  "sig":
    "35fabdae4634eb630880a1896a886e40fd6ea8a60958e30b89b33a93e6235df750097b04f9e13053764251b8bc5dd7e8e0794a3426a90b6bcc7e5ff660",
  "tags": [["p", "16dbf3765ebd1fc55decfe395beff2ea3b2a4e0a8946e7eb578512b555737c99"]],
}

```

4. Broadcast Selectively

Broadcast the kind 1059 event to the recipient's relays only. Delete all the other events.

Code Samples

JavaScript

```

import {bytesToHex} from "@noble/hashes/utils"
import type {EventTemplate, UnsignedEvent, Event} from "nostr-tools"
import {getPublicKey, getEventHash, nip19, nip44, finalizeEvent, generateSecretKey} from "nostr-tools"

type Rumor = UnsignedEvent & {id: string}

const TWO_DAYS = 2 * 24 * 60 * 60

const now = () => Math.round(Date.now() / 1000)
const randomNow = () => Math.round(now() - (Math.random() * TWO_DAYS))

const nip44ConversationKey = (privateKey: Uint8Array, publicKey: string) =>

```

```

nip44.v2.utils.getConversationKey(bytesToHex(privateKey), publicKey)

const nip44Encrypt = (data: EventTemplate, privateKey: Uint8Array, publicKey: string) =>
  nip44.v2.encrypt(JSON.stringify(data), nip44ConversationKey(privateKey, publicKey))

const nip44Decrypt = (data: Event, privateKey: Uint8Array) =>
  JSON.parse(nip44.v2.decrypt(data.content, nip44ConversationKey(privateKey, data.pubkey)))

const createRumor = (event: Partial<UnsignedEvent>, privateKey: Uint8Array) => {
  const rumor = {
    created_at: now(),
    content: "",
    tags: [],
    ...event,
    pubkey: getPublicKey(privateKey),
  } as any

  rumor.id = getEventHash(rumor)

  return rumor as Rumor
}

const createSeal = (rumor: Rumor, privateKey: Uint8Array, recipientPublicKey: string) => {
  return finalizeEvent(
    {
      kind: 13,
      content: nip44Encrypt(rumor, privateKey, recipientPublicKey),
      created_at: randomNow(),
      tags: [],
    },
    privateKey
  ) as Event
}

const createWrap = (event: Event, recipientPublicKey: string) => {
  const randomKey = generateSecretKey()

  return finalizeEvent(
    {
      kind: 1059,
      content: nip44Encrypt(event, randomKey, recipientPublicKey),
      created_at: randomNow(),
      tags: [["p", recipientPublicKey]],
    },
    randomKey
  ) as Event
}

// Test case using the above example
const senderPrivateKey = nip19.decode(`nsec1p0ht6p3wepe47sjrgesyn4m50m6avk2waqudu9r1324cg2c4ufesyp6rdg`).data
const recipientPrivateKey = nip19.decode(`nsec1uyymnx7cgfp40fcskcr2urqnzekc20fj0er6de0q8qvhx34ahazsvs9p36`).data
const recipientPublicKey = getPublicKey(recipientPrivateKey)

const rumor = createRumor(
  {
    kind: 1,
    content: "Are you going to the party tonight?",
  }

```

```
    },  
    senderPrivateKey  
  )  
  
  const seal = createSeal(rumor, senderPrivateKey, recipientPublicKey)  
  const wrap = createWrap(seal, recipientPublicKey)  
  
  // Recipient unwraps with their private key.  
  
  const unwrappedSeal = nip44Decrypt(wrap, recipientPrivateKey)  
  const unsealedRumor = nip44Decrypt(unwrappedSeal, recipientPrivateKey)
```


NIP-46

Nostr Remote Signing

Changes

`remote-signer-key` is introduced, passed in bunker url, clients must differentiate between `remote-signer-pubkey` and `user-pubkey`, must call `get_public_key` after connect, `nip05 login` is removed, `create_account` moved to another NIP.

Rationale

Private keys should be exposed to as few systems - apps, operating systems, devices - as possible as each system adds to the attack surface.

This NIP describes a method for 2-way communication between a remote signer and a Nostr client. The remote signer could be, for example, a hardware device dedicated to signing Nostr events, while the client is a normal Nostr client.

Terminology

- **user:** A person that is trying to use Nostr.
- **client:** A user-facing application that *user* is looking at and clicking buttons in. This application will send requests to *remote-signer*.
- **remote-signer:** A daemon or server running somewhere that will answer requests from *client*, also known as “bunker”.
- **client-keypair/pubkey:** The keys generated by *client*. Used to encrypt content and communicate with *remote-signer*.
- **remote-signer-keypair/pubkey:** The keys used by *remote-signer* to encrypt content and communicate with *client*. This keypair MAY be same as *user-keypair*, but not necessarily.
- **user-keypair/pubkey:** The actual keys representing *user* (that will be used to sign events in response to `sign_event` requests, for example). The *remote-signer* generally has control over these keys.

All pubkeys specified in this NIP are in hex format.

Overview

1. *client* generates `client-keypair`. This keypair doesn't need to be communicated to *user* since it's largely disposable. *client* might choose to store it locally and they should delete it on logout;
2. A connection is established (see below), *remote-signer* learns `client-pubkey`, *client* learns `remote-signer-pubkey`.
3. *client* uses `client-keypair` to send requests to *remote-signer* by p-tagging and encrypting to `remote-signer-pubkey`;
4. *remote-signer* responds to *client* by p-tagging and encrypting to the `client-pubkey`.
5. *client* requests `get_public_key` to learn `user-pubkey`.

Initiating a connection

There are two ways to initiate a connection:

Direct connection initiated by *remote-signer*

remote-signer provides connection token in the form:

`bunker://<remote-signer-pubkey>?relay=<wss://relay-to-connect-on>&relay=<wss://another-relay-to-connect-on>&secret=<optional-secret>`

user passes this token to *client*, which then sends `connect` request to *remote-signer* via the specified relays. Optional secret can be used for single successfully established connection only, *remote-signer* SHOULD ignore new attempts to establish connection with old secret.

Direct connection initiated by the *client*

client provides a connection token using `nostrconnect://` as the protocol, and `client-pubkey` as the origin. Additional information should be passed as query parameters:

- `relay` (required) - one or more relay urls on which the *client* is listening for responses from the *remote-signer*.
- `secret` (required) - a short random string that the *remote-signer* should return as the `result` field of its response.
- `perms` (optional) - a comma-separated list of permissions the *client* is requesting be approved by the *remote-signer*
- `name` (optional) - the name of the *client* application
- `url` (optional) - the canonical url of the *client* application
- `image` (optional) - a small image representing the *client* application

Here's an example:

```
nostrconnect://83f3b2ae6aa368e8275397b9c26cf550101d63ebadb900d19dd4a4429f5ad8f5?relay=wss%3A%2F%2Frelay1.example.com&perms=nip44_e
```

user passes this token to *remote-signer*, which then sends `connect` *response* event to the `client-pubkey` via the specified relays. Client discovers `remote-signer-pubkey` from `connect` response `author`. `secret` value MUST be provided to avoid connection spoofing, *client* MUST validate the `secret` returned by `connect` response.

Request Events `kind: 24133`

```
{
  "kind": 24133,
  "pubkey": <local_keypair_pubkey>,
  "content": <nip44(<request>)>,
  "tags": [[ "p", <remote-signer-pubkey> ]],
}
```

The `content` field is a JSON-RPC-like message that is NIP-44 encrypted and has the following structure:

```
{
  "id": <random_string>,
  "method": <method_name>,
  "params": [array_of_strings]
}
```

- `id` is a random string that is a request ID. This same ID will be sent back in the response payload.
- `method` is the name of the method/command (detailed below).
- `params` is a positional array of string parameters.

Methods/Commands

Each of the following are methods that the *client* sends to the *remote-signer*.

Command	Params	Result
<code>connect</code>	[<remote-signer-pubkey>, <optional_secret>, <optional_requested_permissions>]	"ack" OR <required-secret-value>
<code>sign_event</code>	[<{kind, content, tags, created_at}>]	<code>json_stringified(<signed_event>)</code>
<code>ping</code>	[]	"pong"
<code>get_public_key</code>	[]	<user-pubkey>
<code>nip04_encrypt</code>	[<third_party_pubkey>, <plaintext_to_encrypt>]	<nip04_ciphertext>
<code>nip04_decrypt</code>	[<third_party_pubkey>, <nip04_ciphertext_to_decrypt>]	<plaintext>

Command	Params	Result
nip44_encrypt	[<third_party_pubkey>, <plaintext_to_encrypt>]	<nip44_ciphertext>
nip44_decrypt	[<third_party_pubkey>, <nip44_ciphertext_to_decrypt>]	<plaintext>

Requested permissions

The connect method may be provided with `optional_requested_permissions` for user convenience. The permissions are a comma-separated list of `method[:params]`, i.e. `nip44_encrypt,sign_event:4` meaning permissions to call `nip44_encrypt` and to call `sign_event` with `kind:4`. Optional parameter for `sign_event` is the kind number, parameters for other methods are to be defined later. Same permission format may be used for `perms` field of metadata in `nostrconnect://` string.

Response Events kind:24133

```
{
  "id": <id>,
  "kind": 24133,
  "pubkey": <remote-signer-pubkey>,
  "content": <nip44(<response>)>,
  "tags": [[<"p">, <client-pubkey>]],
  "created_at": <unix timestamp in seconds>
}
```

The content field is a JSON-RPC-like message that is NIP-44 encrypted and has the following structure:

```
{
  "id": <request_id>,
  "result": <results_string>,
  "error": <optional_error_string>
}
```

- `id` is the request ID that this response is for.
- `results` is a string of the result of the call (this can be either a string or a JSON stringified object)
- `error`, *optionally*, it is an error in string form, if any. Its presence indicates an error with the request.

Example flow for signing an event

- `remote-signer-pubkey` is `fa984bd7dbb282f07e16e7ae87b26a2a7b9b90b7246a44771f0cf5ae58018f52`
- `user-pubkey` is also `fa984bd7dbb282f07e16e7ae87b26a2a7b9b90b7246a44771f0cf5ae58018f52`
- `client-pubkey` is `eff37350d839ce3707332348af4549a96051bd695d3223af4aabce4993531d86`

Signature request

```
{
  "kind": 24133,
  "pubkey": "eff37350d839ce3707332348af4549a96051bd695d3223af4aabce4993531d86",
  "content": nip44({
    "id": <random_string>,
    "method": "sign_event",
    "params": [json_stringified(<{
      content: "Hello, I'm signing remotely",
      kind: 1,
      tags: [],
      created_at: 1714078911
    }>)]
  })
}
```

```

  }},
  "tags": [["p", "fa984bd7dbb282f07e16e7ae87b26a2a7b9b90b7246a44771f0cf5ae58018f52"]], // p-tags the
    remote-signer-pubkey
}

```

Response event

```

{
  "kind": 24133,
  "pubkey": "fa984bd7dbb282f07e16e7ae87b26a2a7b9b90b7246a44771f0cf5ae58018f52",
  "content": nip44({
    "id": <random_string>,
    "result": json_stringified(<signed-event>)
  }),
  "tags": [["p", "eff37350d839ce3707332348af4549a96051bd695d3223af4aabc4993531d86"]], // p-tags the client-pubkey
}

```

Diagram

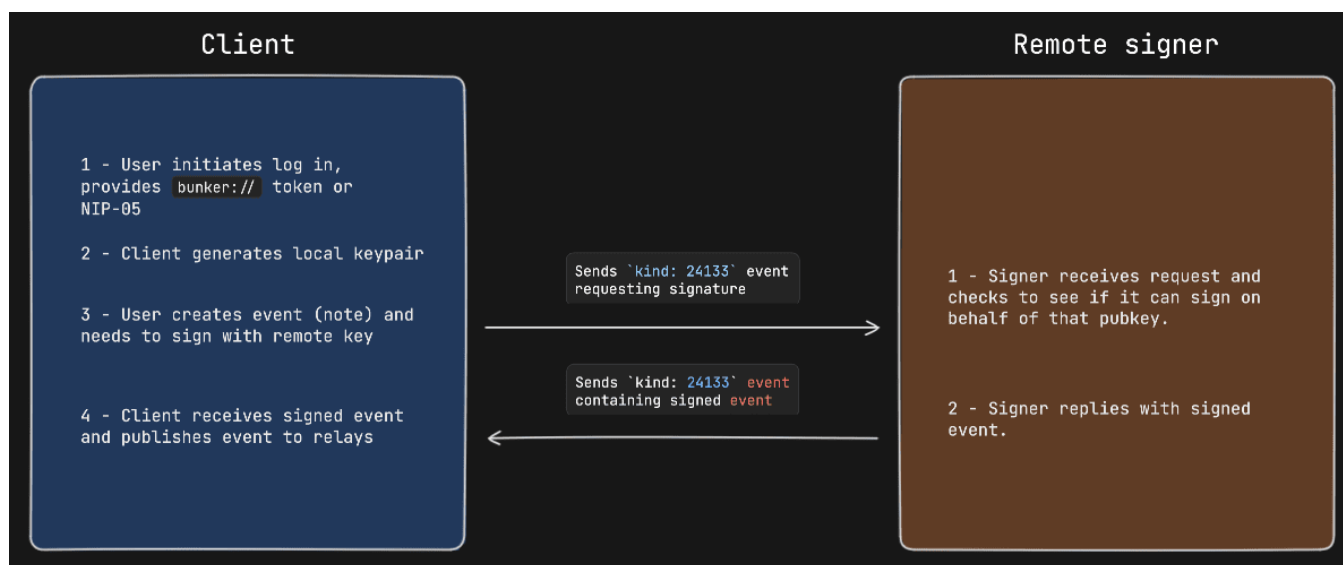


Figure 1: signing-example

Auth Challenges

An Auth Challenge is a response that a *remote-signer* can send back when it needs the *user* to authenticate via other means. The response content object will take the following form:

```

{
  "id": <request_id>,
  "result": "auth_url",
  "error": <URL_to_display_to_end_user>
}

```

client should display (in a popup or new tab) the URL from the `error` field and then subscribe/listen for another response from the *remote-signer* (reusing the same request ID). This event will be sent once the user authenticates in the other window (or will never arrive if the user doesn't authenticate).

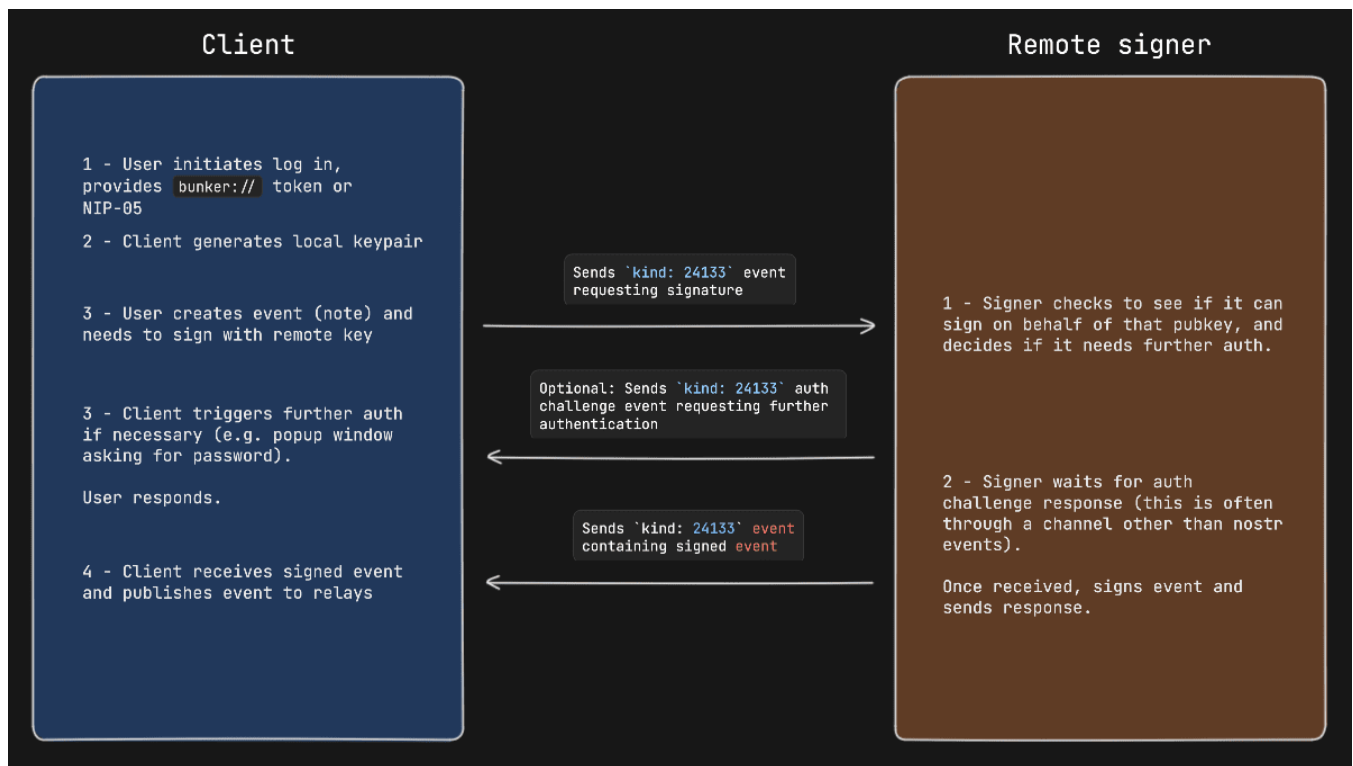


Figure 2: signing-example-with-auth-challenge

Example event signing request with auth challenge

Appendix

Announcing *remote-signer* metadata

remote-signer MAY publish it's metadata by using NIP-05 and NIP-89. With NIP-05, a request to `<remote-signer>/ .well-known/n` MAY return this:

```
{
  "names": {
    "_": <remote-signer-app-pubkey>,
  },
  "nip46": {
    "relays": ["wss://relay1", "wss://relay2" ...],
    "nostrconnect_url": "https://remote-signer-domain.example/<nostrconnect>"
  }
}
```

The `<remote-signer-app-pubkey>` MAY be used to verify the domain from *remote-signer's* NIP-89 event (see below). relays SHOULD be used to construct a more precise `nostrconnect://` string for the specific *remote-signer*. `nostrconnect_url` template MAY be used to redirect users to *remote-signer's* connection flow by replacing `<nostrconnect>` placeholder with an actual `nostrconnect://` string.

Remote signer discovery via NIP-89

remote-signer MAY publish a NIP-89 `kind: 31990` event with `k` tag of 24133, which MAY also include one or more relay tags and MAY include `nostrconnect_url` tag. The semantics of `relay` and `nostrconnect_url` tags are the same as in the section above.

client MAY improve UX by discovering *remote-signers* using their `kind: 31990` events. *client* MAY then pre-generate `nostrconnect://` strings for the *remote-signers*, and SHOULD in that case verify that `kind: 31990` event's author is

mentioned in signer's `nostr.json?name=_` file as `<remote-signer-app-pubkey>`.

NIP-90

Data Vending Machine

draft optional

This NIP defines the interaction between customers and Service Providers for performing on-demand computation. Money in, data out.

Kinds

This NIP reserves the range 5000-7000 for data vending machine use.

Kind	Description
5000-5999	Job request kinds
6000-6999	Job result
7000	Job feedback

Job results always use a kind number that is 1000 higher than the job request kind. (e.g. request: kind:5001 gets a result: kind:6001).

Job request types are defined separately.

Rationale

Nostr can act as a marketplace for data processing, where users request jobs to be processed in certain ways (e.g., “speech-to-text”, “summarization”, etc.), but they don’t necessarily care about “who” processes the data.

This NIP is not to be confused with a 1:1 marketplace; instead, it describes a flow where a user announces a desired output, willingness to pay, and service providers compete to fulfill the job requirement in the best way possible.

Actors

There are two actors in the workflow described in this NIP: * Customers (npubs who request a job) * Service providers (npubs who fulfill jobs)

Job request (kind:5000-5999)

A request to process data, published by a customer. This event signals that a customer is interested in receiving the result of some kind of compute.

```
{
  "kind": 5xxx, // kind in 5000-5999 range
  "content": "",
  "tags": [
    [ "i", "<data>", "<input-type>", "<relay>", "<markers>" ],
    [ "output", "<mime-type>" ],
    [ "relays", "wss://..." ],
    [ "bid", "<nsat-amount>" ],
    [ "t", "bitcoin" ]
  ],
  // other fields...
}
```

All tags are optional.

- i tag: Input data for the job (zero or more inputs)
 - <data>: The argument for the input

- `<input-type>`: The way this argument should be interpreted. MUST be one of:
 - * `url`: A URL to be fetched of the data that should be processed.
 - * `event`: A Nostr event ID.
 - * `job`: The output of a previous job with the specified event ID. The dermination of which output to build upon is up to the service provider to decide (e.g. waiting for a signaling from the customer, waiting for a payment, etc.)
 - * `text`: `<data>` is the value of the input, no resolution is needed
- `<relay>`: If event or job input-type, the relay where the event/job was published, otherwise optional or empty string
- `<marker>`: An optional field indicating how this input should be used within the context of the job
- `output`: Expected output format. Different job request kind defines this more precisely.
- `param`: Optional parameters for the job as key (first argument)/ value (second argument). Different job request kind defines this more precisely. (e.g. ["param", "lang", "es"])
- `bid`: Customer MAY specify a maximum amount (in millisats) they are willing to pay
- `relays`: List of relays where Service Providers SHOULD publish responses to
- `p`: Service Providers the customer is interested in. Other SPs MIGHT still choose to process the job

Encrypted Params

If the user wants to keep the input parameters a secret, they can encrypt the `i` and `param` tags with the service provider's 'p' tag and add it to the content field. Add a tag encrypted as tags. Encryption for private tags will use NIP-04 - Encrypted Direct Message encryption, using the user's private and service provider's public key for the shared secret

```
[
  ["i", "what is the capital of France? ", "text"],
  ["param", "model", "LLaMA-2"],
  ["param", "max_tokens", "512"],
  ["param", "temperature", "0.5"],
  ["param", "top-k", "50"],
  ["param", "top-p", "0.7"],
  ["param", "frequency_penalty", "1"]
]
```

This param data will be encrypted and added to the content field and p tag should be present

```
{
  "content": "BEZY4xvS6HIY7TozIgbE13sAHkdZoXyLRRkZv4fLPh3R7LtvILKAJM5qkC7D6VtMbgIt4iNdMpLtpo...",
  "tags": [
    ["p", "04f74530a6ede6b24731b976b8e78fb449ea61f40ff10e3d869a3030c4edc91f"],
    ["encrypted"]
  ],
  // other fields...
}
```

Job result (kind:6000-6999)

Service providers publish job results, providing the output of the job result. They should tag the original job request event id as well as the customer's pubkey.

```
{
  "pubkey": "<service-provider pubkey>",
  "content": "<payload>",
  "kind": 6xxx,
  "tags": [
    ["request", "<job-request>"],
    ["e", "<job-request-id>", "<relay-hint>"],
    ["i", "<input-data>"],
  ]
}
```



```

    ["p", "<customer's-pubkey>"],
    ["amount", "requested-payment-amount", "<optional-bolt11>"]
  ],
  // other fields...
}

```

- request: The job request event stringified-JSON.
- amount: millisats that the Service Provider is requesting to be paid. An optional third value can be a bolt11 invoice.
- i: The original input(s) specified in the request.

Encrypted Output

If the request has encrypted params, then output should be encrypted and placed in content field. If the output is encrypted, then avoid including i tag with input-data as clear text. Add a tag encrypted to mark the output content as encrypted

```

{
  "pubkey": "<service-provider pubkey>",
  "content": "<encrypted payload>",
  "kind": 6xxx,
  "tags": [
    ["request", "<job-request>"],
    ["e", "<job-request-id>", "<relay-hint>"],
    ["p", "<customer's-pubkey>"],
    ["amount", "requested-payment-amount", "<optional-bolt11>"],
    ["encrypted"]
  ],
  // other fields...
}

```

Job feedback

Service providers can give feedback about a job back to the customer.

```

{
  "kind": 7000,
  "content": "<empty-or-payload>",
  "tags": [
    ["status", "<status>", "<extra-info>"],
    ["amount", "requested-payment-amount", "<bolt11>"],
    ["e", "<job-request-id>", "<relay-hint>"],
    ["p", "<customer's-pubkey>"],
  ],
  // other fields...
}

```

- content: Either empty or a job-result (e.g. for partial-result samples)
- amount tag: as defined in the Job Result section.
- status tag: Service Providers SHOULD indicate what this feedback status refers to. Job Feedback Status defines status. Extra human-readable information can be added as an extra argument.
- NOTE: If the input params requires input to be encrypted, then content field will have encrypted payload with p tag as key.

Job feedback status

status	description
payment-required	Service Provider requires payment before continuing.
processing	Service Provider is processing the job.
error	Service Provider was unable to process the job.
success	Service Provider successfully processed the job.
partial	Service Provider partially processed the job. The .content might include a sample of the partial results.

Any job feedback event MIGHT include results in the .content field, as described in the Job Result section. This is useful for service providers to provide a sample of the results that have been processed so far.

Protocol Flow

- Customer publishes a job request (e.g. kind:5000 speech-to-text).
- Service Providers MAY submit kind:7000 job-feedback events (e.g. payment-required, processing, error, etc.).
- Upon completion, the service provider publishes the result of the job with a kind:6000 job-result event.
- At any point, if there is an amount pending to be paid as instructed by the service provider, the user can pay the included bolt11 or zap the job result event the service provider has sent to the user.

Job feedback (kind:7000) and Job Results (kind:6000-6999) events MAY include an amount tag, this can be interpreted as a suggestion to pay. Service Providers MUST use the payment-required feedback event to signal that a payment is required and no further actions will be performed until the payment is sent.

Customers can always either pay the included bolt11 invoice or zap the event requesting the payment and service providers should monitor for both if they choose to include a bolt11 invoice.

Notes about the protocol flow

The flow is deliberately ambiguous, allowing vast flexibility for the interaction between customers and service providers so that service providers can model their behavior based on their own decisions/perceptions of risk.

Some service providers might choose to submit a payment-required as the first reaction before sending a processing or before delivering results, some might choose to serve partial results for the job (e.g. a sample), send a payment-required to deliver the rest of the results, and some service providers might choose to assess likelihood of payment based on an npub's past behavior and thus serve the job results before requesting payment for the best possible UX.

It's not up to this NIP to define how individual vending machines should choose to run their business.

Cancellation

A job request might be canceled by publishing a kind:5 delete request event tagging the job request event.

Appendix 1: Job chaining

A Customer MAY request multiple jobs to be processed as a chain, where the output of a job is the input of another job. (e.g. podcast transcription -> summarization of the transcription). This is done by specifying as input an event id of a different job with the job type.

Service Providers MAY begin processing a subsequent job the moment they see the prior job's result, but they will likely wait for a zap to be published first. This introduces a risk that Service Provider of job #1 might delay publishing the zap event in order to have an advantage. This risk is up to Service Providers to mitigate or to decide whether the service provider of job #1 tends to have good-enough results so as to not wait for an explicit zap to assume the job was accepted.

This gives a higher level of flexibility to service providers (which sophisticated service providers would take anyway).

Appendix 2: Service provider discoverability

Service Providers MAY use NIP-89 announcements to advertise their support for job kinds:

```
{
  "kind": 31990,
  "pubkey": "<pubkey>",
  "content": "{
    \"name\": \"Translating DVM\",
    \"about\": \"I'm a DVM specialized in translating Bitcoin content.\"
  }",
  "tags": [
    ["k", "5005"], // e.g. translation
    ["t", "bitcoin"] // e.g. optionally advertises it specializes in bitcoin audio transcription that won't confuse
                     "Drivechains" with "Ridechains"
  ],
  // other fields...
}
```

Customers can use NIP-89 to see what service providers their follows use.

Application Features

NIP-52

Calendar Events

draft optional

This specification defines calendar events representing an occurrence at a specific moment or between moments. These calendar events are *addressable* and deletable per NIP-09.

Unlike the term `calendar event` specific to this NIP, the term `event` is used broadly in all the NIPs to describe any Nostr event. The distinction is being made here to discern between the two terms.

Calendar Events

There are two types of calendar events represented by different kinds: *date-based* and *time-based* calendar events.

These tags are common to both types of calendar events:

- `d` (required) a short unique string identifier. Generated by the client creating the calendar event.
- `title` (required) title of the calendar event
- `summary` (optional) brief description of the calendar event
- `image` (optional) url of an image to use for the event
- `location` (optional, repeated) location of the calendar event. e.g. address, GPS coordinates, meeting room name, link to video call
- `g` (optional) geohash to associate calendar event with a searchable physical location
- `p` (optional, repeated) 32-bytes hex pubkey of a participant, optional recommended relay URL, and participant's role in the meeting
- `t` (optional, repeated) hashtag to categorize calendar event
- `r` (optional, repeated) references / links to web pages, documents, video calls, recorded videos, etc.
- `a` (repeated) reference tag to kind 31924 calendar event requesting to be included in Calendar

The following tags are deprecated:

- `name` name of the calendar event. Use only if `title` is not available.

Calendar events are *not* required to be part of a calendar.

Collaborative Calendar Event Requests

Calendar events can include an `a` tag referencing a calendar (kind 31924) to request addition to that calendar. When a calendar event includes such a reference, clients should interpret this as a request to add the event to the referenced calendar by referencing it with an `a` tag.

This enables collaborative calendar management where multiple users can contribute events to calendars they do not own, subject to the calendar owner's approval.

Date-Based Calendar Event

This kind of calendar event starts on a date and ends before a different date in the future. Its use is appropriate for all-day or multi-day events where time and time zone hold no significance. e.g., anniversary, public holidays, vacation days.

It's an *addressable event* of kind: 31922.

The `.` content of these events SHOULD be a description of the calendar event.

Aside from the common tags, this also takes the following tags:

- `start` (required) inclusive start date in ISO 8601 format (YYYY-MM-DD). Must be less than `end`, if it exists.
- `end` (optional) exclusive end date in ISO 8601 format (YYYY-MM-DD). If omitted, the calendar event ends on the same date as `start`.

Example:

```

{
  "id": <32-bytes lowercase hex-encoded SHA-256 of the serialized event data>,
  "pubkey": <32-bytes lowercase hex-encoded public key of the event creator>,
  "created_at": <unix timestamp in seconds>,
  "kind": 31922,
  "content": "<description of calendar event>",
  "tags": [
    ["d", "<random-identifier>"],

    ["title", "<title of calendar event>"],

    // dates
    ["start", "<YYYY-MM-DD>"],
    ["end", "<YYYY-MM-DD>"],

    // location
    ["location", "<location>"],
    ["g", "<geohash>"],

    // participants
    ["p", "<32-bytes hex of a pubkey>", "<optional recommended relay URL>", "<role>"],
    ["p", "<32-bytes hex of a pubkey>", "<optional recommended relay URL>", "<role>"],
  ]
}

```

Time-Based Calendar Event

This kind of calendar event spans between a start time and end time.

It's an *addressable event* of kind:31923.

The .content of these events should be a description of the calendar event. It is required but can be an empty string.

Aside from the common tags, this also takes the following tags:

- start (required) inclusive start Unix timestamp in seconds. Must be less than end, if it exists.
- end (optional) exclusive end Unix timestamp in seconds. If omitted, the calendar event ends instantaneously.
- start_tzid (optional) time zone of the start timestamp, as defined by the IANA Time Zone Database. e.g., America/Costa_Rica
- end_tzid (optional) time zone of the end timestamp, as defined by the IANA Time Zone Database. e.g., America/Costa_Rica. If omitted and start_tzid is provided, the time zone of the end timestamp is the same as the start timestamp.

```

{
  "id": <32-bytes lowercase hex-encoded SHA-256 of the the serialized event data>,
  "pubkey": <32-bytes lowercase hex-encoded public key of the event creator>,
  "created_at": <unix timestamp in seconds>,
  "kind": 31923,
  "content": "<description of calendar event>",
  "tags": [
    ["d", "<random-identifier>"],

    ["title", "<title of calendar event>"],
    ["summary", "<brief description of the calendar event>"],
    ["image", "<string with image URI>"],

    // timestamps
    ["start", "<unix timestamp in seconds>"],
    ["end", "<unix timestamp in seconds>"],
  ]
}

```

```

["start_tzid", "<IANA Time Zone Database identifier>"],
["end_tzid", "<IANA Time Zone Database identifier>"],

// location
["location", "<location>"],
["g", "<geohash>"],

// participants
["p", "<32-bytes hex of a pubkey>", "<optional recommended relay URL>", "<role>"],
["p", "<32-bytes hex of a pubkey>", "<optional recommended relay URL>", "<role>"],
]
}

```

Calendar

A calendar is a collection of calendar events, represented as a custom *addressable list* event using kind 31924. A user can have multiple calendars. One may create a calendar to segment calendar events for specific purposes. e.g., personal, work, travel, meetups, and conferences.

Calendars can accept event requests from other users. When calendar events reference a calendar via an *a* tag, this represents a request for inclusion.

The *.content* of these events should be a detailed description of the calendar. It is required but can be an empty string.

- *d* (required) universally unique identifier. Generated by the client creating the calendar.
- *title* (required) calendar title
- *a* (repeated) reference tag to kind 31922 or 31923 calendar event being responded to

```

{
  "id": <32-bytes lowercase hex-encoded SHA-256 of the the serialized event data>,
  "pubkey": <32-bytes lowercase hex-encoded public key of the event creator>,
  "created_at": <unix timestamp in seconds>,
  "kind": 31924,
  "content": "<description of calendar>",
  "tags": [
    ["d", "<random-identifier>"],
    ["title", "<calendar title>"],
    ["a", "<31922 or 31923>:<calendar event author pubkey>:<d-identifier of calendar event>", "<optional relay url>"],
    ["a", "<31922 or 31923>:<calendar event author pubkey>:<d-identifier of calendar event>", "<optional relay url>"]
  ]
}

```

Calendar Event RSVP

A calendar event RSVP is a response to a calendar event to indicate a user's attendance intention.

If a calendar event tags a pubkey, that can be interpreted as the calendar event creator inviting that user to attend. Clients MAY choose to prompt the user to RSVP for the calendar event.

Any user may RSVP, even if they were not tagged on the calendar event. Clients MAY choose to prompt the calendar event creator to invite the user who RSVP'd. Clients also MAY choose to ignore these RSVPs.

This NIP is intentionally not defining who is authorized to attend a calendar event if the user who RSVP'd has not been tagged. It is up to the calendar event creator to determine the semantics.

This NIP is also intentionally not defining what happens if a calendar event changes after an RSVP is submitted.

The RSVP MUST have an *a* tag of the event coordinates to the calendar event, and optionally an *e* tag of the id of the specific calendar event revision. If an *e* tag is present, clients SHOULD interpret it as an indication that the RSVP is a response to that revision of the calendar event, and MAY interpret it to not necessarily apply to other revisions of the calendar event.

The RSVP MAY tag the author of the calendar event it is in response to using a *p* tag so that clients can easily query all RSVPs that pertain to the author.

The RSVP is an *addressable event* of kind:31925.

The *.* content of these events is optional and should be a free-form note that adds more context to this calendar event response.

The list of tags is as follows:

- *a* (required) coordinates to a kind 31922 or 31923 calendar event being responded to.
- *e* (optional) event id of a kind 31922 or 31923 calendar event being responded to.
- *d* (required) universally unique identifier. Generated by the client creating the calendar event RSVP.
- *status* (required) accepted, declined, or tentative. Determines attendance status to the referenced calendar event.
- *fb* (optional) free or busy. Determines if the user would be free or busy for the duration of the calendar event. This tag must be omitted or ignored if the *status* label is set to declined.
- *p* (optional) pubkey of the author of the calendar event being responded to.

```
{
  "id": <32-bytes lowercase hex-encoded SHA-256 of the the serialized event data>,
  "pubkey": <32-bytes lowercase hex-encoded public key of the event creator>,
  "created_at": <unix timestamp in seconds>,
  "kind": 31925,
  "content": "<note>",
  "tags": [
    ["e", "<kind 31922 or 31923 event id>", "<optional recommended relay URL>"],
    ["a", "<31922 or 31923>:<calendar event author pubkey>:<d-identifier of calendar event>", "<optional recommended relay URL>"],
    ["d", "<random-identifier>"],
    ["status", "<accepted/declined/tentative>"],
    ["fb", "<free/busy>"],
    ["p", "<hex pubkey of kind 31922 or 31923 event>", "<optional recommended relay URL>"]
  ]
}
```

Unsolved Limitations

- No private events

Intentionally Unsupported Scenarios

Recurring Calendar Events

Recurring calendar events come with a lot of complexity, making it difficult for software and humans to deal with. This complexity includes time zone differences between invitees, daylight savings, leap years, multiple calendar systems, one-off changes in schedule or other metadata, etc.

This NIP intentionally omits support for recurring calendar events and pushes that complexity up to clients to manually implement if they desire. i.e., individual calendar events with duplicated metadata represent recurring calendar events.

NIP-53

Live Activities

draft optional

This NIP introduces event kinds to advertise live spaces and the participation of pubkeys in them.

Live Streaming

A special event with kind:30311 “Live Streaming Event” is defined as an *addressable event* whose tags advertise the content and participants of a live stream.

Each p tag SHOULD have a **displayable** marker name for the current role (e.g. Host, Speaker, Participant) of the user in the event and the relay information MAY be empty. This event will be constantly updated as participants join and leave the activity.

For example:

```
{
  "kind": 30311,
  "tags": [
    ["d", "<unique identifier>"],
    ["title", "<name of the event>"],
    ["summary", "<description>"],
    ["image", "<preview image url>"],
    ["t", "hashtag"],
    ["streaming", "<url>"],
    ["recording", "<url>"], // used to place the edited video once the activity is over
    ["starts", "<unix timestamp in seconds>"],
    ["ends", "<unix timestamp in seconds>"],
    ["status", "<planned, live, ended>"],
    ["current_participants", "<number>"],
    ["total_participants", "<number>"],
    ["p", "91cf9..4e5ca", "wss://provider1.com/", "Host", "<proof>"],
    ["p", "14aeb..8dad4", "wss://provider2.com/nostr", "Speaker"],
    ["p", "612ae..e610f", "ws://provider3.com/ws", "Participant"],
    ["relays", "wss://one.com", "wss://two.com", /*...*/],
    ["pinned", "<event id of pinned live chat message>"],
  ],
  "content": "",
  // other fields...
}
```

A distinct d tag should be used for each activity. All other tags are optional.

Providers SHOULD keep the participant list small (e.g. under 1000 users) and, when limits are reached, Providers SHOULD select which participants get named in the event. Clients should not expect a comprehensive list. Once the activity ends, the event can be deleted or updated to summarize the activity and provide async content (e.g. recording of the event).

Clients are expected to subscribe to kind:30311 events in general or for given follow lists and statuses. Clients MAY display participants’ roles in activities as well as access points to join the activity.

Live Activity management clients are expected to constantly update kind:30311 during the event. Clients MAY choose to consider status=live events after 1hr without any update as ended. The starts and ends timestamp SHOULD be updated when the status changes to and from live

The activity MUST be linked to using the NIP-19 naddr code along with the a tag.

Proof of Agreement to Participate

Event owners can add proof as the 5th term in each `p` tag to clarify the participant's agreement in joining the event. The proof is a signed SHA256 of the complete `a` Tag of the event (`kind:pubkey:dTag`) by each `p`'s private key, encoded in hex.

Clients MAY only display participants if the proof is available or MAY display participants as "invited" if the proof is not available.

This feature is important to avoid malicious event owners adding large account holders to the event, without their knowledge, to lure their followers into the malicious owner's trap.

Live Chat Message

Event `kind:1311` is live chat's channel message. Clients MUST include the `a` tag of the activity. An `e` tag denotes the direct parent message this post is replying to.

```
{
  "kind": 1311,
  "tags": [
    ["a", "30311:<Community event author pubkey>:<d-identifier of the community>", "<Optional relay url>"],
  ],
  "content": "Zaps to live streams is beautiful.",
  // other fields...
}
```

`q` tags MAY be used when citing events in the `.content` with NIP-21.

```
["q", "<event-id> or <event-address>", "<relay-url>", "<pubkey-if-a-regular-event>"]
```

Hosts may choose to pin one or more live chat messages by updating the pinned tags in the live event `kind 30311`.

Examples

```
{
  "id": "57f28dbc264990e2c61e80a883862f7c114019804208b14da0bff81371e484d2",
  "pubkey": "1597246ac22f7d1375041054f2a4986bd971d8d196d7997e48973263ac9879ec",
  "created_at": 1687182672,
  "kind": 30311,
  "tags": [
    ["d", "demo-cf-stream"],
    ["title", "Adult Swim Metalocalypse"],
    ["summary", "Live stream from IPTV-ORG collection"],
    ["streaming", "https://adultswim-vodlive.cdn.turner.com/live/metalocalypse/stream.m3u8"],
    ["starts", "1687182672"],
    ["status", "live"],
    ["t", "animation"],
    ["t", "iptv"],
    ["image", "https://i.imgur.com/CaKq6Mt.png"]
  ],
  "content": "",
  "sig":
    "5bc7a60f5688effa5287244a24768cbe0dcd854436090abc3bef172f7f5db1410af4277508dbafc4f70a754a891c90ce3b966a7bc47e7c1eb71ff57640"
}
```

```
{
  "id": "97aa81798ee6c5637f7b21a411f89e10244e195aa91cb341bf49f718e36c8188",
  "pubkey": "3f770d65d3a764a9c5cb503ae123e62ec7598ad035d836e2a810f3877a745b24",
}
```

```

    "created_at": 1687286726,
    "kind": 1311,
    "tags": [
      ["a", "30311:1597246ac22f7d1375041054f2a4986bd971d8d196d7997e48973263ac9879ec:demo-cf-stream", "", "root"]
    ],
    "content": "Zaps to live streams is beautiful.",
    "sig":
      "997f62ddfc0827c121043074d50cfce7a528e978c575722748629a4137c45b75bdbc84170bedc723ef0a5a4c3daebf1fef2e93f5e2ddb98e5d685d022c
  }

```

Meeting Spaces

Meeting spaces contain one or more video/audio rooms where users can join and participate in the streaming.

Meeting Space Event (kind:30312)

A special event with kind:30312 “Space Host” defines the configuration and properties of a virtual interactive space. Each space has a unique identifier and can host multiple events/meetings.

```

{
  "kind": 30312,
  "tags": [
    ["d", "<unique identifier>"], // Required: Room identifier
    ["room", "<name of the room>"], // Required: Display name
    ["summary", "<description>"], // Optional: Room description
    ["image", "<preview image url>"], // Optional: Room image
    ["status", "<open, private, closed>"], // Required: Room accessibility
    ["service", "<url>"], // Required: URL to access the room
    ["endpoint", "<url>"], // Optional: API endpoint for status/info
    ["t", "<hashtag>"], // Optional: Multiple hashtags allowed
    ["p", "<pubkey>", "<url>", "<role>", "<proof>"], // Required: At least one provider
    ["relays", "<url>", "<url>", /*...*/] // Optional: Preferred relays
  ],
  "content": "" // Usually empty, may contain additional metadata
}

```

Space properties: * MUST be either open, private or closed. Closed means the room is not in operation. * MAY specify access control policy for private rooms (e.g. invite-only, payment required) * MAY persist when not in use * MUST have at least one provider with “Host” role * MAY have multiple providers with different roles

Provider roles (p tags): * Host: Full room management capabilities * Moderator: Room moderation capabilities * Speaker: Allowed to present/speak * Optional proof field for role verification

Meeting Room Events (kind:30313)

A special event with kind:30313 represents a scheduled or ongoing meeting within a room. It MUST reference its parent room using the d tag.

```

{
  "kind": 30313,
  "tags": [
    ["d", "<event-unique-identifier>"], // Required: Event identifier
    ["a", "30312:<pubkey>:<room-id>", "wss://nostr.example.com"], // Required: Reference to parent space, 'd' from
    30312
  ]
}

```

```

["title", "<meeting-title>"],          // Required: Meeting title
["summary", "<description>"],          // Optional: Meeting description
["image", "<preview image url>"],      // Optional: Meeting image
["starts", "<unix timestamp>"],        // Required: Start time
["ends", "<unix timestamp>"],          // Optional: End time
["status", "<planned, live, ended>"],  // Required: Meeting status
["total_participants", "<number>"],    // Optional: Total registered
["current_participants", "<number>"],  // Optional: Currently active
["p", "<pubkey>", "<url>", "<role>"],  // Optional: Participant with role
],
"content": "" // Usually empty, may contain additional metadata
}

```

Event properties: * MUST reference parent room via d tag * MUST have a status (planned/live/ended) * MUST have a start time * MAY track participant counts * MAY include participant roles specific to the event

Event management: * Clients SHOULD update event status regularly when live * Events without updates for 1 hour MAY be considered ended * starts and ends timestamps SHOULD be updated when status changes

Examples

```

{
  "kind": 30312,
  "tags": [
    ["d", "main-conference-room"],
    ["room", "Main Conference Hall"],
    ["summary", "Our primary conference space"],
    ["image", "https://example.com/room.jpg"],
    ["status", "open"],
    ["service", "https://meet.example.com/room"],
    ["endpoint", "https://api.example.com/room"],
    ["t", "conference"],
    ["p", "f7234bd4c1394dda46d09f35bd384dd30cc552ad5541990f98844fb06676e9ca", "wss://nostr.example.com/", "Owner"],
    ["p", "14aeb..8dad4", "wss://provider2.com/", "Moderator"],
    ["relays", "wss://relay1.com", "wss://relay2.com"]
  ],
  "content": ""
}

```

```

{
  "kind": 30313,
  "tags": [
    ["d", "annual-meeting-2025"],
    ["a", "30312:f7234bd4c1394dda46d09f35bd384dd30cc552ad5541990f98844fb06676e9ca:main-conference-room",
      "wss://nostr.example.com"]
    ["title", "Annual Company Meeting 2025"],
    ["summary", "Yearly company-wide meeting"],
    ["image", "https://example.com/meeting.jpg"],
    ["starts", "1676262123"],
    ["ends", "1676269323"],
    ["status", "live"],
    ["total_participants", "180"],
    ["current_participants", "175"],

```

```

    ["p", "91cf9..4e5ca", "wss://provider1.com/", "Speaker"],
  ],
  "content": ""
}

```

Room Presence

New kind: 10312 provides an event which signals presence of a listener.

The presence event SHOULD be updated at regular intervals and clients SHOULD filter presence events older than a given time window.

This kind 10312 is a regular replaceable event, as such presence can only be indicated in one room at a time.

```

{
  "kind": 10312,
  "tags": [
    ["a" , "<room-a-tag>", "<relay-hint>", "root"],
    ["hand", "1"] // hand raised flag
  ]
}

```

NIP-84

Highlights

draft optional

This NIP defines kind:9802, a “highlight” event, to signal content a user finds valuable.

Format

The .content of these events is the highlighted portion of the text.

.content might be empty for highlights of non-text based media (e.g. NIP-94 audio / video).

References

Events SHOULD tag the source of the highlight, whether nostr-native or not. a or e tags should be used for nostr events and r tags for URLs.

When tagging a URL, clients generating these events SHOULD do a best effort of cleaning the URL from trackers or obvious non-useful information from the query string.

Attribution

Clients MAY include one or more p tags, tagging the original authors of the material being highlighted; this is particularly useful when highlighting non-nostr content for which the client might be able to get a nostr pubkey somehow (e.g. prompting the user or reading a <link rel="me" href="nostr:nprofile1..." /> tag on the document). A role MAY be included as the last value of the tag.

```
{
  "tags": [
    ["p", "<pubkey-hex>", "<relay-url>", "author"],
    ["p", "<pubkey-hex>", "<relay-url>", "author"],
    ["p", "<pubkey-hex>", "<relay-url>", "editor"]
  ],
  // other fields...
}
```

Context

Clients MAY include a context tag, useful when the highlight is a subset of a paragraph and displaying the surrounding content might be beneficial to give context to the highlight.

Quote Highlights

A `comment` tag may be added to create a quote highlight. This **MUST** be rendered like a quote repost with the highlight as the quoted note.

This is to prevent the creation and multiple notes (`highlight` + `kind 1`) for a single highlight action, which looks bad in micro-blogging clients where these notes may appear in succession.

`p-tag` mentions **MUST** have a `mention` attribute to distinguish it from authors and editors.

`r-tag` urls from the comment **MUST** have a `mention` attribute to distinguish from the highlighted source url. The source url **MUST** have the `source` attribute.

NIP-15

Nostr Marketplace

draft optional

Based on Diagon-Alley.

Implemented in NostrMarket and Plebeian Market.

Terms

- merchant - seller of products with NOSTR key-pair
- customer - buyer of products with NOSTR key-pair
- product - item for sale by the merchant
- stall - list of products controlled by merchant (a merchant can have multiple stalls)
- marketplace - clientside software for searching stalls and purchasing products

Nostr Marketplace Clients

Merchant admin

Where the merchant creates, updates and deletes stalls and products, as well as where they manage sales, payments and communication with customers.

The merchant admin software can be purely clientside, but for convenience and uptime, implementations will likely have a server client listening for NOSTR events.

Marketplace

Marketplace software should be entirely clientside, either as a stand-alone app, or as a purely frontend webpage. A customer subscribes to different merchant NOSTR public keys, and those merchants stalls and products become listed and searchable. The marketplace client is like any other ecommerce site, with basket and checkout. Marketplaces may also wish to include a customer support area for direct message communication with merchants.

Merchant publishing/updating products (event)

A merchant can publish these events: | Kind | | Description | | _____ | _____ | _____
| | 0 | set_meta | The merchant description (similar
with any nostr public key). | | 30017 | set_stall | Create or update a stall. | | 30018 | set_product | Create or
update a product. | | 4 | direct_message | Communicate with the customer. The messages can be plain-text or
JSON. | | 5 | delete | Delete a product or a stall. |

Event 30017: Create or update a stall.

Event Content

```
{
  "id": <string, id generated by the merchant. Sequential IDs (`0`, `1`, `2`...) are discouraged>,
  "name": <string, stall name>,
  "description": <string (optional), stall description>,
  "currency": <string, currency used>,
  "shipping": [
    {
      "id": <string, id of the shipping zone, generated by the merchant>,
      "name": <string (optional), zone name>,
      "cost": <float, base cost for shipping. The currency is defined at the stall level>,
      "regions": [<string, regions included in this zone>]
    }
  ]
}
```

Fields that are not self-explanatory: - shipping: - an array with possible shipping zones for this stall. - the customer MUST choose exactly one of those shipping zones. - shipping to different zones can have different costs. For some goods (digital for example) the cost can be zero. - the id is an internal value used by the merchant. This value must be sent back as the customer selection. - each shipping zone contains the base cost for orders made to that shipping zone, but a specific shipping cost per product can also be specified if the shipping cost for that product is higher than what's specified by the base cost.

Event Tags

```
{
  "tags": [["d", <string, id of stall>],
  // other fields...
}
```

- the d tag is required, its value MUST be the same as the stall id.

Event 30018: Create or update a product

Event Content

```
{
  "id": <string, id generated by the merchant (sequential ids are discouraged)>,
  "stall_id": <string, id of the stall to which this product belong to>,
  "name": <string, product name>,
  "description": <string (optional), product description>,
  "images": <[string]>, array of image URLs, optional>,
  "currency": <string, currency used>,
  "price": <float, cost of product>,
  "quantity": <int or null, available items>,
  "specs": [
    [<string, spec key>, <string, spec value>]
  ],
  "shipping": [
    {
      "id": <string, id of the shipping zone (must match one of the zones defined for the stall)>,
      "cost": <float, extra cost for shipping. The currency is defined at the stall level>
    }
  ]
}
```

```
}
```

Fields that are not self-explanatory: - `quantity` can be null in the case of items with unlimited availability, like digital items, or services - `specs`: - an optional array of key pair values. It allows for the Customer UI to present product specifications in a structure mode. It also allows comparison between products - eg: `[["operating_system", "Android 12.0"], ["screen_size", "6.4 inches"], ["connector_type", "USB Type C"]]`

`_Open_`: better to move ``spec`` in the ``tags`` section of the event?

- `shipping`:
 - an *optional* array of extra costs to be used per shipping zone, only for products that require special shipping costs to be added to the base shipping cost defined in the stall
 - the `id` should match the `id` of the shipping zone, as defined in the `shipping` field of the stall
 - to calculate the total cost of shipping for an order, the user will choose a shipping option during checkout, and then the client must consider this costs:
 - * the base cost from the stall for the chosen shipping option
 - * the result of multiplying the product units by the shipping costs specified in the product, if any.

Event Tags

```
"tags": [  
  ["d", <string, id of product>],  
  ["t", <string (optional), product category>],  
  ["t", <string (optional), product category>],  
  // other fields...  
>],  
...
```

- the `d` tag is required, its value **MUST** be the same as the product `id`.
- the `t` tag is as searchable tag, it represents different categories that the product can be part of (food, fruits). Multiple `t` tags can be present.

Checkout events

All checkout events are sent as JSON strings using NIP-04.

The merchant and the customer can exchange JSON messages that represent different actions. Each JSON message **MUST** have a `type` field indicating the what the JSON represents. Possible types:

Message Type	Sent By	Description
0	Customer	New Order
1	Merchant	Payment Request
2	Merchant	Order Status Update

Step 1: customer order (event)

The below JSON goes in content of NIP-04.

```
{
  "id": <string, id generated by the customer>,
  "type": 0,
  "name": <string (optional), ???>,
  "address": <string (optional), for physical goods an address should be provided>,
  "message": <string (optional), message for merchant>,
  "contact": {
    "nostr": <32-bytes hex of a pubkey>,
    "phone": <string (optional), if the customer wants to be contacted by phone>,
    "email": <string (optional), if the customer wants to be contacted by email>
  },
  "items": [
    {
      "product_id": <string, id of the product>,
      "quantity": <int, how many products the customer is ordering>
    }
  ],
  "shipping_id": <string, id of the shipping zone>
}
```

Open: is contact.nostr required?

Step 2: merchant request payment (event)

Sent back from the merchant for payment. Any payment option is valid that the merchant can check.

The below JSON goes in content of NIP-04.

payment_options/type include:

- url URL to a payment page, stripe, paypal, btcpayserver, etc
- btc onchain bitcoin address
- ln bitcoin lightning invoice
- lnurl bitcoin lnurl-pay

```
{
  "id": <string, id of the order>,
  "type": 1,
  "message": <string, message to customer, optional>,
  "payment_options": [
    {
      "type": <string, option type>,
      "link": <string, url, btc address, ln invoice, etc>
    },
    {
      "type": <string, option type>,
      "link": <string, url, btc address, ln invoice, etc>
    },
    {
      "type": <string, option type>,
      "link": <string, url, btc address, ln invoice, etc>
    }
  ]
}
```

```
}
```

Step 3: merchant **verify payment/shipped (event)**

Once payment has been received and processed.

The below JSON goes in content of NIP-04.

```
{
  "id": <string, id of the order>,
  "type": 2,
  "message": <string, message to customer>,
  "paid": <bool: has received payment>,
  "shipped": <bool: has been shipped>,
}
```

Customize Marketplace

Create a customized user experience using the `naddr` from NIP-19. The use of `naddr` enables easy sharing of marketplace events while incorporating a rich set of metadata. This metadata can include relays, merchant profiles, and more. Subsequently, it allows merchants to be grouped into a market, empowering the market creator to configure the marketplace's user interface and user experience, and share that marketplace. This customization can encompass elements such as market name, description, logo, banner, themes, and even color schemes, offering a tailored and unique marketplace experience.

Event 30019: Create or update marketplace UI/UX

Event Content

```
{
  "name": <string (optional), market name>,
  "about": <string (optional), market description>,
  "ui": {
    "picture": <string (optional), market logo image URL>,
    "banner": <string (optional), market logo banner URL>,
    "theme": <string (optional), market theme>,
    "darkMode": <bool, true/false>
  },
  "merchants": [array of pubkeys (optional)],
  // other fields...
}
```

This event leverages `naddr` to enable comprehensive customization and sharing of marketplace configurations, fostering a unique and engaging marketplace environment.

Auctions

Event 30020: Create or update a product sold as an auction

Event Content:

```

{
  "id": <String, UUID generated by the merchant. Sequential IDs (`0`, `1`, `2`...) are discouraged>,
  "stall_id": <String, UUID of the stall to which this product belong to>,
  "name": <String, product name>,
  "description": <String (optional), product description>,
  "images": <[String], array of image URLs, optional>,
  "starting_bid": <int>,
  "start_date": <int (optional) UNIX timestamp, date the auction started / will start>,
  "duration": <int, number of seconds the auction will run for, excluding eventual time extensions that might happen>,
  "specs": [
    [<String, spec key>, <String, spec value>]
  ],
  "shipping": [
    {
      "id": <String, UUID of the shipping zone. Must match one of the zones defined for the stall>,
      "cost": <float, extra cost for shipping. The currency is defined at the stall level>
    }
  ]
}

```

[!NOTE] Items sold as an auction are very similar in structure to fixed-price items, with some important differences worth noting.

- The `start_date` can be set to a date in the future if the auction is scheduled to start on that date, or can be omitted if the start date is unknown/hidden. If the start date is not specified, the auction will have to be edited later to set an actual date.
- The auction runs for an initial number of seconds after the `start_date`, specified by `duration`.

Event 1021: Bid

```

{
  "content": <int, amount of sats>,
  "tags": [["e", <event ID of the auction to bid on>]],
  // other fields...
}

```

Bids are simply events of kind 1021 with a `content` field specifying the amount, in the currency of the auction. Bids must reference an auction.

[!NOTE] Auctions can be edited as many times as desired (they are “addressable events”) by the author - even after the `start_date`, but they cannot be edited after they have received the first bid! This is enforced by the fact that bids reference the event ID of the auction (rather than the product UUID), which changes with every new version of the auctioned product. So a bid is always attached to one “version”. Editing the auction after a bid would result in the new product losing the bid!

Event 1022: Bid confirmation

Event Content:

```
{
  "status": <String, "accepted" | "rejected" | "pending" | "winner">,
  "message": <String (optional)>,
  "duration_extended": <int (optional), number of seconds>
}
```

Event Tags:

```
"tags": [["e" <event ID of the bid being confirmed>], ["e", <event ID of the auction>]],
```

Bids should be confirmed by the merchant before being considered as valid by other clients. So clients should subscribe to *bid confirmation* events (kind 1022) for every auction that they follow, in addition to the actual bids and should check that the pubkey of the bid confirmation matches the pubkey of the merchant (in addition to checking the signature).

The content field is a JSON which includes *at least* a status. winner is how the *winning bid* is replied to after the auction ends and the winning bid is picked by the merchant.

The reasons for which a bid can be marked as rejected or pending are up to the merchant's implementation and configuration - they could be anything from basic validation errors (amount too low) to the bidder being blacklisted or to the bidder lacking sufficient *trust*, which could lead to the bid being marked as pending until sufficient verification is performed. The difference between the two is that pending bids *might* get approved after additional steps are taken by the bidder, whereas rejected bids can not be later approved.

An additional message field can appear in the content JSON to give further context as of why a bid is rejected or pending.

Another thing that can happen is - if bids happen very close to the end date of the auction - for the merchant to decide to extend the auction duration for a few more minutes. This is done by passing a `duration_extended` field as part of a bid confirmation, which would contain a number of seconds by which the initial duration is extended. So the actual end date of an auction is always `start_date + duration + (SUM(c.duration_extended) FOR c in all confirmations)`.

Customer support events

Customer support is handled over whatever communication method was specified. If communicating via nostr, NIP-04 is used.

Additional

Standard data models can be found [here](#)

NIP-99

Classified Listings

draft optional

This NIP defines `kind:30402`: an addressable event to describe classified listings that list any arbitrary product, service, or other thing for sale or offer and includes enough structured metadata to make them useful.

The specification supports a broad range of use cases physical goods, services, work opportunities, rentals, free give-aways, personals, etc. To promote interoperability between clients implementing NIP-99 for e-commerce, you can find the extension proposal here which standardizes the e-commerce use case while maintaining the specification's lightweight and flexible nature. While NIP-15 provides a strictly structured marketplace specification, NIP-99 has emerged as a simpler and more flexible alternative.

The structure of these events is very similar to NIP-23 long-form content events.

Draft / Inactive Listings

`kind:30403` has the same structure as `kind:30402` and is used to save draft or inactive classified listings.

Content

The `.content` field should be a description of what is being offered and by whom. These events should be a string in Markdown syntax.

Author

The `.pubkey` field of these events are treated as the party creating the listing.

Metadata

- For “tags” / “hashtags” (i.e. categories or keywords of relevance for the listing) the “t” event tag should be used.
- For images, whether included in the markdown content or not, clients SHOULD use `image` tags as described in NIP-58. This allows clients to display images in carousel format more easily.

The following tags, used for structured metadata, are standardized and SHOULD be included. Other tags may be added as necessary.

- “title”, a title for the listing
- “summary”, for short tagline or summary for the listing
- “published_at”, for the timestamp (in unix seconds – converted to string) of the first time the listing was published.
- “location”, for the location.
- “price”, for the price of the thing being listed. This is an array in the format [“price”, “<number>”, “<currency>”, “<frequency>”].

- "price" is the name of the tag
- "<number>" is the amount in numeric format (but included in the tag as a string)
- "<currency>" is the currency unit in 3-character ISO 4217 format or ISO 4217-like currency code (e.g. "btc", "eth").
- "<frequency>" is optional and can be used to describe recurring payments. SHOULD be in noun format (hour, day, week, month, year, etc.)
- "status" (optional), the status of the listing. SHOULD be either "active" or "sold".

price examples

- \$50 one-time payment ["price", "50", "USD"]
- €15 per month ["price", "15", "EUR", "month"]
- £50,000 per year ["price", "50000", "GBP", "year"]

Other common tags that might be useful.

- "g", a geohash for more precise location

Example Event

```
{
  "kind": 30402,
  "created_at": 1675642635,
  // Markdown content
  "content": "Lorem
    [ipsum][nostr:nevent1qqst8cujk046negxgwm5ynqwn53t8aqjr6afd8g59nfqwxpdhy1pcpzamhxue69uhhyetw9ujuetcv9khqmr99e3k7mg8arnc9]
    dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna
    aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo
    consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
    pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id
    est laborum.\n\nRead more at
    nostr:naddr1qqzkjurnw4ks29thwden5te0wfjkccte9ehx7um5wghx7un8qgs2d90kkcq3nk2jry62dyf50k0h36rhpdt594my40w9pka1876jxgrqsqqqa2
  "tags": [
    ["d", "lorem-ipsum"],
    ["title", "Lorem Ipsum"],
    ["published_at", "1296962229"],
    ["t", "electronics"],
    ["image", "https://url.to.img", "256x256"],
    ["summary", "More lorem ipsum that is a little more than the title"],
    ["location", "NYC"],
    ["price", "100", "USD"],
    [
      "e",
      "b3e392b11f5d4f28321cedd09303a748acfd0487aea5a7450b3481c60b6e4f87",
      "wss://relay.example.com"
    ],
    [
      "a",
      "30023:a695f6b60119d9521934a691347d9f78e8770b56da16bb255ee286ddf9fda919:ipsum",
      "wss://relay.nostr.org"
    ]
  ],
  "pubkey": "...",
  "id": "..."
}
```


NIP-54

Wiki

draft optional

This NIP defines kind:30818 (an *addressable event*) for descriptions (or encyclopedia entries) of particular subjects, and it's expected that multiple people will write articles about the exact same subjects, with either small variations or completely independent content.

Articles are identified by lowercase, normalized ascii d tags.

Articles

```
{
  "content": "A wiki is a hypertext publication collaboratively edited and managed by its own audience.",
  "tags": [
    ["d", "wiki"],
    ["title", "Wiki"],
  ]
}
```

d tag normalization rules

- Any non-letter character MUST be converted to a -.
- All letters MUST be converted to lowercase.

Content

The content should be AsciiDoc with two extra functionalities: **wikilinks** and **nostr:... links**.

Unlike normal AsciiDoc links `http://example.com[]` that link to external webpages, wikilinks `[]` link to other articles in the wiki. In this case, the wiki is the entirety of Nostr. Clicking on a wikilink should cause the client to ask relays for events with d tags equal to the target of that wikilink.

Wikilinks can take these two forms:

1. `[[Target Page]]` – in this case it will link to the page `target-page` (according to d tag normalization rules above) and be displayed as `Target Page`;
2. `[[target page|see this]]` – in this case it will link to the page `target-page`, but will be displayed as `see this`.

`nostr:... links`, as per NIP-21, should link to profiles or arbitrary Nostr events. Although it is not recommended to link to specific versions of articles – instead the *wikilink* syntax should be preferred, since it should be left to the reader and their client to decide what version of any given article they want to read.

Optional extra tags

- `title`: for when the display title should be different from the d tag.
- `summary`: for display in lists.
- `a` and `e`: for referencing the original event a wiki article was forked from.

Merge Requests

Event kind:818 represents a request to merge from a forked article into the source. It is directed to a pubkey and references the original article and the modified event.

[INSERT EVENT EXAMPLE]

Redirects

Event kind:30819 is also defined to stand for “wiki redirects”, i.e. if one thinks Shell structure should redirect to Thin-shell structure they can issue one of these events instead of replicating the content. These events can be used for automatically redirecting between articles on a client, but also for generating crowdsourced “disambiguation” pages (common in Wikipedia).

[INSERT EVENT EXAMPLE]

How to decide what article to display

As there could be many articles for each given name, some kind of prioritization must be done by clients. Criteria for this should vary between users and clients, but some means that can be used are described below:

Reactions

NIP-25 reactions are very simple and can be used to create a simple web-of-trust between wiki article writers and their content. While just counting a raw number of “likes” is unproductive, reacting to any wiki article event with a + can be interpreted as a recommendation for that article specifically and a partial recommendation of the author of that article. When 2 or 3-level deep recommendations are followed, suddenly a big part of all the articles may have some form of tagging.

Relays

NIP-51 lists of relays can be created with the kind 10102 and then used by wiki clients in order to determine where to query articles first and to rank these differently in relation to other events fetched from other relays.

Contact lists

NIP-02 contact lists can form the basis of a recommendation system that is then expanded with relay lists and reaction lists through nested queries. These lists form a good starting point only because they are so widespread.

Wiki-related contact lists

NIP-51 lists can also be used to create a list of users that are trusted only in the context of wiki authorship or wiki curationship.

Forks

Wiki-events can tag other wiki-events with a fork marker to specify that this event came from a different version. Both a and e tags SHOULD be used and have the fork marker applied, to identify the exact version it was forked from.

Deference

Wiki-events can tag other wiki-events with a defer marker to indicate that it considers someone else’s entry as a “better” version of itself. If using a defer marker both a and e tags SHOULD be used.

This is a stronger signal of trust than a + reaction.

This marker is useful when a user edits someone else’s entry; if the original author includes the editor’s changes and the editor doesn’t want to keep / maintain an independent version, the link tag could effectively be considered a “deletion” of the editor’s version and putting that pubkey’s WoT weight behind the original author’s version.

Why AsciiDoc?

Wikitext is garbage and Markdown is not powerful enough (besides being too freeform and unspecified and prone to generate incompatibilities in the future).

Asciidoc has a strict spec, multiple implementations in many languages, and support for features that are very much necessary in a wiki article, like *sidebars*, *tables* (with rich markup inside cells), many levels of *headings*, *footnotes*, *superscript* and *subscript* markup and *description lists*. It is also arguably easier to read in its plaintext format than Markdown (and certainly much better than Wikitext).

Appendix 1: Merge requests

Users can request other users to get their entries merged into someone else's entry by creating a kind:818 event.

```
{
  "content": "I added information about how to make hot ice-creams",
  "kind": 818,
  "tags": [
    [ "a", "30818:<destination-pubkey>:hot-ice-creams", "<relay-url>" ],
    [ "e", "<version-against-which-the-modification-was-made>", "<relay-url>" ],
    [ "p", "<destination-pubkey>" ],
    [ "e", "<version-to-be-merged>", "<relay-url>", "source" ]
  ]
}
```

. content: an optional explanation detailing why this merge is being requested. a tag: tag of the article which should be modified (i.e. the target of this merge request). e tag: optional version of the article in which this modifications is based e tag with source marker: the ID of the event that should be merged. This event id MUST be of a kind:30818 as defined in this NIP.

The destination-pubkey is the pubkey being requested to merge something into their article can create [NIP-25] reactions that tag the kind:818 event with + or -

NIP-34

git stuff

draft optional

This NIP defines all the ways code collaboration using and adjacent to git can be done using Nostr.

Repository announcements

Git repositories are hosted in Git-enabled servers, but their existence can be announced using Nostr events. By doing so the author asserts themselves as a maintainer and expresses a willingness to receive patches, bug reports and comments in general, unless `t` tag `personal-fork` is included.

```
{
  "kind": 30617,
  "content": "",
  "tags": [
    ["d", "<repo-id>"], // usually kebab-case short name
    ["name", "<human-readable project name>"],
    ["description", "brief human-readable project description"],
    ["web", "<url for browsing>", ...], // a webpage url, if the git server being used provides such a thing
    ["clone", "<url for git-cloning>", ...], // a url to be given to `git clone` so anyone can clone it
    ["relays", "<relay-url>", ...], // relays that this repository will monitor for patches and issues
    ["r", "<earliest-unique-commit-id>", "euc"],
    ["maintainers", "<other-recognized-maintainer>", ...],
    ["t", "personal-fork"], // optionally indicate author isn't a maintainer
    ["t", "<arbitrary string>"], // hashtags labelling the repository
  ]
}
```

The tags `web`, `clone`, `relays`, `maintainers` can have multiple values.

The `r` tag annotated with the `"euc"` marker should be the commit ID of the earliest unique commit of this repo, made to identify it among forks and group it with other repositories hosted elsewhere that may represent essentially the same project. In most cases it will be the root commit of a repository. In case of a permanent fork between two projects, then the first commit after the fork should be used.

Except `d`, all tags are optional.

Repository state announcements

An optional source of truth for the state of branches and tags in a repository.

```
{
  "kind": 30618,
  "content": "",
  "tags": [
    ["d", "<repo-id>"], // matches the identifier in the corresponding repository announcement
    ["refs/<heads|tags>/<branch-or-tag-name>", "<commit-id>"],
    ["HEAD", "ref: refs/heads/<branch-name>"]
  ]
}
```

The `refs` tag may appear multiple times, or none.

If no `refs` tags are present, the author is no longer tracking repository state using this event. This approach enables the author to restart tracking state at a later time unlike NIP-09 deletion requests.

The `refs` tag can be optionally extended to enable clients to identify how many commits ahead a ref is:

```
{
  "tags": [
    ["refs/<heads|tags>/<branch-or-tag-name>", "<commit-id>", "<shorthand-parent-commit-id>",
      "<shorthand-grandparent>", ...],
  ]
}
```

Patches and Pull Requests (PRs)

Patches and PRs can be sent by anyone to any repository. Patches and PRs to a specific repository SHOULD be sent to the relays specified in that repository's announcement event's "relays" tag. Patch and PR events SHOULD include an a tag pointing to that repository's announcement address.

Patches SHOULD be used if each event is under 60kb, otherwise PRs SHOULD be used.

Patches

Patches in a patch set SHOULD include a NIP-10 e reply tag pointing to the previous patch.

The first patch revision in a patch revision SHOULD include a NIP-10 e reply to the original root patch.

```
{
  "kind": 1617,
  "content": "<patch>", // contents of <git format-patch>
  "tags": [
    ["a", "30617:<base-repo-owner-pubkey>:<base-repo-id>"],
    ["r", "<earliest-unique-commit-id-of-repo>"] // so clients can subscribe to all patches sent to a local git repo
    ["p", "<repository-owner>"],
    ["p", "<other-users>"], // optionally send the patch to another user to bring it to their attention

    ["t", "root"], // omitted for additional patches in a series
    // for the first patch in a revision
    ["t", "root-revision"],

    // optional tags for when it is desirable that the merged patch has a stable commit id
    // these fields are necessary for ensuring that the commit resulting from applying a patch
    // has the same id as it had in the proposer's machine -- all these tags can be omitted
    // if the maintainer doesn't care about these things
    ["commit", "<current-commit-id>"],
    ["r", "<current-commit-id>"] // so clients can find existing patches for a specific commit
    ["parent-commit", "<parent-commit-id>"],
    ["commit-pgp-sig", "-----BEGIN PGP SIGNATURE-----..."], // empty string for unsigned commit
    ["committer", "<name>", "<email>", "<timestamp>", "<timezone offset in minutes>"],
  ]
}
```

The first patch in a series MAY be a cover letter in the format produced by `git format-patch`.

Pull Requests

The PR or PR update tip SHOULD be successfully pushed to `refs/nostr/<[PR|PR-Update]-event-id>` in all repositories listed in its `clone` tag before the event is signed.

An attempt SHOULD be made to push this ref to all repositories listed in the repository's announcement event's "clone" tag, for which there is reason to believe the user might have write access. This includes each grasp server which can be identified using this method: `clone` tag includes `[http|https]://<grasp-path>/<valid-npub>/<string>.git` and `relays` tag includes `[ws/wss]://<grasp-path>`.

Clients MAY fallback to creating a 'personal-fork' repository announcement listing other grasp servers, e.g. from the `User grasp` list, for the purpose of serving the specified commit(s).

```
{
  "kind": 1618,
  "content": "<markdown text>",
  "tags": [
    ["a", "30617:<base-repo-owner-pubkey>:<base-repo-id>"],
    ["r", "<earliest-unique-commit-id-of-repo>"] // so clients can subscribe to all PRs sent to a local git repo
    ["p", "<repository-owner>"],
    ["p", "<other-user>"], // optionally send the PR to another user to bring it to their attention

    ["subject", "<PR-subject>"],
    ["t", "<PR-label>"], // optional
    ["t", "<another-PR-label>"], // optional

    ["c", "<current-commit-id>"], // tip of the PR branch
    ["clone", "<clone-url>", ...], // at least one git clone url where commit can be downloaded
    ["branch-name", "<branch-name>"], // optional recommended branch name

    ["e", "<root-patch-event-id>"], // optionally indicate PR is a revision of an existing patch, which should be
      closed
    ["merge-base", "<commit-id>"], // optional: the most recent common ancestor with the target branch
  ]
}
```

Pull Request Updates

A PR Update changes the tip of a referenced PR event.

```
{
  "kind": 1619,
  "content": "",
  "tags": [
    ["a", "30617:<base-repo-owner-pubkey>:<base-repo-id>"],
    ["r", "<earliest-unique-commit-id-of-repo>"] // so clients can subscribe to all PRs sent to a local git repo
    ["p", "<repository-owner>"],
    ["p", "<other-user>"], // optionally send the PR to another user to bring it to their attention

    // NIP-22 tags
    ["E", "<pull-request-event-id>"],
    ["P", "<pull-request-author>"],

    ["c", "<current-commit-id>"], // updated tip of PR
    ["clone", "<clone-url>", ...], // at least one git clone url where commit can be downloaded
    ["merge-base", "<commit-id>"], // optional: the most recent common ancestor with the target branch
  ]
}
```

Issues

Issues are Markdown text that is just human-readable conversational threads related to the repository: bug reports, feature requests, questions or comments of any kind. Like patches, these SHOULD be sent to the relays specified in that repository's announcement event's "relays" tag.

Issues may have a subject tag, which clients can utilize to display a header. Additionally, one or more t tags may be included to provide labels for the issue.

```
{
  "kind": 1621,
  "content": "<markdown text>",
```

```

"tags": [
  ["a", "30617:<base-repo-owner-pubkey>:<base-repo-id>"],
  ["p", "<repository-owner>"],
  ["subject", "<issue-subject>"],
  ["t", "<issue-label>"],
  ["t", "<another-issue-label>"]
]
}

```

Replies

Replies to either a kind:1621 (*issue*), kind:1617 (*patch*) or kind:1618 (*pull request*) event should follow NIP-22 comment.

Status

Root Patches, PRs and Issues have a Status that defaults to ‘Open’ and can be set by issuing Status events.

```

{
  "kind": 1630, // Open
  "kind": 1631, // Applied / Merged for Patches; Resolved for Issues
  "kind": 1632, // Closed
  "kind": 1633, // Draft
  "content": "<markdown text>",
  "tags": [
    ["e", "<issue-or-PR-or-original-root-patch-id-hex>", "", "root"],
    ["e", "<accepted-revision-root-id-hex>", "", "reply"], // for when revisions applied
    ["p", "<repository-owner>"],
    ["p", "<root-event-author>"],
    ["p", "<revision-author>"],

    // optional for improved subscription filter efficiency
    ["a", "30617:<base-repo-owner-pubkey>:<base-repo-id>", "<relay-url>"],
    ["r", "<earliest-unique-commit-id-of-repo>"]

    // optional for `1631` status
    ["q", "<applied-or-merged-patch-event-id>", "<relay-url>", "<pubkey>"], // for each
    // when merged
    ["merge-commit", "<merge-commit-id>"]
    ["r", "<merge-commit-id>"]
    // when applied
    ["applied-as-commits", "<commit-id-in-master-branch>", ...]
    ["r", "<applied-commit-id>"] // for each
  ]
}

```

The most recent Status event (by created_at date) from either the issue/patch author or a maintainer is considered valid.

The Status of a patch-revision is to either that of the root-patch, or 1632 (*Closed*) if the root-patch’s Status is 1631 (*Applied/Merged*) and the patch-revision isn’t tagged in the 1631 (*Applied/Merged*) event.

User grasp list

List of grasp servers the user generally wishes to use for NIP-34 related activity. It is similar in function to the NIP-65 relay list and NIP-B7 blossom list.

The event SHOULD include a list of g tags with grasp service websocket URLs in order of preference.

```
{
  "kind": 10317,
  "content": "",
  "tags": [
    ["g", "<grasp-service-websocket-url>"], // zero or more grasp sever urls
  ]
}
```

Possible things to be added later

- inline file comments kind (we probably need one for patches and a different one for merged files)

NIP-94

File Metadata

draft optional

The purpose of this NIP is to allow an organization and classification of shared files. So that relays can filter and organize in any way that is of interest. With that, multiple types of filesharing clients can be created. NIP-94 support is not expected to be implemented by “social” clients that deal with kind:1 notes or by longform clients that deal with kind:30023 articles.

Event format

This NIP specifies the use of the 1063 event kind, having in content a description of the file content, and a list of tags described below:

- url the url to download the file
- m a string indicating the data type of the file. The MIME types format must be used, and they should be lowercase.
- x containing the SHA-256 hexencoded string of the file.
- ox containing the SHA-256 hexencoded string of the original file, before any transformations done by the upload server
- size (optional) size of file in bytes
- dim (optional) size of file in pixels in the form <width>x<height>
- magnet (optional) URI to magnet file
- i (optional) torrent infohash
- blurhash(optional) the blurhash to show while the file is being loaded by the client
- thumb (optional) url of thumbnail with same aspect ratio
- image (optional) url of preview image with same dimensions
- summary (optional) text excerpt
- alt (optional) description for accessibility
- fallback (optional) zero or more fallback file sources in case url fails
- service (optional) service type which is serving the file (eg. NIP-96)

```
{
  "kind": 1063,
  "tags": [
    ["url", <string with URI of file>],
    ["m", <MIME type>],
    ["x", <Hash SHA-256>],
    ["ox", <Hash SHA-256>],
    ["size", <size of file in bytes>],
    ["dim", <size of file in pixels>],
    ["magnet", <magnet URI> ],
    ["i", <torrent infohash>],
    ["blurhash", <value>],
    ["thumb", <string with thumbnail URI>, <Hash SHA-256>],
    ["image", <string with preview URI>, <Hash SHA-256>],
    ["summary", <excerpt>],
    ["alt", <description>]
  ],
  "content": "<caption>",
  // other fields...
}
```

Suggested use cases

- A relay for indexing shared files. For example, to promote torrents.

- A pinterest-like client where people can share their portfolio and inspire others.
- A simple way to distribute configurations and software updates.

Warning unrecommended: deprecated in favor of NIP-B7

NIP-96

HTTP File Storage Integration

draft optional

Introduction

This NIP defines a REST API for HTTP file storage servers intended to be used in conjunction with the nostr network. The API will enable nostr users to upload files and later reference them by url on nostr notes.

The spec DOES NOT use regular nostr events through websockets for storing, requesting nor retrieving data because, for simplicity, the server will not have to learn anything about nostr relays.

Server Adaptation

File storage servers wishing to be accessible by nostr users should opt-in by making available an https route at `/.well-known/nostr/nip96.json` with `api_url`:

```
{
  // Required
  // File upload and deletion are served from this url
  // Also downloads if "download_url" field is absent or empty string
  "api_url": "https://your-file-server.example/custom-api-path",
  // Optional
  // If absent, downloads are served from the api_url
  "download_url": "https://a-cdn.example/a-path",
  // Optional
  // Note: This field is not meant to be set by HTTP Servers.
  // Use this if you are a nostr relay using your /.well-known/nostr/nip96.json
  // just to redirect to someone else's http file storage server's /.well-known/nostr/nip96.json
  // In this case, "api_url" field must be an empty string
  "delegated_to_url": "https://your-file-server.example",
  // Optional
  "supported_nips": [60],
  // Optional
  "tos_url": "https://your-file-server.example/terms-of-service",
  // Optional
  "content_types": ["image/jpeg", "video/webm", "audio/*"],
  // Optional
  "plans": {
    // "free" is the only standardized plan key and
    // clients may use its presence to learn if server offers free storage
    "free": {
      "name": "Free Tier",
      // Default is true
      // All plans MUST support NIP-98 uploads
      // but some plans may also allow uploads without it
      "is_nip98_required": true,
      "url": "https://...", // plan's landing page if there is one
      "max_byte_size": 10485760,
      // Range in days / 0 for no expiration
      // [7, 0] means it may vary from 7 days to unlimited persistence,
      // [0, 0] means it has no expiration
      // early expiration may be due to low traffic or any other factor
    }
  }
}
```

```

    "file_expiration": [14, 90],
    "media_transformations": {
      "image": [
        "resizing"
      ]
    }
  }
}
}

```

Relay Hints

Note: This section is not meant to be used by HTTP Servers.

A nostr relay MAY redirect to someone else's HTTP file storage server by adding a `/.well-known/nostr/nip96.json` with `"delegated_to_url"` field pointing to the url where the server hosts its own `/.well-known/nostr/nip96.json`. In this case, the `"api_url"` field must be an empty string and all other fields must be absent.

If the nostr relay is also an HTTP file storage server, it must use the `"api_url"` field instead.

List of Supporting File Storage Servers

See <https://github.com/aljazceru/awesome-nostr#nip-96-file-storage-servers>.

Auth

When indicated, clients must add an NIP-98 Authorization header (**optionally** with the encoded payload tag set to the base64-encoded 256-bit SHA-256 hash of the file - not the hash of the whole request body).

Upload

POST `$api_url` as `multipart/form-data`.

AUTH required

List of form fields:

- **file**: **REQUIRED** the file to upload
- **caption**: **RECOMMENDED** loose description;
- **expiration**: UNIX timestamp in seconds. Empty string if file should be stored forever. The server isn't required to honor this.
- **size**: File byte size. This is just a value the server can use to reject early if the file size exceeds the server limits.
- **alt**: **RECOMMENDED** strict description text for visibility-impaired users.
- **media_type**: "avatar" or "banner". Informs the server if the file will be used as an avatar or banner. If absent, the server will interpret it as a normal upload, without special treatment.
- **content_type**: mime type such as "image/jpeg". This is just a value the server can use to reject early if the mime type isn't supported.
- **no_transform**: "true" asks server not to transform the file and serve the uploaded file as is, may be rejected.

Others custom form data fields may be used depending on specific server support. The server isn't required to store any metadata sent by clients.

The filename embedded in the file may not be honored by the server, which could internally store just the SHA-256 hash value as the file name, ignoring extra metadata. The hash is enough to uniquely identify a file, that's why it will be used on the download and delete routes.

The server **MUST** link the user's pubkey string as the owner of the file so to later allow them to delete the file.

`no_transform` can be used to replicate a file to multiple servers for redundancy, clients can use the server list to find alternative servers which might contain the same file. When uploading a file and requesting `no_transform` clients should check that the hash matches in the response in order to detect if the file was modified.

Response codes

- 200 OK: File upload exists, but is successful (Existing hash)
- 201 Created: File upload successful (New hash)
- 202 Accepted: File upload is awaiting processing, see Delayed Processing section
- 413 Payload Too Large: File size exceeds limit
- 400 Bad Request: Form data is invalid or not supported.
- 403 Forbidden: User is not allowed to upload or the uploaded file hash didnt match the hash included in the Authorization header payload tag.
- 402 Payment Required: Payment is required by the server, **this flow is undefined**.

The upload response is a json object as follows:

```
{
  // "success" if successful or "error" if not
  "status": "success",
  // Free text success, failure or info message
  "message": "Upload successful.",
  // Optional. See "Delayed Processing" section
  "processing_url": "...",
  // This uses the NIP-94 event format but DO NOT need
  // to fill some fields like "id", "pubkey", "created_at" and "sig"
  //
  // This holds the download url ("url"),
  // the ORIGINAL file hash before server transformations ("ox")
  // and, optionally, all file metadata the server wants to make available
  //
  // nip94_event field is absent if unsuccessful upload
  "nip94_event": {
    // Required tags: "url" and "ox"
    "tags": [
      // Can be same from /.well-known/nostr/nip96.json's "download_url" field
      // (or "api_url" field if "download_url" is absent or empty) with appended
      // original file hash.
      //
      // Note we appended .png file extension to the `ox` value
      // (it is optional but extremely recommended to add the extension as it will help nostr clients
      // with detecting the file type by using regular expression)
      //
      // Could also be any url to download the file
      // (using or not using the /.well-known/nostr/nip96.json's "download_url" prefix),
      // for load balancing purposes for example.
      ["url",
        "https://your-file-server.example/custom-api-path/719171db19525d9d08dd69cb716a18158a249b7b3b3ec4bbdec5698dca104b7b.png"],
      // SHA-256 hash of the ORIGINAL file, before transformations.
      // The server MUST store it even though it represents the ORIGINAL file because
      // users may try to download/delete the transformed file using this value
      ["ox", "719171db19525d9d08dd69cb716a18158a249b7b3b3ec4bbdec5698dca104b7b"],
      // Optional. SHA-256 hash of the saved file after any server transformations.
      // The server can but does not need to store this value.
      ["x", "543244319525d9d08dd69cb716a18158a249b7b3b3ec4bbde5435543acb34443"],
      // Optional. Recommended for helping clients to easily know file type before downloading it.
      ["m", "image/png"],
      // Optional. Recommended for helping clients to reserve an adequate UI space to show the file before
      // downloading it.
      ["dim", "800x600"]
      // ... other optional NIP-94 tags
    ],
  },
}
```

```

    "content": ""
  },
  // ... other custom fields (please consider adding them to this NIP or to NIP-94 tags)
}

```

Note that if the server didn't apply any transformation to the received file, both `nip94_event.tags.*.ox` and `nip94_event.tags.*.x` fields will have the same value. The server **MUST** link the saved file to the SHA-256 hash of the **original** file before any server transformations (the `nip94_event.tags.*.ox` tag value). The **original** file's SHA-256 hash will be used to identify the saved file when downloading or deleting it.

clients may upload the same file to one or many servers. After successful upload, the client may optionally generate and send to any set of nostr relays a NIP-94 event by including the missing fields.

Alternatively, instead of using NIP-94, the client can share or embed on a nostr note just the above url.

clients may also use the tags from the `nip94_event` to construct an `imeta` tag

Delayed Processing

Sometimes the server may want to place the uploaded file in a processing queue for deferred file processing.

In that case, the server **MUST** serve the original file while the processing isn't done, then swap the original file for the processed one when the processing is over. The upload response is the same as usual but some optional metadata like `nip94_event.tags.*.x` and `nip94_event.tags.*.size` won't be available.

The expected resulting metadata that is known in advance should be returned on the response. For example, if the file processing would change a file from "jpg" to "webp", use ".webp" extension on the `nip94_event.tags.*.url` field value and set "image/webp" to the `nip94_event.tags.*.m` field. If some metadata are unknown before processing ends, omit them from the response.

The upload response **MAY** include a `processing_url` field informing a temporary url that may be used by clients to check if the file processing is done.

If the processing isn't done, the server should reply at the `processing_url` url with **200 OK** and the following JSON:

```

{
  // It should be "processing". If "error" it would mean the processing failed.
  "status": "processing",
  "message": "Processing. Please check again later for updated status.",
  "percentage": 15 // Processing percentage. An integer between 0 and 100.
}

```

When the processing is over, the server replies at the `processing_url` url with **201 Created** status and a regular successful JSON response already mentioned before (now **without** a `processing_url` field), possibly including optional metadata at `nip94_event.tags.*` fields that weren't available before processing.

File compression

File compression and other transformations like metadata stripping can be applied by the server. However, for all file actions, such as download and deletion, the **original** file SHA-256 hash is what identifies the file in the url string.

Download

```
GET $api_url/<sha256-hash>(.ext)
```

The primary file download url informed at the upload's response field `nip94_event.tags.*.url` can be that or not (it can be any non-standard url the server wants). If not, the server still **MUST** also respond to downloads at the standard url mentioned on the previous paragraph, to make it possible for a client to try downloading a file on any NIP-96 compatible server by knowing just the SHA-256 file hash.

Note that the "<sha256-hash>" part is from the **original** file, **not** from the **transformed** file if the uploaded file went through any server transformation.

Supporting “.ext”, meaning “file extension”, is required for servers. It is optional, although recommended, for clients to append it to the path. When present it may be used by servers to know which Content-Type header to send (e.g.: “Content-Type”: “image/png” for “.png” extension). The file extension may be absent because the hash is the only needed string to uniquely identify a file.

Example: `$api_url/719171db19525d9d08dd69cb716a18158a249b7b3b3ec4bbdec5698dca104b7b.png`

Media Transformations

servers may respond to some media transformation query parameters and ignore those they don't support by serving the original media file without transformations.

Image Transformations

Resizing Upon upload, servers may create resized image variants, such as thumbnails, respecting the original aspect ratio. clients may use the `w` query parameter to request an image version with the desired pixel width. servers can then serve the variant with the closest width to the parameter value or an image variant generated on the fly.

Example: `$api_url/<sha256-hash>.png?w=32`

Deletion

`DELETE $api_url/<sha256-hash>(.ext)`

AUTH required

Note that the `/<sha256-hash>` part is from the **original** file, **not** from the **transformed** file if the uploaded file went through any server transformation.

The extension is optional as the file hash is the only needed file identification.

The server should reject deletes from users other than the original uploader with the appropriate http response code (403 Forbidden).

It should be noted that more than one user may have uploaded the same file (with the same hash). In this case, a delete must not really delete the file but just remove the user's pubkey from the file owners list (considering the server keeps just one copy of the same file, because multiple uploads of the same file results in the same file hash).

The successful response is a 200 OK one with just basic JSON fields:

```
{
  "status": "success",
  "message": "File deleted."
}
```

Listing files

`GET $api_url?page=x&count=y`

AUTH required

Returns a list of files linked to the authenticated users pubkey.

Example Response:

```
{
  "count": 1, // server page size, eg. max(1, min(server_max_page_size, arg_count))
  "total": 1, // total number of files
  "page": 0, // the current page number
  "files": [
    {
      "tags": [
```

```

    ["ox", "719171db19525d9d08dd69cb716a18158a249b7b3b3ec4bbdec5698dca104b7b"],
    ["x", "5d2899290e0e69bcd809949ee516a4a1597205390878f780c098707a7f18e3df"],
    ["size", "123456"],
    ["alt", "a meme that makes you laugh"],
    ["expiration", "1715691139"],
    // ...other metadata
  ],
  "content": "haha funny meme", // caption
  "created_at": 1715691130 // upload timestamp
}
]
}

```

files contains an array of NIP-94 events

Query args

- page page number (offset=page*count)
- count number of items per page

Selecting a Server

Note: HTTP File Storage Server developers may skip this section. This is meant for client developers.

A File Server Preference event is a kind 10096 replaceable event meant to select one or more servers the user wants to upload files to. Servers are listed as server tags:

```

{
  "kind": 10096,
  "content": "",
  "tags": [
    ["server", "https://file.server.one"],
    ["server", "https://file.server.two"]
  ],
  // other fields...
}

```


NIP-B7

Blossom media

draft optional

This NIP specifies how Nostr clients can use Blossom for handling media.

Blossom is a set of standards (called BUDs) for dealing with servers that store files addressable by their SHA-256 sums. Nostr clients may make use of all the BUDs for allowing users to upload files, manage their own files and so on, but most importantly Nostr clients SHOULD make use of BUD-03 to fetch kind:10063 lists of servers for each user:

```
{
  "id": "e4bee088334cb5d38cff1616e964369c37b6081be997962ab289d6c671975d71",
  "pubkey": "781208004e09102d7da3b7345e64fd193cd1bc3fce8fdae6008d77f9cabcd036",
  "content": "",
  "kind": 10063,
  "created_at": 1708774162,
  "tags": [
    ["server", "https://blossom.self.hosted"],
    ["server", "https://cdn.blossom.cloud"]
  ],
  "sig":
    "cc5efa74f59e80622c77cacf4dd62076bcb7581b45e9acff471e7963a1f4d8b3406addb5ee1ac9673487480e57d20e523428e60ffcc7e7a904ac882cf"
}
```

Whenever a Nostr client finds a URL in an event published by a given user and that URL ends a 64-character hex string (with or without an ending file extension) and that URL is not available anymore, that means that string is likely a representation of a sha256 and that the user may have a kind:10063 list of Blossom servers published.

Given that, the client SHOULD look into the kind:10063 list for other Blossom servers and lookup for the same 64-character hex string in them, by just using the hex string as a path (optionally with the file extension at the end), producing a URL like `https://blossom.self.hosted/<hex-string>.png`.

When downloading such files Nostr clients SHOULD verify that the sha256-hash of its contents matches the 64-character hex string.

More information can be found at BUD-03.

More complex interactions

Clients may use other facilities exposed by Blossom servers (for example, for checking if a file exists in a Blossom server, instead of actually downloading it) which are better documented in the BUDs.

NIP-78

Arbitrary custom app data

draft optional

The goal of this NIP is to enable remoteStorage-like capabilities for custom applications that do not care about interoperability.

Even though interoperability is great, some apps do not want or do not need interoperability, and it wouldn't make sense for them. Yet Nostr can still serve as a generalized data storage for these apps in a "bring your own database" way, for example: a user would open an app and somehow input their preferred relay for storage, which would then enable these apps to store application-specific data there.

Nostr event

This NIP specifies the use of event kind **30078** (an *addressable* event) with a `d` tag containing some reference to the app name and context – or any other arbitrary string. `content` and other tags can be anything or in any format.

Some use cases

- User personal settings on Nostr clients (and other apps unrelated to Nostr)
- A way for client developers to propagate dynamic parameters to users without these having to update
- Personal private data generated by apps that have nothing to do with Nostr, but allow users to use Nostr relays as their personal database

Security

NIP-06

Basic key derivation from mnemonic seed phrase

draft optional

BIP39 is used to generate mnemonic seed words and derive a binary seed from them.

BIP32 is used to derive the path `m/44'/1237'/<account>' /0/0` (according to the Nostr entry on SLIP44).

A basic client can simply use an account of `0` to derive a single key. For more advanced use-cases you can increment account, allowing generation of practically infinite keys from the 5-level path with hardened derivation.

Other types of clients can still get fancy and use other derivation paths for their own other purposes.

Test vectors

mnemonic: leader monkey parrot ring guide accident before fence cannon height naive bean

private key (hex): 7f7ff03d123792d6ac594bfa67bf6d0c0ab55b6b1fdb6249303fe861f1ccba9a

nsec: nsec10allq0gjx7fddtzef0ax00mdps9t2kmtrldkyjfs8l5xruwvh2dq0lhhkp

public key (hex): 17162c921dc4d2518f9a101db33695df1afb56ab82f5ff3e5da6eec3ca5cd917

npub: npub1zutzeysacnf9rru6zqwmxd54mud0k44tst6l70ja5mhv8jjumytsd2x7nu

mnemonic: what bleak badge arrange retreat wolf trade produce cricket blur garlic valid proud rude strong choose

busy staff weather area salt hollow arm fade

private key (hex): c15d739894c81a2fcfd3a2df85a0d2c0dbc47a280d092799f144d73d7ae78add

nsec: nsec1c9wh8xy5eqdzln7n5t0ctgxjcrdug73gp5yj0x03gntn67h83twssdfhel

public key (hex): d41b22899549e1f3d335a31002cfd382174006e166d3e658e3a5eecdb6463573

npub: npub16sdj9zv4f8sl85e45vgq9n7nsgt5qphpvmf7vk8r5hhvmdjxx4es8rq74h

NIP-49

Private Key Encryption

draft optional

This NIP defines a method by which clients can encrypt (and decrypt) a user's private key with a password.

Symmetric Encryption Key derivation

PASSWORD = Read from the user. The password should be unicode normalized to NFKC format to ensure that the password can be entered identically on other computers/clients.

LOG_N = Let the user or implementer choose one byte representing a power of 2 (e.g. 18 represents 262,144) which is used as the number of rounds for scrypt. Larger numbers take more time and more memory, and offer better protection:

LOG_N	MEMORY REQUIRED	APPROX TIME ON FAST COMPUTER
16	64 MiB	100 ms
18	256 MiB	
20	1 GiB	2 seconds
21	2 GiB	
22	4 GiB	

SALT = 16 random bytes

SYMMETRIC_KEY = scrypt(password=PASSWORD, salt=SALT, log_n=LOG_N, r=8, p=1)

The symmetric key should be 32 bytes long.

This symmetric encryption key is temporary and should be zeroed and discarded after use and not stored or reused for any other purpose.

Encrypting a private key

The private key encryption process is as follows:

PRIVATE_KEY = User's private (secret) secp256k1 key as 32 raw bytes (not hex or bech32 encoded!)

KEY_SECURITY_BYTE = one of:

- 0x00 - if the key has been known to have been handled insecurely (stored unencrypted, cut and paste unencrypted, etc)
- 0x01 - if the key has NOT been known to have been handled insecurely (stored unencrypted, cut and paste unencrypted, etc)
- 0x02 - if the client does not track this data

ASSOCIATED_DATA = KEY_SECURITY_BYTE

NONCE = 24 byte random nonce

CIPHERTEXT = XChaCha20-Poly1305(plaintext=PRIVATE_KEY, associated_data=ASSOCIATED_DATA, nonce=NONCE, key=SYMMETRIC_KEY)

VERSION_NUMBER = 0x02

CIPHERTEXT_CONCATENATION = concat(VERSION_NUMBER, LOG_N, SALT, NONCE, ASSOCIATED_DATA, CIPHERTEXT)

ENCRYPTED_PRIVATE_KEY = bech32_encode('ncryptsec', CIPHERTEXT_CONCATENATION)

The output prior to bech32 encoding should be 91 bytes long.

The decryption process operates in the reverse.

Test Data

Password Unicode Normalization

The following password input: “ÅΩİ” - Unicode Codepoints: U+212B U+2126 U+1E9B U+0323 - UTF-8 bytes: [0xE2, 0x84, 0xAB, 0xE2, 0x84, 0xA6, 0xE1, 0xBA, 0x9B, 0xCC, 0xA3]

Should be converted into the unicode normalized NFKC format prior to use in scrypt: “ÅΩİ” - Unicode Codepoints: U+00C5 U+03A9 U+1E69 - UTF-8 bytes: [0xC3, 0x85, 0xCE, 0xA9, 0xE1, 0xB9, 0xA9]

Encryption

The encryption process is non-deterministic due to the random nonce.

Decryption

The following encrypted private key:

```
ncryptsec1qgg9947r1pvqu76pj5ecreduf9jxhse1q2nae2kghhvd5g7dgjtcxfqtd67p9m0w57lspw8gsq6yphnm8623ns18xn9j4jdzz8
```

When decrypted with password=‘nostr’ and log_n=16 yields the following hex-encoded private key:

```
3501454135014541350145413501453fefb02227e449e57cf4d3a3ce05378683
```

Discussion

On Key Derivation

Passwords make poor cryptographic keys. Prior to use as a cryptographic key, two things need to happen:

1. An encryption key needs to be deterministically created from the password such that it has a uniform functionally random distribution of bits, such that the symmetric encryption algorithm’s assumptions are valid, and
2. A slow irreversible algorithm should be injected into the process, so that brute-force attempts to decrypt by trying many passwords are severely hampered.

These are achieved using a password-based key derivation function. We use scrypt, which has been proven to be maximally memory hard and which several cryptographers have indicated to the author is better than argon2 even though argon2 won a competition in 2015.

On the symmetric encryption algorithm

XChaCha20-Poly1305 is typically favored by cryptographers over AES and is less associated with the U.S. government. It (or its earlier variant without the ‘X’) is gaining wide usage, is used in TLS and OpenSSH, and is available in most modern crypto libraries.

Recommendations

It is not recommended that users publish these encrypted private keys to nostr, as cracking a key may become easier when an attacker can amass many encrypted private keys.

It is recommended that clients zero out the memory of passwords and private keys before freeing that memory.

NIP-98

HTTP Auth

draft optional

This NIP defines an ephemeral event used to authorize requests to HTTP servers using nostr events.

This is useful for HTTP services which are built for Nostr and deal with Nostr user accounts.

Nostr event

A kind 27235 (In reference to RFC 7235) event is used.

The content SHOULD be empty.

The following tags MUST be included.

- u - absolute URL
- method - HTTP Request Method

Example event:

```
{
  "id": "fe964e758903360f28d8424d092da8494ed207cba823110be3a57dfe4b578734",
  "pubkey": "63fe6318dc58583cfe16810f86dd09e18bfd76aabc24a0081ce2856f330504ed",
  "content": "",
  "kind": 27235,
  "created_at": 1682327852,
  "tags": [
    ["u", "https://api.snort.social/api/v1/n5sp/list"],
    ["method", "GET"]
  ],
  "sig":
    "5ed9d8ec958bc854f997bdc24ac337d005af372324747efe4a00e24f4c30437ff4dd8308684bed467d9d6be3e5a517bb43b1732cc7d33949a3aaf86705"
}
```

Servers **MUST** perform the following checks in order to validate the event: 1. The kind **MUST** be 27235. 2. The `created_at` timestamp **MUST** be within a reasonable time window (suggestion 60 seconds). 3. The `u` tag **MUST** be exactly the same as the absolute request URL (including query parameters). 4. The `method` tag **MUST** be the same HTTP method used for the requested resource.

When the request contains a body (as in POST/PUT/PATCH methods) clients SHOULD include a SHA256 hash of the request body in a payload tag as hex ([`"payload", "<sha256-hex>"`]), servers MAY check this to validate that the requested payload is authorized.

If one of the checks was to fail the server SHOULD respond with a 401 Unauthorized response code.

Servers MAY perform additional implementation-specific validation checks.

Request Flow

Using the Authorization HTTP header, the kind 27235 event MUST be base64 encoded and use the Authorization scheme Nostr

Example HTTP Authorization header:

Authorization: Nostr

evJpZCI6ImZl0TY0ZTc1ODkwMzM2MGYyOG04NDI0ZDA5MmRhOD05NGVkmjA3Y2JhODIzMTEwYmUzYTU3ZGZlNGI1Nza3MzoiLCJwdWJrZXkiOiI2M2ZlNiMxOGRjNTg1OD

Reference Implementations

- C# ASP.NET AuthenticationHandler NostrAuth.cs

Developers

NIP-07

window.nostr capability for web browsers

draft optional

The window.nostr object may be made available by web browsers or extensions and websites or web-apps may make use of it after checking its availability.

That object must define the following methods:

```
async window.nostr.getPublicKey(): string // returns a public key as hex
async window.nostr.signEvent(event: { created_at: number, kind: number, tags: string[], content: string }): Event
// takes an event object, adds `id`, `pubkey` and `sig` and returns it
```

Aside from these two basic above, the following functions can also be implemented optionally:

```
async window.nostr.nip04.encrypt(pubkey, plaintext): string // returns ciphertext and iv as specified in nip-04
(deprecated)
async window.nostr.nip04.decrypt(pubkey, ciphertext): string // takes ciphertext and iv as specified in nip-04
(deprecated)
async window.nostr.nip44.encrypt(pubkey, plaintext): string // returns ciphertext as specified in nip-44
async window.nostr.nip44.decrypt(pubkey, ciphertext): string // takes ciphertext as specified in nip-44
```

Recommendation to Extension Authors

To make sure that the window.nostr is available to nostr clients on page load, the authors who create Chromium and Firefox extensions should load their scripts by specifying "run_at": "document_end" in the extension's manifest.

Implementation

See <https://github.com/aljazceru/awesome-nostr#nip-07-browser-extensions>.

NIP-31

Dealing with unknown event kinds

draft optional

When creating a new custom event kind that is part of a custom protocol and isn't meant to be read as text (like `kind:1`), clients should use an `alt` tag to write a short human-readable plaintext summary of what that event is about.

The intent is that social clients, used to display only `kind:1` notes, can still show something in case a custom event pops up in their timelines. The content of the `alt` tag should provide enough context for a user that doesn't know anything about this event kind to understand what it is.

These clients that only know `kind:1` are not expected to ask relays for events of different kinds, but users could still reference these weird events on their notes, and without proper context these could be nonsensical notes. Having the fallback text makes that situation much better – even if only for making the user aware that they should try to view that custom event elsewhere.

`kind:1`-centric clients can make interacting with these event kinds more functional by supporting NIP-89.

NIP-89

Recommended Application Handlers

draft optional

This NIP describes kind:31989 and kind:31990: a way to discover applications that can handle unknown event-kinds.

Rationale

Nostr's discoverability and transparent event interaction is one of its most interesting/novel mechanics. This NIP provides a simple way for clients to discover applications that handle events of a specific kind to ensure smooth cross-client and cross-kind interactions.

Parties involved

There are three actors to this workflow:

- application that handles a specific event kind (note that an application doesn't necessarily need to be a distinct entity and it could just be the same pubkey as user A)
 - Publishes kind:31990, detailing how apps should redirect to it
- user A, who recommends an app that handles a specific event kind
 - Publishes kind:31989
- user B, who seeks a recommendation for an app that handles a specific event kind
 - Queries for kind:31989 and, based on results, queries for kind:31990

Events

Recommendation event

```
{
  "kind": 31989,
  "pubkey": <recommender-user-pubkey>,
  "tags": [
    ["d", <supported-event-kind>],
    ["a", "31990:app1-pubkey:<d-identifier>", "wss://relay1", "ios"],
    ["a", "31990:app2-pubkey:<d-identifier>", "wss://relay2", "web"]
  ],
  // other fields...
}
```

The d tag in kind:31989 is the supported event kind this event is recommending.

Multiple a tags can appear on the same kind:31989.

The second value of the tag SHOULD be a relay hint. The third value of the tag SHOULD be the platform where this recommendation might apply.

Handler information

```
{
  "kind": 31990,
  "pubkey": "<application-pubkey>",
  "content": "<optional-kind:0-style-metadata>",
  "tags": [
    ["d", <random-id>],
    ["k", <supported-event-kind>],
    ["web", "https://.../a/<bech32>", "nevent"],
    ["web", "https://.../p/<bech32>", "nprofile"],
  ]
}
```

```

    ["web", "https://.../e/<bech32>"],
    ["ios", ".../<bech32>"]
  ],
  // other fields...
}

```

- content is an optional metadata-like stringified JSON object, as described in NIP-01. This content is useful when the pubkey creating the kind:31990 is not an application. If content is empty, the kind:0 of the pubkey should be used to display application information (e.g. name, picture, web, LUD16, etc.)
- k tags' value is the event kind that is supported by this kind:31990. Using a k tag(s) (instead of having the kind of the d tag) provides:
 - Multiple k tags can exist in the same event if the application supports more than one event kind and their handler URLs are the same.
 - The same pubkey can have multiple events with different apps that handle the same event kind.
- bech32 in a URL MUST be replaced by clients with the NIP-19-encoded entity that should be loaded by the application.

Multiple tags might be registered by the app, following NIP-19 nomenclature as the second value of the array.

A tag without a second value in the array SHOULD be considered a generic handler for any NIP-19 entity that is not handled by a different tag.

Client tag

When publishing events, clients MAY include a client tag. Identifying the client that published the note. This tag is a tuple of name, address identifying a handler event and, a relay hint for finding the handler event. This has privacy implications for users, so clients SHOULD allow users to opt-out of using this tag.

```

{
  "kind": 1,
  "tags": [
    ["client", "My Client", "31990:app1-pubkey:<d-identifier>", "wss://relay1"]
  ]
  // other fields...
}

```

User flow

A user A who uses a non-kind:1-centric nostr app could choose to announce/recommend a certain kind-handler application.

When user B sees an unknown event kind, e.g. in a social-media centric nostr client, the client would allow user B to interact with the unknown-kind event (e.g. tapping on it).

The client MIGHT query for the user's and the user's follows handler.

Example

User A recommends a kind:31337-handler

User A might be a user of Zapstr, a kind:31337-centric client (tracks). Using Zapstr, user A publishes an event recommending Zapstr as a kind:31337-handler.

```

{
  "kind": 31989,
  "tags": [
    ["d", "31337"],
    ["a", "31990:1743058db7078661b94aaf4286429d97ee5257d14a86d6bfa54cb0482b876fb0:abcd", "<relay-url>", "web"]
  ],
  // other fields...
}

```

```
}
```

User B interacts with a kind:31337-handler

User B might see in their timeline an event referring to a kind:31337 event (e.g. a kind:1 tagging a kind:31337).

User B's client, not knowing how to handle a kind:31337 might display the event using its alt tag (as described in NIP-31). When the user clicks on the event, the application queries for a handler for this kind:

```
["REQ", <id>, { "kinds": [31989], "#d": ["31337"], "authors": [<user>, <users-contact-list>] }]
```

User B, who follows User A, sees that kind:31989 event and fetches the a-tagged event for the app and handler information.

User B's client sees the application's kind:31990 which includes the information to redirect the user to the relevant URL with the desired entity replaced in the URL.

Alternative query bypassing kind:31989

Alternatively, users might choose to query directly for kind:31990 for an event kind. Clients SHOULD be careful doing this and use spam-prevention mechanisms or querying high-quality restricted relays to avoid directing users to malicious handlers.

```
["REQ", <id>, { "kinds": [31990], "#k": [<desired-event-kind>], "authors": [...] }]
```

Conclusion

Thank you for exploring the nostr-book. My hope is that this reorganized collection of Nostr Notes in Progress (NIPs) has provided you with a clearer and more structured understanding of the Nostr protocol. By grouping similar NIPs together, the aim was to create a logical flow that enhances comprehension and makes the information more accessible to everyone.

As we wrap up this book, remember that the journey with Nostr doesn't end here. The protocol is continuously evolving, and your engagement and contributions are crucial for its growth and refinement. I encourage you to participate in the discussions, contribute your ideas, and help in developing this open and decentralized platform.

Once again, all the credit for the content in this book goes to the original authors of the NIPs. This compilation is merely a tool to assist in navigating their innovative work. Whether you're a developer, researcher, or enthusiast, your insights and enthusiasm are what will propel Nostr forward.

Let's keep the spirit of innovation and collaboration alive. Here's to building a more connected and decentralized future together!